CBSE 2026-27

EduSpark

SCIENCE

ASTRA SERIES

CLASS 10 • CBSE • BOARD EXAM 2027

$E=mc^2$

60 DAYS MEIN PURA SYLLABUS **KHATAM!**



INSIDE THIS BOOK

- **Complete Notes**
  With Real Images & Diagrams
- **Timelines & Flowcharts**
  For Better Understanding
- **MCQs, Assertion-Reason**
  Case-Based Questions
- **All 3 Parts Covered**
  Physics | Chemistry | Biology
- **English + Hindi Explanation**
  Easy to Understand
- **60-Day Study Plan**
  With Mock Tests & Revision Days

By Nitesh Bhaiya

- **Concepts** That Stick
- **PYQ Analysis** Eduspark board trends
- **Memes** Concepts that stick

Chemistry | Physics | Biology
13 chapters, 60 levels, NCERT based

ASTRA SERIES
Free e-book by Nitesh Prakash

IMPRINT

PUBLISHER'S INFORMATION

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ASTRA SERIES

SCIENCE CLASS 10

CBSE | 60 Days Complete Plan | Session 2026-27

by Nitesh Prakash

BOOK DETAILS

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ERRATA	Found a mistake? Write to us and we will fix it in the next edition.

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Dear student, this book was made with one wish: that you walk into the exam hall without fear. Every chapter and question has been checked against the latest NCERT textbook and the CBSE syllabus for 2026-27. Question tags marked Eduspark PYQ Analysis are based on our own analysis, not an official CBSE record. If you spot a mistake, tell us and we will fix it in the next edition.

Nitesh Prakash

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FROM THE AUTHOR'S DESK

NOTE FROM NITESH BHAIYA

A few words before you start Level 1.

Paheli and Bujho say: Level 1 is waiting!



NITESH PRAKASH

Author and Educator, Eduspark

"Paheli samjhayegi, Bujho poochega, aur tum seekhoge."

CBSE CLASS 10 | SCIENCE | 60 DAYS

Dear student,

Main Nitesh hoon, and this book is my promise to you: one clear day at a time, no topic left half-done.

Every year I meet students who work hard for months and still walk into the Science paper nervous. Not because they are weak, but because their notes are scattered, their guide is too thick and nobody showed them what the board actually asks. Chemistry ke equations, Physics ke ray diagrams, Biology ke labelled diagrams: sab ek saath feels like too much.

So I built ASTRA differently. The full NCERT syllabus for 2026-27 is split into 60 days. Each day is one level: read the concept, laugh a little with Paheli and Bujho, practise the exact kind of questions the board loves, and collect your XP. Boss Levels test a full chapter, Swahaa pages help you revise it, and the Grand Mock on Day 60 shows you how ready you are.

Keep your NCERT textbook beside this book, solve every question honestly before you check the answer, and be truthful in the self-assessment tracker, because it is only for you. Every R and RA you mark is a mark you are about to win back.

Some days will feel easy and some will feel heavy. That is normal. Bas rukna nahi hai. This e-book is free for every student, so share it with a friend who needs it, and if you spot a mistake, tell us. I believe in you, and I am proud of you already.

With love and best wishes,

Nitesh Prakash

AUTHOR, ASTRA SERIES
EDUSPARK

> *Roz ek level. 60 din. Phir board exam bas ek aur level hai, aur woh bhi tum clear karoge.*
>
> NITESH BHAIYA

1 ONE DAY, ONE LEVEL
Finish today's level fully before you open tomorrow's.

2 HONEST PRACTICE
Attempt first, check later. Your mistakes are your map.

3 REVISE ON TIME
Use every Swahaa page. Revision turns effort into marks.

BEFORE YOU START

HOW TO USE THIS BOOK

One complete CBSE Class 10 Science plan. One day = one level = 4 pages.

60	13
LEVELS	CHAPTERS
8000	4
TOTAL XP	PAGES A DAY

CHEMISTRY Days 1–20 | Pages 9–92

PHYSICS Days 21–40 | Pages 93–176

BIOLOGY Days 41–60 | Pages 177–261

1 YOUR PATH THROUGH THE BOOK

STEP 1 — Read the concept
Theory in NCERT order with diagrams, tables, formulas and Class Tricks.
EVERY DAY

STEP 2 — Paheli-Bujho and memes
Paheli explains, Bujho asks your doubts. A few memes lock the idea in.
EVERY DAY

STEP 3 — Practice
MCQ, A-R, case-based, short and long answers. Then check with the answer key.
EVERY DAY

STEP 4 — Boss Level
A full chapter test on the last day of the chapter. No notes, timer on.
CHAPTER END

STEP 5 — Level Complete and XP
Tick the 5-item checklist and claim the XP you earned on the Level Complete card.
EVERY DAY

STEP 6 — Swahaa revision page
One-page chapter recap, the chapter XP you earned and your book progress.
CHAPTER END

STEP 7 — Grand Mock
A full 80-mark CBSE board-pattern paper on all 13 chapters, with answer key.
DAY 60

2 BOXES YOU WILL MEET

+125 XP
XP chip on each Day banner: the XP that Level is worth.

CLASS TRICK
Mnemonics and shortcuts to recall formulas and lists fast.

PAHELI **BUJHO**
The dialogue box: concept explained through real student doubts.

EDUSPARK PYQ ANALYSIS
How often a topic has come in board papers. Prioritise these.

EXTRA KNOWLEDGE
Beyond the syllabus, for curiosity and olympiads. Optional.

BOSS
Chapter test day. Clear it before you start the next chapter.

SWAHAA
Chapter celebration and recap page. Revise from it later.

LEVEL COMPLETE
End-of-day card: checklist, score and XP collected.

3 THE XP SYSTEM

1 CBSE mark = **100 XP**

1	60	13	8000
Day = 1 Level	Levels in all	Swahaa pages	XP in the book

Chemistry	2500
Physics	2500
Biology	3000
BOOK TOTAL	**8000 XP**

Each chapter's XP is listed in the Index (page 5). Collect all 8000 XP and you have covered the full board syllabus.

4 SUGGESTED DAILY ROUTINE

- **5 min** — **Warm up.** Revise yesterday's tricks and mistakes.
- **20 min** — **Theory.** Read the notes, diagrams and the Paheli and Bujho dialogue.
- **5 min** — **Tricks and memes.** Say every mnemonic aloud once.
- **20 min** — **Practice.** Solve all questions without looking at answers.
- **10 min** — **Check and claim.** Use the answer key, tick the checklist, note your score on the Self Assessment page (page 8) and collect your XP.

About 60 minutes a day. 60 days. Board ready.

GOLDEN RULES

1. Never skip a day. If you miss one, do two the next day.
2. Attempt first, check later. Mark every mistake honestly.
3. Clear the Boss Level before you open the next chapter.

ASTRA SERIES | SCIENCE CLASS 10

CONTENTS & BOOK MAP

60 Days, 13 chapters, 3 parts, **8000 XP**
Every day = 4 pages: theory, tricks, practice, answer key
Boss Level ends each chapter, Swahaa page celebrates it

Cover 1 | Publisher's Info 2 | Author's Note 3 | How to Use 4 | Contents 5 | OMR Guide 6 | Admit Card 7 | Self Assessment 8

CHEMISTRY
Days 1–20 • Pages 9–92 • 2500 XP

CH 1 Chemical Reactions and Equations +500 XP
Day 1 Chemical Reactions and Equations 9
Day 2 Combination and Decomposition 13
Day 3 Displacement and Redox Reactions 17
Day 4 Energy, Corrosion and Rancidity 21
★ Chapter 1 Swahaa 25

CH 2 Acids, Bases and Salts +625 XP
Day 5 Acids, Bases and Salts 26
Day 6 Ions, Strength and the pH Scale 30
Day 7 pH in Daily Life and Salts 34
Day 8 Chemicals from Common Salt 38
Day 9 Chapter 2 Boss Level BOSS 42
★ Chapter 2 Swahaa 46

CH 3 Metals and Non-metals +625 XP
Day 10 Metals and their Reactions 47
Day 11 Ionic Bonds and Compounds 51
Day 12 Extraction of Metals 55
Day 13 Corrosion and Alloys 59
Day 14 Chapter 3 Boss Level BOSS 63
★ Chapter 3 Swahaa 67

CH 4 Carbon and its Compounds +750 XP
Day 15 Carbon and its Compounds 68
Day 16 Hydrocarbons and Homologous Series 72
Day 17 Functional Groups and IUPAC Names 76
Day 18 Properties of Carbon Compounds 80
Day 19 Soaps and Detergents 84
Day 20 Chapter 4 Boss + Chemistry Mock BOSS + MOCK 88
★ Chapter 4 Swahaa 92

PHYSICS
Days 21–40 • Pages 93–176 • 2500 XP

CH 5 Light: Reflection and Refraction +900 XP
Day 21 Light and Reflection 93
Day 22 Ray Diagrams and Mirror Images 97
Day 23 Mirror Formula and Numericals 101
Day 24 Refraction and Refractive Index 105
Day 25 Spherical Lenses and Images 109
Day 26 Lens Formula and Power 113
Day 27 Chapter 5 Boss Level BOSS 117
★ Chapter 5 Swahaa 121

CH 6 The Human Eye and the Colourful World +300 XP
Day 28 The Human Eye 122
Day 29 Colourful World + Boss BOSS 126
★ Chapter 6 Swahaa 130

CH 7 Electricity +700 XP
Day 30 Current and Potential Difference 131
Day 31 Ohm's Law and Resistivity 135
Day 32 Resistors in Series 139
Day 33 Parallel and Mixed Circuits 143
Day 34 Heating Effect and Electric Power 147
Day 35 Chapter 7 Boss Level BOSS 151
★ Chapter 7 Swahaa 155

CH 8 Magnetic Effects of Electric Current +600 XP
Day 36 Magnets and Magnetic Field Lines 156
Day 37 Field of a Straight Wire and a Loop 160
Day 38 Solenoid and Electromagnet 164
Day 39 Fleming's Rule and Home Circuits 168
Day 40 Chapter 8 Boss + Physics Mock BOSS + MOCK 172
★ Chapter 8 Swahaa 176

BIOLOGY
Days 41–60 • Pages 177–261 • 3000 XP

CH 9 Life Processes +900 XP
Day 41 Life Processes and Photosynthesis 177
Day 42 Heterotrophic Nutrition and Digestion 181
Day 43 Respiration and Breathing 185
Day 44 Transportation in Humans 189
Day 45 Plant Transport and Excretion 193
Day 46 Chapter 9 Boss Level BOSS 197
★ Chapter 9 Swahaa 201

CH 10 Control and Coordination +575 XP
Day 47 Nervous System and the Human Brain 202
Day 48 Coordination in Plants 206
Day 49 Hormones in Animals and Feedback 210
Day 50 Chapter 10 Boss Level BOSS 214
★ Chapter 10 Swahaa 218

CH 11 How do Organisms Reproduce? +575 XP
Day 51 Asexual Reproduction and Variation 219
Day 52 Plant Reproduction and the Flower 223
Day 53 Reproduction in Human Beings 227
Day 54 Chapter 11 Boss Level BOSS 231
★ Chapter 11 Swahaa 235

CH 12 Heredity +450 XP
Day 55 Mendel and the Monohybrid Cross 236
Day 56 Dihybrid Cross and Sex Determination 240
Day 57 Chapter 12 Boss Level BOSS 244
★ Chapter 12 Swahaa 248

CH 13 Our Environment +500 XP
Day 58 Ecosystem and Food Chains 249
Day 59 Our Environment II + Boss BOSS 253
Day 60 Science Grand Mock, 80 marks GRAND MOCK 257
★ Ch 13 Swahaa + Book Complete 261

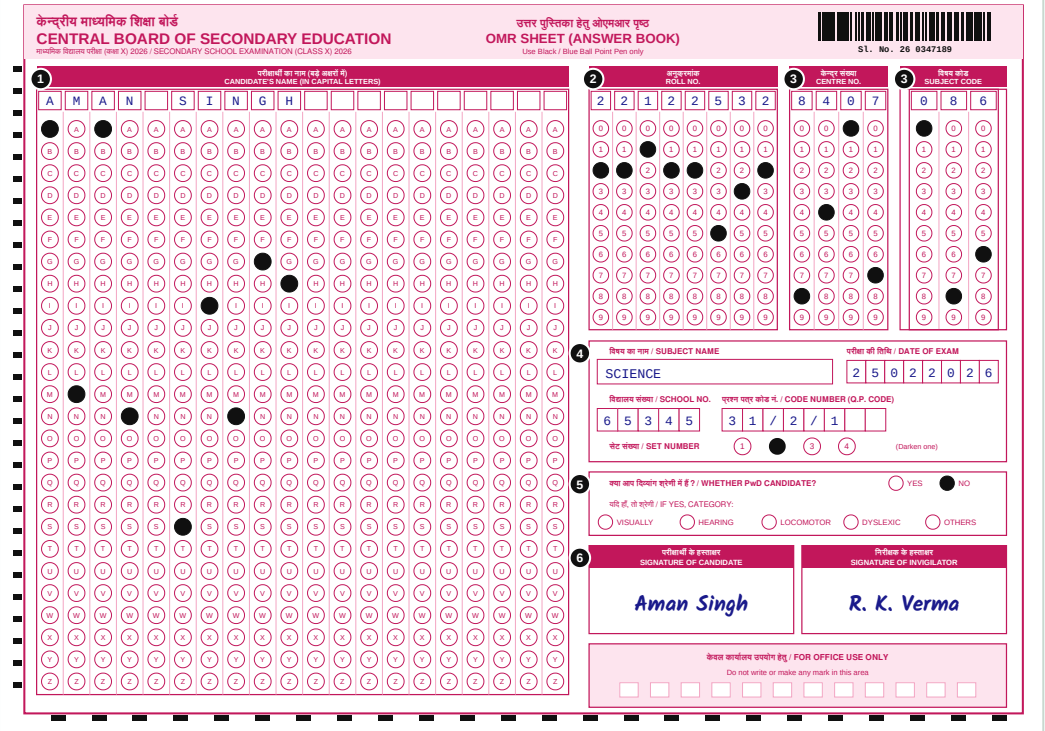
BOSS Full chapter test | GRAND MOCK 80-mark paper | ★ Swahaa: recap + XP | Total **8000 XP** | **Page 262**: review QR + ASTRA books

BEFORE THE BOARD EXAM

HOW TO FILL THE OMR SHEET

One wrong bubble can hold back your result. Fill it right, first time.

CBSE CLASS X BOARD EXAM



Sample OMR sheet, filled correctly. Black circled numbers match the steps below. *Details shown are imaginary.*

BEFORE YOU START

Pen
Blue or black ballpoint only. No gel, pencil or sketch pen.

Admit card
Keep it beside the sheet. Copy roll no., centre no. and subject code from it.

Darken fully
Fill the whole bubble, dark and inside the circle. One bubble per column.

Check twice
Match every digit before you sign. A wrong roll no. can hold your result.

No damage
Do not fold, staple or use whitener on the sheet.

1 FILL IT STEP BY STEP

1 Name
Write it in CAPITAL letters, one letter per box, exactly as on your admit card. Leave one box blank between words and darken the matching letter bubble below each box.

2 Roll No.
Copy all 8 digits from the admit card into the boxes, then darken one bubble per column (0 to 9).

3 Centre No. and Subject Code
Fill them the same way, digit by digit. Science is **086**.

4 Subject, date, school no.
Copy from the admit card. Take the **Set Number** and **Q.P. Code** from the question paper cover, every "/" included.

5 PwD
Darken YES and your category only if it applies to you; otherwise darken NO.

6 Signatures
Sign only in your own box. The invigilator signs in theirs after checking your entries.

2 CHECK BEFORE YOU HAND IT IN

WHERE EACH ENTRY COMES FROM

Admit card	Name, Roll No., Centre No., School No., Subject, Subject Code, Date of Exam
Q.P. cover	Set Number, Code Number (Q.P. Code)
Yourself	PwD status, your signature
Invigilator	Invigilator's signature, after checking the sheet

MISTAKES THAT GET AN OMR REJECTED

1. Roll No. written but bubbles left empty
2. Two bubbles darkened in one column
3. Q.P. Code copied without the "/" marks
4. Pencil, gel pen, whitener or overwriting used
5. Sheet folded, torn or written on in the office area

DARKENING A BUBBLE ● Correct ✓ Wrong ⊗ Wrong ⊙ Wrong ◑ Wrong — Blue or black ballpoint pen only

IMPORTANT: READ BEFORE THE EXAM

YOUR ADMIT CARD

Know every entry on it. Circled numbers show what you copy onto the OMR sheet.

CBSE CLASS X EXAM 2026

केन्द्रीय माध्यमिक शिक्षा बोर्ड, दिल्ली

CENTRAL BOARD OF SECONDARY EDUCATION, DELHI

ADMIT CARD FOR SECONDARY EXAMINATION 2026

Digital CBSE

6 ROLL NO.	22122532	DOB	14/08/2010	9 SCHOOL NO.	65345	8 CENTRE NO.	8407

Roll No. in Words	7 TWO CRORE TWENTY ONE LAKH TWENTY TWO THOUSAND FIVE HUNDRED THIRTY TWO
Candidate's Name	4 AMAN SINGH
Mother's Name	SUNITA DEVI
Father's / Guardian's Name	5 RAJESH KUMAR SINGH
Examination Centre	8407 : SAMPLE PUBLIC SCHOOL, SECTOR 12, NEW TOWN
PwD Category	NO
Admit Card ID	AC26X0347189

PHOTOGRAPH

2 SUB CODE	1 SUBJECT NAME	MEDIUM	3 DATE	INVIGILATOR SIGN
041	MATHEMATICS STANDARD	ENGLISH	17/02/2026	
184	ENGLISH LNG & LIT.	ENGLISH	21/02/2026	
086	**SCIENCE**	**ENGLISH**	**25/02/2026**	
002	HINDI COURSE-A	HINDI	02/03/2026	
087	SOCIAL SCIENCE	ENGLISH	07/03/2026	

ON THE QUESTION PAPER — **Series:** ZXYW4 | 10 **Set No.** 2 | 11 **Q.P. Code** 31/2/1

Aman Singh — Signature of Candidate

Signature of Principal with Seal

S. Bhardwaj — Controller of Examinations

Circles 1 to 11: entries you copy onto the OMR sheet (page 3). 10 and 11 come from the question paper.

Details shown are imaginary.

! BEFORE AND ON EXAM DAY

WHAT TO CHECK

1. Spelling of your name, mother's and father's name.
2. Photo is yours; date of birth is correct.
3. Every subject and code is right (Science = 086).
4. Centre name and address; visit it a day before.
5. Principal's signature and school seal are present.
6. Any mistake: report to your school at once, never on exam day.

EXAM DAY REPORTING

- 9:30 — Reach the centre
- 10:00 — Entry closes
- 10:15 — Question paper given, 15 min reading
- 10:30 — Start writing
- 1:30 — Exam ends (3 hours)

Always follow the times printed on your own admit card.

DOS AND DON'TS

- ✓ Carry the admit card and school ID card.
- ✓ Blue or black ballpoint pens in a transparent pouch.
- ✓ School uniform; transparent water bottle.
- ✗ No phone, smartwatch, calculator or any device.
- ✗ No chits, and nothing written on the admit card.
- ✗ Never write your name or roll no. inside the answer book.

KNOW WHERE YOU STAND

SELF ASSESSMENT

Check yourself after every level. Your score today shows you what to revise tomorrow.

WHY ASSESS YOURSELF?

1. You find your weak topics early, while there is still time to fix them.
2. Marks stop being a surprise: you know exactly where you stand before the board exam.
3. Revision becomes targeted. You re-read only what you actually got wrong.
4. Seeing your scores improve day by day builds real exam confidence.
5. It builds the habit of checking your own answers like an examiner.

HOW TO SELF-EVALUATE

1. Finish the day's practice questions (or the Boss Level) without looking at the answers.
2. Check every answer with the answer key and the marking points.
3. Write your score in the tracker below (marks scored / total marks).
4. Put a remark: E, R or RA (see the key). Be honest, nobody else is grading.
5. Revise R and RA days within a week and write the date in Revised on.

E — EASY
Concept clear, almost all correct. Move ahead.

R — REVISE
Small mistakes. Revise the tricks and wrong questions once more.

RA — RE-ATTEMPT
Many mistakes. Study the topic again and redo all questions.

MY 60-DAY TRACKER

CHEMISTRY — Days 1–20

DAY	SCORE	E / R / RA	REVISED ON
1			
2			
3			
4			
5			
6			
7			
8			
9 B			
10			
11			
12			
13			
14 B			
15			
16			
17			
18			
19			
20			

PHYSICS — Days 21–40

DAY	SCORE	E / R / RA	REVISED ON
21			
22			
23			
24			
25			
26			
27 B			
28			
29			
30			
31			
32			
33			
34			
35 B			
36			
37			
38			
39			
40			

BIOLOGY — Days 41–60

DAY	SCORE	E / R / RA	REVISED ON
41			
42			
43			
44			
45			
46 B			
47			
48			
49			
50 B			
51			
52			
53			
54 B			
55			
56			
57 B			
58			
59 B			
60	GRAND MOCK / 80		

MY XP PROGRESS TRACKER

Write the XP you earned per chapter (1 CBSE mark = 100 XP). Chemistry 2500, Physics 2500, Biology 3000.

Ch 1 Chemical Reactions _____ / 500	Ch 2 Acids, Bases, Salts _____ / 625	Ch 3 Metals and Non-metals _____ / 625	Ch 4 Carbon Compounds _____ / 750	Ch 5 Light _____ / 900	Ch 6 Human Eye _____ / 300	Ch 7 Electricity _____ / 700
Ch 8 Magnetic Effects _____ / 600	Ch 9 Life Processes _____ / 900	Ch 10 Control and Coordination _____ / 575	Ch 11 How do Organisms Reproduce _____ / 575	Ch 12 Heredity _____ / 450	Ch 13 Our Environment _____ / 500	TOTAL Science, 60 Levels _____ / 8000

B *Boss Level day (chapter test). Shaded rows: Day 20 and 40 (Boss + Mock), Day 60 Science Grand Mock.*

DAY 01 · LEVEL 1

CHEMISTRY | CHAPTER 1 | PART 1 · ASTRA SERIES

CHEMICAL REACTIONS & EQUATIONS

Goal: identify reactions, write and balance equations.



by Nitesh Prakash

01 Reactions | 02 5 Signs | 03 Equations | 04 Balancing | 05 States & Conditions | +100 XP today | Level 1 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 PHYSICAL VS CHEMICAL CHANGE

PHYSICAL	CHEMICAL
× No new substance; only shape, size or state changes	✔ A **new substance** with new properties forms
Usually reversible	Usually irreversible
Melting ice, dissolving sugar, breaking glass	Milk → curd, rusting, cooking, burning paper

2 WHAT IS A CHEMICAL REACTION?

DEFINITION

- Substances change into **new substances** with **new properties**
- Old bonds break and **new bonds form**
- Atoms are only **rearranged**, never created or destroyed

REACTANTS (what you start with (left side)) → bonds break and re-form → **PRODUCTS** (what is formed (right side))

$2Mg_{(s)} + O_{2(g)} \rightarrow 2MgO_{(s)}$

Reactants: magnesium, oxygen | Product: magnesium oxide



MEME BREAK

*Sodium (catches fire in water) + chlorine (poisonous gas) = the salt in your dal! New properties = **chemical reaction**.*



BUJHO *Doodh garam kiya, reaction hua?*

PAHELI *Nahi! Wo physical change hai. Doodh khatta hona chemical hai!*



EXOTHERMIC ★★★	ENDOTHERMIC
Heat **given out**; write + **Heat on the product side**	Heat **absorbed**; + **Heat on the reactant side** (or Δ on arrow)
Burning Mg, coal, natural gas; respiration; CaO + water	Heating $CaCO_3$; $Ba(OH)_2$ + NH_4Cl; AgCl in sunlight



★★★ *Nitesh Bhaiya ki guarantee hai — Exo vs Endo dimaag mein set kar lo!*

3 5 SIGNS OF A REACTION

Change in state	$NH_{3(g)} + HCl_{(g)} \rightarrow NH_4Cl_{(s)}$ two gases → white solid (fumes)
Change in colour ★★★	$Fe_{(s)} + CuSO_{4(aq)} \rightarrow FeSO_{4(aq)} + Cu_{(s)}$ blue solution → light green; brown Cu coats the iron nail
Gas evolved ★☆☆	$Zn_{(s)} + H_2SO_{4(aq)} \rightarrow ZnSO_{4(aq)} + H_{2(g)}\uparrow$ H_2 burns with a "pop" sound
Temperature change ★★☆	$CaO_{(s)} + H_2O_{(l)} \rightarrow Ca(OH)_{2(aq)}$ + **Heat** quicklime + water: beaker gets hot
Precipitate ★★★	$BaCl_{2(aq)} + Na_2SO_{4(aq)} \rightarrow BaSO_{4(s)}\downarrow + 2NaCl_{(aq)}$ **white** ppt of $BaSO_4$; NaCl stays dissolved



BUJHO *Gas nikli, matlab reaction pakka?*

PAHELI *Sign hai, proof nahi! Asli test: naya substance bana ya nahi?*





★★★ **PYQ FAVOURITE** $BaCl_2 + Na_2SO_4 \rightarrow$ white **$BaSO_4$** (SQP 2023, SQP 2025, CBSE 2019). Also asked: $Pb(NO_3)_2 + 2KI \rightarrow$ yellow **PbI_2**. *Ye pakka aayega, Nitesh Bhaiya ki guarantee!*

Activity: Burning magnesium ribbon ★★★

1. Clean Mg ribbon with **sandpaper** (removes MgO).
2. Burn it held in **tongs**; collect the ash on a **watch glass**.

$2Mg_{(s)} + O_{2(g)} \rightarrow 2MgO_{(s)}$ + **Heat** + **Light**

Bright white flame, white ash; exothermic. Don't stare!



★★★ *Board ka favourite activity — Nitesh Bhaiya ki guarantee, isko yaad rakhna hi hai!*

4 CHEMICAL EQUATIONS

WORD (names only): Magnesium + Oxygen → Magnesium oxide

↓ replace names with formulae

SKELETAL (unbalanced): $Mg + O_2 \rightarrow MgO$ O: 2 vs 1 ×

↓ equalise atoms with coefficients

BALANCED (atoms equal): $2Mg_{(s)} + O_{2(g)} \rightarrow 2MgO_{(s)}$ ✔

LAW OF CONSERVATION OF MASS

- Mass is **neither created nor destroyed** → atoms of each element: **LHS = RHS**
- So every equation **must be balanced**

$2H_{2(g)} + O_{2(g)} \rightarrow 2H_2O_{(l)}$
H: 4 = 4, O: 2 = 2 ✔

4 g + 32 g (reactants) = 36 g (product)

Sentence → Symbols decoder

If the question says…	Write
reacts with, combines with, is added to, burns in, "and" between products	+
to form, to produce, to give, yields	→

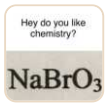
Solved Example 1: Sentence → equation

Q. "Hydrogen gas combines with nitrogen to form ammonia."
Skeletal: $H_2 + N_2 \rightarrow NH_3$ H 2 vs 3, N 2 vs 1 ×

$3H_{2(g)} + N_{2(g)} \rightarrow 2NH_{3(g)}$
Balanced: H 6 = 6, N 2 = 2 ✔

Formula toolkit

Diatomic gases are always written with "2" when free: H_2, N_2, O_2, F_2, Cl_2, Br_2, I_2. *"HON" + saare halogens jodi me!*



Criss-cross valencies for the right formula: $Ca^{2+} + OH^- \rightarrow Ca(OH)_2$

5 BALANCING ★★★ MOST ASKED

- **Hit-and-trial:** put **coefficients** in front of formulae
- Stop when every element has equal atoms on both sides



★★★ *Balancing pakka aayega — Nitesh Bhaiya ki guarantee, dimaag mein fit kar lo!*

Golden rules

1. Change only **coefficients**, never **subscripts**.
2. Balance a group (SO_4, NO_3, OH) that stays the same on both sides as **one unit**.
3. Order: metals → non-metals → H → O last. Smallest whole numbers.

× WRONG	✔ RIGHT
$H_2O \rightarrow H_2O_2$ to "fix" O	$2H_2O$ (H_2O_2 = new substance!)
$Mg + O \rightarrow MgO$	$2Mg + O_2 \rightarrow 2MgO$
CaOH	$Ca(OH)_2$

Solved Example 3: Burning of methane ★★☆

$CH_4 + O_2 \rightarrow CO_2 + H_2O$
H: 4 vs 2 × → write $2H_2O$; then O on right = 4 → write $2O_2$

$CH_{4(g)} + 2O_{2(g)} \rightarrow CO_{2(g)} + 2H_2O_{(l)} + \text{Heat}$
C 1 = 1, H 4 = 4, O 4 = 4 ✔

Solved Example 2: Iron + steam

1. Skeletal equation. Treat each formula as a sealed box.
   $Fe + H_2O \rightarrow Fe_3O_4 + H_2$

Atom	LHS	RHS
Fe	1	3 ×
H	2	2 ✔
O	1	4 ×

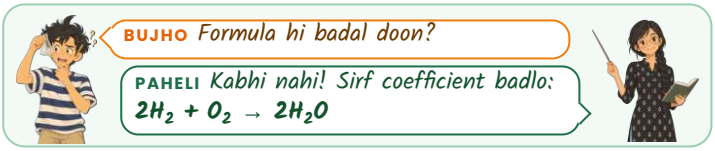
Start with the biggest formula, Fe_3O_4, and its biggest number: O = 4.

2. O: put 4 before H_2O. Now H = 8 on left, so 4 before H_2.
   $Fe + 4H_2O \rightarrow Fe_3O_4 + 4H_2$
3. Fe: 1 vs 3, so put 3 before Fe. Check: Fe 3 = 3, H 8 = 8, O 4 = 4 ✔
4. Add states (steam is a gas).
   $3Fe_{(s)} + 4H_2O_{(g)} \rightarrow Fe_3O_{4(s)} + 4H_{2(g)}$



MEME BREAK

LEFT: BALANCED (SWOLE)
$3Fe + 4H_2O \rightarrow Fe_3O_4 + 4H_2$
RIGHT: SKELETAL (CHEEMS)
$Fe + H_2O \rightarrow Fe_3O_4 + H_2$
Coefficients lagao, equation ko gym bhejo!



6 INFORMATIVE EQUATIONS

(a) Physical states ★★☆

Symbol	Meaning	Symbol	Meaning
(s)	solid	(g)	gas; ↑ = gas given off
(l)	liquid	(aq)	dissolved in water
↓	precipitate (insoluble solid separates out)		

Water: $H_2O(l)$ as liquid, $H_2O(g)$ as steam.

(b) Heat & conditions

- **+ Heat** on the product side means exothermic; on the reactant side, it means endothermic
- Δ, sunlight, electricity, pressure, catalyst → **on the arrow**

$CaCO_{3(s)} \xrightarrow{\Delta} CaO_{(s)} + CO_{2(g)}$
limestone breaks down on heating

$2AgCl_{(s)} \xrightarrow{\text{Sunlight}} 2Ag_{(s)} + Cl_{2(g)}$
white silver chloride turns grey in sunlight

$N_{2(g)} + 3H_{2(g)} \xrightarrow[200\text{ atm, }450\text{–}500\,^\circ C]{\text{Fe (catalyst)}} 2NH_{3(g)}$
Haber process: making ammonia

A **catalyst** changes (usually speeds up) the rate and is **not used up**.

Solved Example 4: Statement → balanced equation

(a) Hydrogen sulphide burns in air to give water and sulphur dioxide. NCERT

$2H_2S(g) + 3O_2(g) \rightarrow 2H_2O(l) + 2SO_2(g)$

H 4 = 4, S 2 = 2, O 6 = 6 ✔

(b) Barium chloride reacts with aluminium sulphate to give aluminium chloride and a precipitate of barium sulphate. CBSE 2019 ★★★

$3BaCl_2(aq) + Al_2(SO_4)_3(aq) \rightarrow 2AlCl_3(aq) + 3BaSO_4(s)\downarrow$

Ba 3 = 3, Cl 6 = 6, Al 2 = 2, SO_4 3 = 3 ✔

Exam tips

- ✔ "Balanced equation" asked: an unbalanced one loses marks even with correct formulae.
- ✔ "With physical states": add (s), (l), (g) or (aq) to every formula.
- ✔ Use an arrow (→), never "=". Recount every atom at the end.
- ✔ Colours: PbI_2 yellow, $BaSO_4$ white, MgO white, $CuSO_4$ blue, $FeSO_4$ light green.

TRY YOURSELF: BALANCE THESE

Pehle khud solve karo, phir answer key check karo!



(i) $H_2 + Cl_2 \rightarrow HCl$

(ii) $Na + H_2O \rightarrow NaOH + H_2$

(iii) $KClO_3 \rightarrow KCl + O_2$

(iv) $Fe_2O_3 + Al \rightarrow Al_2O_3 + Fe$

(v) $C_3H_8 + O_2 \rightarrow CO_2 + H_2O$

Day 1 mind map: revise in 2 minutes

REACTION
Reactants → products; new substance; atoms rearranged

5 SIGNS
Change in state, colour or temperature; gas; precipitate

CHEMICAL REACTIONS & EQUATIONS

3 EQUATIONS
Word → skeletal → balanced

BALANCING ★★★
Conservation of mass; coefficients only

INFORMATIVE
(s) (l) (g) (aq) ↑ ↓; + Heat (exo: product side, endo: reactant side); Δ, sunlight, catalyst on the arrow

Short answer questions

1. Dilute HCl is added to zinc granules. How will you prove that a chemical change has taken place? Give two arguments. CBSE SQP 2023
2. Write balanced equations for: (A) steam passed over red-hot iron, (B) natural gas burnt in air, (C) glucose reacting with oxygen in our body cells. CBSE 2025

7 PRACTICE MCQS

Copy-paste nahi chalega! Khud solve karo, answer key page 4 par hai.



1. Barium chloride solution is mixed with sodium sulphate solution. Which product is formed in the solid state? CBSE SQP 2023 ★★★

(a) Barium chloride (b) Barium sulphate
(c) Sodium chloride (d) Sodium sulphate

2. In the same reaction, which substance stays dissolved in the solution? CBSE SQP 2025 ★★★

(a) $BaCl_2$ (b) $BaSO_4$
(c) Na_2SO_4 (d) NaCl

3. Find *p*, *q*, *r* in the balanced equation: $p\ Pb(NO_3)_2(s) \rightarrow q\ PbO(s) + r\ NO_2(g) + O_2(g)$ CBSE SQP 2024 ★★☆

(a) 2, 2, 4 (b) 2, 4, 2
(c) 2, 4, 4 (d) 4, 2, 2

4. A magnesium ribbon is burnt over a watch glass. Which is the correct observation? CBSE SQP 2022 ★★★

(a) A brown powder of magnesium oxide is formed.
(b) A colourless gas that turns lime water milky is evolved.
(c) The ribbon burns with a brilliant white light.
(d) A reddish-brown gas smelling of burning sulphur is evolved.

5. Which statement is NOT true about the burning of magnesium ribbon in air? CBSE 2025 ★★★

(a) It burns with a dazzling white flame.
(b) A white powder is formed on burning.
(c) It is an endothermic reaction.
(d) It is an example of a combination reaction.

6. Dilute hydrochloric acid is added to granulated zinc in a test tube. The correct observation is: ★☆☆

(a) The surface of the metal becomes shiny.
(b) The reaction mixture turns milky.
(c) A pungent-smelling gas is given off.
(d) A colourless, odourless gas is evolved.

7. We balance a chemical equation because of the: ★★★

(a) law of constant proportions (b) law of conservation of mass
(c) law of multiple proportions (d) Avogadro's law

8. The balanced form of $Al + O_2 \rightarrow Al_2O_3$ is: ★★★

(a) $2Al + 3O_2 \rightarrow Al_2O_3$ (b) $4Al + 3O_2 \rightarrow 2Al_2O_3$
(c) $4Al + 2O_2 \rightarrow 2Al_2O_3$ (d) $2Al + O_2 \rightarrow Al_2O_3$

9. "+ Heat" written on the product side of an equation shows that the reaction is: ★★★

(a) endothermic (b) exothermic
(c) a precipitation reaction (d) not balanced

10. **Assertion (A):** $3Fe(s) + 4H_2O(g) \rightarrow Fe_3O_4(s) + 4H_2(g)$ is a balanced equation for the action of steam on iron.
Reason (R): The law of conservation of mass holds good for a chemical equation. CBSE 2020 ★★★

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

Answer key: MCQs

Q	Ans	Why
1	(b)	$BaSO_4$ is the insoluble white precipitate.
2	(d)	NaCl stays dissolved in water.
3	(a)	$2Pb(NO_3)_2 \rightarrow 2PbO + 4NO_2 + O_2$
4	(c)	Dazzling white light; white MgO powder.
5	(c)	It gives out heat and light: exothermic.
6	(d)	Hydrogen gas: colourless, odourless.
7	(b)	Atoms can't be created or destroyed.
8	(b)	Al 4 = 4, O 6 = 6.
9	(b)	Heat given out = exothermic.
10	(a)	Balanced because mass is conserved.

Try-yourself answers

(i) $H_2 + Cl_2 \rightarrow 2HCl$
(ii) $2Na + 2H_2O \rightarrow 2NaOH + H_2$
(iii) $2KClO_3 \rightarrow 2KCl + 3O_2$
(iv) $Fe_2O_3 + 2Al \rightarrow Al_2O_3 + 2Fe$
(v) $C_3H_8 + 5O_2 \rightarrow 3CO_2 + 4H_2O$

Model answers: short answer

1 Zinc + dilute HCl: two proofs of a chemical change

- **Gas evolved:** colourless H_2 bubbles; burns with a "pop"
- **Temperature change:** test tube becomes warm
- Zinc dissolves; new substance $ZnCl_2$ forms

$$Zn_{(s)} + 2HCl_{(aq)} \rightarrow ZnCl_{2(aq)} + H_{2(g)}$$

2 Balanced equations

$$3Fe_{(s)} + 4H_2O_{(g)} \rightarrow Fe_3O_{4(s)} + 4H_{2(g)}$$
(A) steam over red-hot iron

$$CH_{4(g)} + 2O_{2(g)} \rightarrow CO_{2(g)} + 2H_2O_{(l)} + \text{Heat}$$
(B) natural gas (methane) burning

$$C_6H_{12}O_{6(aq)} + 6O_{2(aq)} \rightarrow 6CO_{2(aq)} + 6H_2O_{(l)} + \text{Energy}$$
(C) respiration in our cells

Chapter 1 hit-list: Eduspark PYQ analysis

Topic / reaction	Day	Rating
Balancing equations	1	★★★
Oxidation & reduction (redox)	3	★★★
Exothermic vs endothermic	1	★★★
$BaCl_2 + Na_2SO_4$ (white ppt)	1	★★★
$Fe + CuSO_4$ (colour change)	1–2	★★★
Burning Mg ribbon	1	★★★
AgCl / AgBr in sunlight	2	★★☆
Corrosion & rancidity	3	★★☆
Heating $Pb(NO_3)_2$ (brown NO_2)	2	★★☆



★★★ Ye sab pakka aayenge — Nitesh Bhaiya ki guarantee hai! Dimaag mein set kar lo.

Day 1 checklist: tick before you sleep

- ☐ I can spot a chemical change and name its 5 signs
- ☐ I can turn a word statement into a skeletal equation
- ☐ I can balance by hit-and-trial without changing any formula
- ☐ I can add (s), (l), (g), (aq), heat and conditions
- ☐ I solved all 10 MCQs and checked them with the key

My score: ____ / 5

5/5 = Jalwa hai! 4 se kam? Page 1–2 phir se padho.



LEVEL 1 COMPLETE +100 XP earned. Next: Level 2



BUJHO Samajh gaya! Naya substance bana toh reaction, aur balancing mein sirf coefficient badalte hain!

PAHELI Shabaash! Kal Level 2: types of reactions.

Real life: Diwali crackers

- Fuse lit → **fuel + oxidiser** react instantly
- Heat, light & gas released; fast gas = **bang**
- Metal salts glow: strontium **red**, barium **green**, sodium **yellow**

DAILY-LIFE REACTIONS

Milk → curd, rusting, cooking, digestion, respiration, cut apple turning brown

Coming up: Day 2

Types of reactions: combination, decomposition, and the famous displacement drama where iron kicks copper out of $CuSO_4$. Poor Cu!

Valency chart: keep handy

Val.	Positive ions	Negative ions
1	Na^+, K^+, Ag^+, H^+, NH_4^+	Cl^-, OH^-, NO_3^-, HCO_3^-
2	Mg^{2+}, Ca^{2+}, Zn^{2+}, Cu^{2+}, Fe^{2+}, Ba^{2+}, Pb^{2+}	O^{2-}, S^{2-}, SO_4^{2-}, CO_3^{2-}
3	Al^{3+}, Fe^{3+}	PO_4^{3-}



NITESH SIR SAYS

Kal milte hai Day 2 pe.
Consistency hi jalwa hai!



★ How we starred (Eduspark PYQ analysis): we counted how often each reaction or topic has come in Chapter 1 board and sample-paper questions. Year tags appear only where the paper year is known.

DAY 02 · LEVEL 2

CHEMISTRY | CHAPTER 1 | PART 2 · ASTRA SERIES

COMBINATION & DECOMPOSITION

Goal: classify reactions and master the five NCERT activities.



by Nitesh Prakash

01 Combination | 02 Thermal | 03 Electrolytic | 04 Photolytic | 05 Comparison | +100 XP today | Level 2 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 COMBINATION REACTION ★★★ MOST ASKED

DEFINITION

- Two or more reactants combine to form a **single product**
- General form: **A + B → AB**
- Most combination reactions **release heat** (exothermic)

REACTANTS		PRODUCT
two or more (A + B)	→ join together	only one product (AB)

Activity: Quicklime + water ★★★

1. Take a small amount of **calcium oxide** (quicklime) in a beaker.
2. Add water slowly. It reacts **vigorously** with a hissing sound; the beaker becomes **hot**.

$$CaO_{(s)} + H_2O_{(l)} \rightarrow Ca(OH)_{2(aq)} + \text{Heat}$$

Quicklime + water → **slaked lime**; an exothermic reaction



★★★ *Quicklime + paani = slaked lime + garmi. Ye pakka aayega, Nitesh Bhaiya ki guarantee!*

Whitewashing: why walls shine

A solution of slaked lime is applied on walls. It reacts slowly with CO_2 in air and forms a thin layer of calcium carbonate, which gives a **shiny finish** after 2–3 days.

$$Ca(OH)_{2(aq)} + CO_{2(g)} \rightarrow CaCO_{3(s)} + H_2O_{(l)}$$

Ca 1 = 1, C 1 = 1, O 4 = 4, H 2 = 2 ✔

Note: this step gives two products, so it is not itself a combination reaction. Marble and limestone are also $CaCO_3$.

Limewater vs limewash: limewater is the clear, filtered solution of $Ca(OH)_2$ (turns milky with CO_2); limewash (milk of lime) is a suspension of $Ca(OH)_2$ used for whitewashing.

More combination examples

Type	Balanced equation
Element + element	$C_{(s)} + O_{2(g)} \rightarrow CO_{2(g)} + \text{Heat}$ burning of coal
Element + element	$2H_{2(g)} + O_{2(g)} \rightarrow 2H_2O_{(l)}$ formation of water
Compound + element	$2CO_{(g)} + O_{2(g)} \rightarrow 2CO_{2(g)}$
Compound + compound	$NH_{3(g)} + HCl_{(g)} \rightarrow NH_4Cl_{(s)}$

2 DECOMPOSITION REACTION ★★★ MOST ASKED

DEFINITION

- A **single reactant** breaks down into **two or more simpler products**
- General form: **AB → A + B** (the reverse of combination)
- Needs energy to break bonds, so most are **endothermic**

Three ways to supply the energy

Type	Energy	Example
Thermal	heat (Δ)	$CaCO_3$, $FeSO_4$, $Pb(NO_3)_2$
Electrolytic	electricity	water → $H_2 + O_2$
Photolytic	light	AgCl, AgBr in sunlight

3 THERMAL DECOMPOSITION

Activity: Heating ferrous sulphate crystals ★★★

1. Heat a few **green** ferrous sulphate crystals in a dry boiling tube.
2. The crystals first lose water and change colour, then decompose: **smell of burning sulphur**.

$$2FeSO_{4(s)} \xrightarrow{\Delta} Fe_2O_{3(s)} + SO_{2(g)} + SO_{3(g)}$$

Fe 2 = 2, S 2 = 2, O 8 = 8 ✔

Green crystals → **reddish-brown** ferric oxide; SO_2 gives the sulphur smell.

Precautions: hold the tube with a test-tube holder, keep its mouth away from yourself and others, heat gently, and smell the gas only by wafting.

Activity: Heating lead nitrate ★★★

Heat white lead nitrate powder in a boiling tube: a crackling sound and **brown fumes** of nitrogen dioxide.

$$2Pb(NO_3)_{2(s)} \xrightarrow{\Delta} 2PbO_{(s)} + 4NO_{2(g)} + O_{2(g)}$$

Pb 2 = 2, N 4 = 4, O 12 = 12 ✔; residue PbO is yellow

Limestone: heart of the cement industry ★★☆

$$CaCO_{3(s)} \xrightarrow{\Delta} CaO_{(s)} + CO_{2(g)}$$

limestone → quicklime + carbon dioxide

Quicklime is used in the manufacture of cement.



BUJHO *Lead nitrate garam kiya, brown dhuaan nikla. Kuch jal gaya kya?*

PAHELI *Kuch nahi jala! Brown fumes* NO_2 *gas ke hain, aur peela residue PbO hai.*



4 ELECTROLYTIC DECOMPOSITION ★★★ MOST ASKED

Activity: Electrolysis of water

1. Fix two **carbon electrodes** in a plastic mug of water; add a few drops of **dilute sulphuric acid**.
2. Invert two water-filled test tubes over the electrodes and pass current from a 6 V battery.
3. Bubbles form at both electrodes and push the water out of the tubes.

$$2H_2O_{(l)} \xrightarrow{\text{Electricity}} 2H_{2(g)} + O_{2(g)}$$

H 4 = 4, O 2 = 2 ✔

Electrode	Gas	Volume	Test
Cathode (−)	H_2	2	burns with a "pop"
Anode (+)	O_2	1	candle burns brighter



BUJHO *Dono test tubes mein gas barabar honi chahiye na?*

PAHELI *Nahi!* H_2O *mein H ke 2 atom, O ka 1. Isliye* H_2 *ka volume* ***double*** *hota hai.*



Why add acid? Other examples

- Pure water is a **poor conductor**; a little dilute acid makes it conduct
- Electricity is the energy that breaks the compound

$$2NaCl_{(l)} \xrightarrow{\text{Electricity}} 2Na_{(s)} + Cl_{2(g)}$$

molten sodium chloride (used to get sodium metal)

$$2Al_2O_{3(l)} \xrightarrow{\text{Electricity}} 4Al_{(s)} + 3O_{2(g)}$$

molten alumina gives aluminium; Al 4 = 4, O 6 = 6 ✔



MEME BREAK

Bijli ka aashirwad mila, paani tootkar H_2 *aur* O_2 *ban gaya! Hydrogen double, oxygen single.*

5 PHOTOLYTIC DECOMPOSITION ★★☆

Activity: Silver chloride in sunlight ★★☆

1. Take about 2 g of **white** silver chloride in a china dish.
2. Keep it in sunlight for some time: it turns **grey**.

$$2AgCl_{(s)} \xrightarrow{\text{Sunlight}} 2Ag_{(s)} + Cl_{2(g)}$$

white → grey (silver metal); Ag 2 = 2, Cl 2 = 2 ✔

$$2AgBr_{(s)} \xrightarrow{\text{Sunlight}} 2Ag_{(s)} + Br_{2(g)}$$

pale yellow silver bromide behaves the same way

Where is photolysis used?

- ✔ **Black-and-white photography:** AgBr on film darkens where light falls.
- ✔ AgCl and AgBr are stored in **dark coloured bottles** so light cannot decompose them.



MEME BREAK

Safed AgCl dhoop mein rakha, thodi der mein grey! Silver ban gaya, Cl_2 *ud gayi.*

6 COMBINATION VS DECOMPOSITION

COMBINATION	DECOMPOSITION
Two or more reactants → **one product**	One reactant → **two or more products**
A + B → AB	AB → A + B
Usually **exothermic** (heat released)	Usually **endothermic** (heat, light or electricity absorbed)
$CaO + H_2O \rightarrow Ca(OH)_2$	$CaCO_3 \rightarrow CaO + CO_2$



★★★ *Combination ulta = decomposition! Electrolysis ka 2 : 1 pakka aayega, Nitesh Bhaiya ki guarantee.*

× WRONG	✔ RIGHT
Energy: heat only	Heat, electricity **or** light
O_2 collects at the cathode	H_2 at cathode, O_2 at anode
$FeSO_4$ residue is green	Reddish-brown Fe_2O_3
Brown fumes = smoke	Brown fumes = NO_2 gas

Exam tips

- ✔ Name the type **and** the energy: "thermal decomposition (heat)".
- ✔ Write Δ, Sunlight or Electricity over the arrow.
- ✔ Learn the colours: $FeSO_4$ green, Fe_2O_3 reddish-brown, NO_2 brown, AgCl white → grey.

Day 2 mind map: revise in 2 minutes

COMBINATION
A + B → AB; one product; usually exothermic ($CaO + H_2O$)

THERMAL ★★★
Heat: $CaCO_3$, $FeSO_4$ ($SO_2 + SO_3$), $Pb(NO_3)_2$ (brown NO_2)

TYPES OF REACTIONS I

ELECTROLYTIC ★★★
Electricity: $2H_2O \rightarrow 2H_2 + O_2$; volume 2 : 1

PHOTOLYTIC
Sunlight: AgCl, AgBr turn grey (black-and-white photography)

7 PRACTICE MCQS

Main? Haan tum! Pehle khud solve karo, answer key page 4 par hai.



1. Quicklime reacts vigorously with water and releases heat. The product formed and the type of reaction are: ★★★

(a) $Ca(OH)_2$; decomposition
(b) $Ca(OH)_2$; combination
(c) $CaCO_3$; combination
(d) CaO_2; displacement

2. On heating ferrous sulphate crystals, the gases evolved are: ★★★

(a) SO_2 only
(b) SO_2 and SO_3
(c) SO_3 and O_2
(d) H_2S and SO_2

3. The colour of the residue left after strongly heating ferrous sulphate is: ★★☆

(a) green
(b) white
(c) reddish-brown
(d) black

4. On heating lead nitrate, the products formed are: CBSE 2024 ★★★

(a) NO, PbO and O_2
(b) NO_2, Pb and O_2
(c) NO_2, PbO and N_2
(d) NO_2, PbO and O_2

5. In the electrolysis of water, the gas collected at the cathode and the volume ratio of hydrogen to oxygen are: ★★★

(a) oxygen; 2 : 1
(b) hydrogen; 2 : 1
(c) hydrogen; 1 : 2
(d) oxygen; 1 : 2

6. Silver chloride is stored in dark coloured bottles because it: ★★☆

(a) reacts with glass
(b) decomposes in light
(c) absorbs moisture
(d) is volatile

7. Which of these is a decomposition reaction? ★★★

(a) $2H_2 + O_2 \rightarrow 2H_2O$
(b) $CaCO_3 \rightarrow CaO + CO_2$
(c) $C + O_2 \rightarrow CO_2$
(d) $CaO + H_2O \rightarrow Ca(OH)_2$

8. A few drops of dilute sulphuric acid are added to water before its electrolysis to: CBSE 2020 ★★☆

(a) increase its conductivity
(b) produce more oxygen
(c) act as a reactant
(d) keep the water cool

9. **Assertion (A):** Decomposition reactions are generally endothermic.
Reason (R): Energy in the form of heat, light or electricity is needed to break down the reactant. ★★★

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

10. **Assertion (A):** Whitewashed walls develop a shiny finish after two to three days.
Reason (R): Calcium hydroxide reacts slowly with carbon dioxide in air to form calcium carbonate. ★★★

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

Short and long answer questions

1 A student heats white lead nitrate powder in a boiling tube. (a) State two observations. (b) Write the balanced equation with states. (c) Name the type of reaction.

2 In the electrolysis of water, why is the volume of gas in one test tube double that in the other? Name both gases and give a test for each. CBSE 2020

3 What is slaked lime? Write the equation for its formation. How does it give walls a shiny finish after whitewashing?

Case-based question (4 marks) ★★★

A white solid **X**, found as limestone and marble, is heated strongly. It gives a white solid **Y** and a gas **Z** that turns lime water milky. **Y** reacts vigorously with water to give **W**, which is used for whitewashing.

1 Identify X, Y, Z and W.

2 Write the balanced equation for the reaction of Y with water and name its type.

3 Is the heating of X exothermic or endothermic? Give a reason.

4 Why does a whitewashed wall shine after two to three days? Write the equation.

TRY YOURSELF: NAME THE TYPE

Combination ya decomposition? Aur energy kaunsi?

(i) $2KClO_3 \xrightarrow{\Delta} 2KCl + 3O_2$

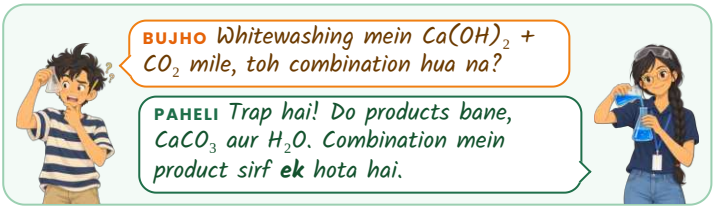
(ii) $N_2 + 3H_2 \rightarrow 2NH_3$

(iii) $2H_2O_2 \xrightarrow{Light} 2H_2O + O_2$

(iv) $2Mg + O_2 \rightarrow 2MgO$

(v) $2HgO \xrightarrow{\Delta} 2Hg + O_2$

(vi) $CaO + CO_2 \rightarrow CaCO_3$



Revision sheet: colours, gases, smells

Reaction	What you see
CaO + water	hissing; beaker gets hot
Heating $FeSO_4$	green → reddish-brown; burning sulphur smell
Heating $Pb(NO_3)_2$	brown NO_2 fumes; yellow PbO residue
Heating $CaCO_3$	CO_2 turns lime water milky
Water electrolysis	bubbles; $H_2 : O_2 = 2 : 1$
AgCl in sunlight	white → grey

Answer key: MCQs

Q	Ans	Why
1	(b)	$CaO + H_2O \rightarrow Ca(OH)_2$; one product.
2	(b)	$2FeSO_4 \rightarrow Fe_2O_3 + SO_2 + SO_3$
3	(c)	Ferric oxide, Fe_2O_3, is reddish-brown.
4	(d)	$2Pb(NO_3)_2 \rightarrow 2PbO + 4NO_2 + O_2$
5	(b)	H_2 at the cathode; twice the volume of O_2.
6	(b)	Light decomposes AgCl to Ag + Cl_2.
7	(b)	One reactant gives two products.
8	(a)	Pure water conducts poorly.
9	(a)	Energy is needed to break bonds.
10	(a)	The $CaCO_3$ layer gives the shine.

Try-yourself answers

(i) Thermal decomposition
(ii) Combination
(iii) Photolytic decomposition
(iv) Combination
(v) Thermal decomposition
(vi) Combination

Case-based answers

1. X = $CaCO_3$ (limestone); Y = CaO (quicklime); Z = CO_2; W = $Ca(OH)_2$ (slaked lime).
2. $CaO_{(s)} + H_2O_{(l)} \rightarrow Ca(OH)_{2(aq)}$ + **Heat**. Type: combination.
3. Endothermic: heat must be supplied to decompose $CaCO_3$.
4. $Ca(OH)_2 + CO_2 \rightarrow CaCO_3 + H_2O$; the thin $CaCO_3$ layer shines.

Model answers: short and long

1 Heating lead nitrate

- **Observations:** brown fumes of NO_2; a yellow residue of PbO (crackling sound)
- **Type:** thermal decomposition

$$2Pb(NO_3)_{2(s)} \xrightarrow{\Delta} 2PbO_{(s)} + 4NO_{2(g)} + O_{2(g)}$$

2 Electrolysis of water: why 2 : 1

- Each H_2O molecule gives 2 H atoms for 1 O atom, so 2 volumes of H_2 form for 1 volume of O_2
- **H_2** (cathode): burns with a pop sound; **O_2** (anode): a candle burns brighter in it

3 Slaked lime and whitewashing

- Slaked lime is **calcium hydroxide**, $Ca(OH)_2$
- On walls it reacts with CO_2 to form a thin, shiny $CaCO_3$ layer

$$CaO_{(s)} + H_2O_{(l)} \rightarrow Ca(OH)_{2(aq)} + \text{Heat}$$

Real life: decomposition around you

- **Cement:** limestone is heated in kilns to make quicklime
- **Digestion:** starch in food breaks down into simple sugars
- **Metals:** sodium and aluminium are extracted by electrolysis
- **Car airbags:** sodium azide decomposes in a flash to fill the bag with N_2

Day 2 checklist: tick before you sleep

- [ ] I can define combination and decomposition with one example each
- [ ] I can write the CaO + water and whitewashing equations
- [ ] I can write the heating equations of ferrous sulphate, lead nitrate and limestone
- [ ] I can explain electrolysis of water (2 : 1) and AgCl in sunlight
- [ ] I solved all 10 MCQs and checked them with the key

My score: ____ / 5

5/5 = Jalwa hai! 4 se kam? Page 1–2 phir se padho.



LEVEL 2 COMPLETE +100 XP earned. Next: Level 3



BUJHO *Samajh gaya! Jodna matlab combination, aur todna matlab decomposition!*

PAHELI *Shabaash! Kal Level 3: displacement aur redox ki baari.*

Name → formula bank

Common name	Chemical name	Formula
Quicklime	calcium oxide	CaO
Slaked lime	calcium hydroxide	$Ca(OH)_2$
Limestone, marble	calcium carbonate	$CaCO_3$
Green vitriol	ferrous sulphate	$FeSO_4$
Ferric oxide	iron(III) oxide	Fe_2O_3
Litharge	lead(II) oxide	PbO

EXTRA FROM CLASS

Gentle heating: $CuSO_4.5H_2O$ (blue) → $CuSO_4$ (white) + $5H_2O$
Strong heating (above about 650 °C, beyond NCERT): $2CuSO_4 \rightarrow 2CuO + 2SO_2 + O_2$; black CuO is left ($SO_3$ formed first breaks into $SO_2 + O_2$).
Cu 2 = 2, S 2 = 2, O 8 = 8 ✔

Coming up: Day 3

Displacement drama: iron kicks copper out of $CuSO_4$, plus double displacement, precipitates, and oxidation and reduction.



NITESH SIR SAYS
Har reaction ko roz ek baar likho aur colours yaad rakho. ***Kal milte hai Day 3 pe!***

★ How we starred (Eduspark PYQ analysis): we counted how often each reaction or topic has come in Chapter 1 board and sample-paper questions. Year tags appear only where the paper year is known.

DAY 03 · LEVEL 3

CHEMISTRY | CHAPTER 1 | PART 3 · ASTRA SERIES

DISPLACEMENT & REDOX REACTIONS

Goal: displacement, precipitation, oxidation and reduction.



by Nitesh Prakash

01 Displacement | 02 Reactivity | 03 Precipitation | 04 Redox | 05 Agents

+100 XP today | Level 3 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 DISPLACEMENT REACTION

DEFINITION

- A **more reactive** element pushes out a **less reactive** element from its compound
- General form: **A + BC → AC + B** (A is more reactive than B)

NCERT Activity 1.9: Iron nail in $CuSO_4$ ★★★

$$Fe_{(s)} + CuSO_{4(aq)} \rightarrow FeSO_{4(aq)} + Cu_{(s)}$$

copper sulphate (blue) → iron sulphate (light green)

- Blue colour of the solution **fades to light green**
- A **reddish-brown coat** of copper forms on the nail
- Iron is **more reactive** than copper, so it displaces Cu

More examples (both NCERT):

$$Zn_{(s)} + CuSO_{4(aq)} \rightarrow ZnSO_{4(aq)} + Cu_{(s)}$$

zinc displaces copper; blue solution turns colourless

$$Pb_{(s)} + CuCl_{2(aq)} \rightarrow PbCl_{2(aq)} + Cu_{(s)}$$

lead is above copper, so it displaces copper



MEME BREAK

*Iron nail $CuSO_4$ mein ghusa aur copper ko bahar nikal diya. Copper: "Kya tapleek hai aapko?" Iron: "Main **zyada reactive** hoon!"*



BUJHO *Toh copper bhi iron ko $FeSO_4$ se nikal dega na?*

PAHELI *Bilkul nahi! Cu kam reactive hai. Cu + $FeSO_4$ → **koi reaction nahi.***



2 REACTIVITY SERIES: WHO WINS?

MOST REACTIVE → LEAST REACTIVE

K | Na | Ca | Mg | Al | Zn | Fe | Pb | [H] | Cu | Hg | Ag | Au

- ✔ A metal **higher** in the series displaces a metal **below** it
- ✔ A metal **lower** in the series cannot displace one above it
- ✔ Metals above hydrogen displace H_2 from dilute acids

CLASS TRICK

Katrina Ne Car Mangi Alto Zen Ferrari Saturn Phir Bhi Hoyee Kyu Mili Silver Audi = K, Na, Ca, Mg, Al, Zn, Fe, Sn, Pb, H, Cu, Hg, Ag, Au. Class version includes Tin (Sn) between Fe and Pb; NCERT's list skips Sn, both are correct.



REACTIVITY CHECK

*Zn + $CuSO_4$: reaction ho gaya, sab khush. Cu + $ZnSO_4$: kuch nahi hua. **Series yaad karo!***

3 DOUBLE DISPLACEMENT REACTION

DEFINITION

- Two compounds **exchange their ions** to form two new compounds
- General form: **AB + CD → AD + CB**
- **Three types:** (1) **precipitation**: forms an **insoluble solid** (precipitate, ppt), shown with ↓ or (s), e.g. $BaSO_4$ below; (2) **neutralisation**: HCl + NaOH → NaCl + H_2O; (3) **gas-forming** (vinegar + baking soda): $CH_3COOH + NaHCO_3 \rightarrow CH_3COONa + H_2O + CO_2\uparrow$

NCERT Activity 1.10: $BaCl_2$ + Na_2SO_4 ★★★

$$Na_2SO_{4(aq)} + BaCl_{2(aq)} \rightarrow BaSO_{4(s)}\downarrow + 2NaCl_{(aq)}$$

Ba^{2+} and Na^+ swap partners

- A **white**, insoluble solid of $BaSO_4$ settles down
- NaCl stays dissolved in water

$$Pb(NO_3)_{2(aq)} + 2KI_{(aq)} \rightarrow PbI_{2(s)}\downarrow + 2KNO_{3(aq)}$$

yellow ppt of lead iodide

Precipitate colours: learn by heart

Precipitate	Formed from	Colour
$BaSO_4$	$BaCl_2 + Na_2SO_4$	**White**
PbI_2	$Pb(NO_3)_2 + KI$	**Yellow**
AgCl	$AgNO_3 + NaCl$	**White** (curdy)
$CaCO_3$	$Ca(OH)_2 + CO_2$ (limewater)	**White** (milky)
CuS	$CuSO_4 + H_2S$	**Black**



BUJHO *Ppt matlab? Sab kuch paani mein ghul jaata hai na?*

PAHELI *Nahi! $BaSO_4$ ghulta nahi, **white solid** neeche baith jaata hai.*





★★★ *$BaCl_2$ + Na_2SO_4 ka white ppt aur PbI_2 ka yellow ppt: **Nitesh Bhaiya ki guarantee, board mein aayega!***

4 OXIDATION AND REDUCTION

OXIDATION	REDUCTION
✔ Gain of oxygen	✔ Loss of oxygen
✔ Loss of hydrogen	✔ Gain of hydrogen
$2Cu + O_2 \rightarrow 2CuO$	$CuO + H_2 \rightarrow Cu + H_2O$

NCERT Activity 1.11: Heating copper powder ★★★

$$2Cu_{(s)} + O_{2(g)} \xrightarrow{\Delta} 2CuO_{(s)}$$

brown copper powder → **black** copper(II) oxide

Now pass **hydrogen gas** over the hot black CuO:

$$CuO_{(s)} + H_{2(g)} \xrightarrow{\Delta} Cu_{(s)} + H_2O_{(l)}$$

black coating turns **brown** again: copper is back

DEFINITION

- **Redox reaction**: one substance is **oxidised** while another is **reduced** in the same reaction
- Oxidation and reduction always happen **together**

OXIDATION: H_2 gains oxygen → H_2O

$$CuO_{(s)} + H_{2(g)} \xrightarrow{\Delta} Cu_{(s)} + H_2O_{(l)}$$

REDUCTION: CuO loses oxygen → Cu



MEME BREAK

*CuO ne apna oxygen H_2 ko de diya. CuO: "Ouch, mera oxygen!" Isi ko kehte hain **reduction**.*



BUJHO *Ek hi reaction mein oxidation aur reduction dono kaise?*

PAHELI *Jo O deta hai, wo reduce hota hai; jo O leta hai, wo oxidise. **Dono saath-saath!***



Solved Example: Displacement and redox together ★★☆

$$Fe_2O_{3(s)} + 2Al_{(s)} \rightarrow 2Fe_{(l)} + Al_2O_{3(s)} + \text{Heat}$$

Fe 2 = 2, Al 2 = 2, O 3 = 3 ✔

- Al is more reactive than Fe, so it **displaces** iron
- Fe_2O_3 loses O: **reduced**; Al gains O: **oxidised**
- This thermite reaction is used to **join railway tracks**

EXTRA FROM CLASS

Non-metals displace too: a more reactive halogen pushes out a less reactive one (**F_2 > Cl_2 > Br_2 > I_2**). **$Cl_2 + 2KBr \rightarrow 2KCl + Br_2$**; the solution turns orange-brown.

Real life: redox all around you

- **Respiration:** glucose is oxidised to CO_2 and water, giving energy
- **Burning fuel:** carbon is oxidised to CO_2
- **Extracting zinc:** ZnO is reduced by carbon (coke)

More redox examples: spot O and H ★★★

$$ZnO_{(s)} + C_{(s)} \rightarrow Zn_{(s)} + CO_{(g)}$$

ZnO loses O: **reduced**; C gains O: **oxidised**

$$MnO_{2(s)} + 4HCl_{(aq)} \rightarrow MnCl_{2(aq)} + 2H_2O_{(l)} + Cl_{2(g)}\uparrow$$

MnO_2 loses O: **reduced**; HCl loses H: **oxidised**

$$H_2S_{(g)} + Cl_{2(g)} \rightarrow 2HCl_{(g)} + S_{(s)}$$

H_2S loses H: **oxidised**; Cl_2 gains H: **reduced**

$$4Na_{(s)} + O_{2(g)} \rightarrow 2Na_2O_{(s)}$$

Na gains O: **oxidised**; O_2 is reduced

5 OXIDISING AND REDUCING AGENTS

OXIDISING AGENT — gives oxygen or removes hydrogen; **itself gets reduced**

→ always a pair

REDUCING AGENT — removes oxygen or gives hydrogen; **itself gets oxidised**

Identify: substance oxidised, reduced and the agents ★★★

Reaction	Oxidised (RA)	Reduced (OA)
$CuO + H_2 \rightarrow Cu + H_2O$	H_2	CuO
$ZnO + C \rightarrow Zn + CO$	C	ZnO
$MnO_2 + 4HCl \rightarrow MnCl_2 + 2H_2O + Cl_2$	HCl	MnO_2
$H_2S + Cl_2 \rightarrow 2HCl + S$	H_2S	Cl_2

RA = reducing agent, OA = oxidising agent. The substance oxidised is the reducing agent; the substance reduced is the oxidising agent.



BUJHO *Oxidising agent toh khud oxidise hota hoga na?*

PAHELI *Ulta! Wo **dusron ko oxidise** karta hai aur khud reduce hota hai.*



× WRONG	✔ RIGHT
Oxidising agent gets oxidised	Oxidising agent gets **reduced**
In $CuO + H_2$, Cu is oxidised	CuO is **reduced**; H_2 is oxidised
With MnO_2, HCl is the oxidising agent	MnO_2 is the oxidising agent



★★★ *Oxidised ya reduced pehchano, agent batao: Nitesh Bhaiya ki guarantee, har saal ka sawaal!*

BEYOND NCERT: THE ELECTRON VIEW

- **Oxidation** = loss of electrons; **reduction** = gain of electrons
- In $Fe + Cu^{2+} \rightarrow Fe^{2+} + Cu$, iron loses $2e^-$ and Cu^{2+} gains $2e^-$
- So the iron nail reaction is a displacement **and** a redox reaction

Trick: oxidising agent oxygen deta hai, khud reduce. Reducing agent oxygen leta hai, khud oxidise!



6 EXAM TIPS AND QUICK RECAP

Exam tips

- ✔ Displacement: always name the **colour change** and the coating (brown Cu).
- ✔ Write ↓ or (s) for every precipitate and name its colour.
- ✔ For redox, first mark who **gains or loses O/H**, then name the agents.
- ✔ Check balancing: MnO_2 + **4**HCl gives **2**H_2O.

TRY YOURSELF: OXIDISED OR REDUCED?

Har reaction mein oxidised, reduced aur dono agents likho. Answer page 4 par!



(i) $PbO + C \rightarrow Pb + CO$ (ii) $2Mg + CO_2 \rightarrow 2MgO + C$

(iii) $Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$

(iv) $SO_2 + 2H_2S \rightarrow 3S + 2H_2O$

7 PRACTICE MCQS

Answer dekh ke solve kiya toh horse bhi hasega! Khud karo, key page 4 par hai.



1. An iron nail is put into each of two test tubes: P with copper(II) sulphate solution and Q with silver nitrate solution. Which observation is correct? CBSE SQP 2025 ★★★

(a) P: blue coating on the nail; Q: no reaction
(b) P: brown coating on the nail; Q: silver coating on the nail
(c) No reaction in either test tube
(d) P: no reaction; Q: silver coating on the nail

2. Which reaction is different from the other three? CBSE 2024 ★★★

(a) $NaCl + AgNO_3 \rightarrow AgCl + NaNO_3$
(b) $CaO + H_2O \rightarrow Ca(OH)_2$
(c) $KNO_3 + H_2SO_4 \rightarrow KHSO_4 + HNO_3$
(d) $ZnCl_2 + H_2S \rightarrow ZnS + 2HCl$

3. Pick the correct statement about the reaction $CuO + H_2 \rightarrow Cu + H_2O$. CBSE Compartment 2024 ★★★

(a) CuO is oxidised and H_2 is reduced.
(b) H_2 is oxidised and CuO is reduced.
(c) CuO is the reducing agent.
(d) H_2 is the oxidising agent.

4. In $MnO_2 + 4HCl \rightarrow MnCl_2 + 2H_2O + Cl_2$, the oxidising agent is: ★★★

(a) MnO_2 (b) HCl
(c) $MnCl_2$ (d) Cl_2

5. Lead powder is added to blue copper sulphate solution. After a few hours the solution becomes colourless. This is an example of a: CBSE SQP 2024 ★★☆

(a) combination reaction (b) decomposition reaction
(c) displacement reaction (d) double displacement reaction

6. Lead nitrate solution is mixed with potassium iodide solution. The yellow precipitate formed is of: ★★★

(a) potassium nitrate (b) lead iodide
(c) lead nitrate (d) potassium iodide

7. Copper powder heated in a china dish turns black on the surface. The black substance is: ★★☆

(a) copper(II) oxide (b) copper sulphate
(c) copper carbonate (d) copper hydroxide

8. In the reaction $ZnO + C \rightarrow Zn + CO$, the reducing agent is: ★★☆

(a) ZnO (b) C
(c) Zn (d) CO

9. **Assertion (A):** When an iron nail is dipped in copper sulphate solution, the blue colour of the solution fades.
Reason (R): Iron is more reactive than copper and displaces copper from copper sulphate solution. ★★★

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

10. **Assertion (A):** In the reaction $CuO + H_2 \rightarrow Cu + H_2O$, hydrogen acts as the oxidising agent.
Reason (R): Hydrogen removes oxygen from CuO. ★★★

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

Short and long answer practice

1. Why does the colour of copper sulphate solution change when an iron nail is dipped in it? NCERT
2. H_2S is passed through blue $CuSO_4$ solution: the colour fades and a black ppt forms. Name the reaction type and the ppt, explain the fading, and write the equation. CBSE 2020
3. A shiny brown element X turns black when heated in air. Name X and the black compound. Write the equation, and show how the change can be reversed. NCERT ★★★



★★★ Iron nail + $CuSO_4$ ka colour change: Nitesh Bhaiya ki guarantee, ye pakka aayega!

Day 3 mind map: revise in 2 minutes

DISPLACEMENT
A + BC → AC + B; more reactive metal wins

DOUBLE DISPLACEMENT
AB + CD → AD + CB; ions swap; ppt often forms

TYPES OF REACTIONS II & REDOX

OXIDATION
Gain of O or loss of H

REDUCTION
Loss of O or gain of H

AGENTS ★★★
Oxidising agent is reduced itself; reducing agent is oxidised itself. Oxidation and reduction together = redox.

Did you know? Copper wire in $AgNO_3$

$Cu_{(s)} + 2AgNO_{3(aq)} \rightarrow Cu(NO_3)_{2(aq)} + 2Ag_{(s)}$
shiny silver crystals form; solution turns blue

Answer key: MCQs

Q	Ans	Why
1	(b)	Fe displaces Cu (brown) and Ag (silver).
2	(b)	Combination; the other three swap ions.
3	(b)	H_2 gains O; CuO loses O.
4	(a)	MnO_2 loses oxygen, so it is reduced.
5	(c)	Pb is above Cu, so it displaces Cu.
6	(b)	PbI_2 is the yellow precipitate.
7	(a)	$2Cu + O_2 \rightarrow 2CuO$ (black).
8	(b)	C takes oxygen from ZnO.
9	(a)	R correctly explains the colour change.
10	(d)	H_2 is the reducing agent; R is true.

Try-yourself answers

No.	Oxidised (RA)	Reduced (OA)
(i)	C	PbO
(ii)	Mg	CO_2
(iii)	CO	Fe_2O_3
(iv)	H_2S	SO_2

Model answers: short and long answer

1 Iron nail in $CuSO_4$

- Iron is **more reactive** than copper, so it displaces copper
- Blue $CuSO_4$ changes to light green $FeSO_4$; brown Cu deposits on the nail

$$Fe_{(s)} + CuSO_{4(aq)} \rightarrow FeSO_{4(aq)} + Cu_{(s)}$$

2 H_2S and $CuSO_4$

- **Double displacement** (precipitation) reaction; black ppt = **copper sulphide** (CuS)
- Colour fades because blue $CuSO_4$ is used up as Cu^{2+} ions leave as CuS

$$CuSO_{4(aq)} + H_2S_{(g)} \rightarrow CuS_{(s)}\downarrow + H_2SO_{4(aq)}$$

3 X = copper (Cu); black compound = copper(II) oxide (CuO)

$$2Cu_{(s)} + O_{2(g)} \xrightarrow{\Delta} 2CuO_{(s)}$$

oxidation of copper

$$CuO_{(s)} + H_{2(g)} \xrightarrow{\Delta} Cu_{(s)} + H_2O_{(l)}$$

reversed by passing H_2 over hot CuO (reduction)

BOARD ANSWER FRAME: REDOX

Balanced equation with states → mark who gains or loses O/H → name the agents (OA is reduced, RA is oxidised).



NITESH SIR SAYS

Jo oxygen le, wo oxidise; jo oxygen de, wo reduce. Bas itna yaad rakho!



Day 3 checklist: tick before you sleep

- ☐ I can explain displacement using the reactivity series
- ☐ I can write the $BaSO_4$ and PbI_2 precipitation equations
- ☐ I can define oxidation and reduction using oxygen and hydrogen
- ☐ I can name the oxidising and reducing agents in a redox reaction
- ☐ I solved all 10 MCQs and checked them with the key

My score: ____ / 5

5/5 = Jalwa hai! 4 se kam? Page 1–2 phir se padho.



LEVEL 3 COMPLETE +100 XP earned. Next: Level 4



BUJHO *Samajh gaya! Zyada reactive metal kam reactive ko nikalta hai, aur oxidising agent khud reduce hota hai!*

PAHELI *Shabaash! Kal Level 4: exothermic, corrosion aur rancidity.*

Formula sheet: must-write equations of Day 3

$$Fe_{(s)} + CuSO_{4(aq)} \rightarrow FeSO_{4(aq)} + Cu_{(s)}$$

$$Na_2SO_{4(aq)} + BaCl_{2(aq)} \rightarrow BaSO_{4(s)}\downarrow + 2NaCl_{(aq)}$$

$$Pb(NO_3)_{2(aq)} + 2KI_{(aq)} \rightarrow PbI_{2(s)}\downarrow + 2KNO_{3(aq)}$$

$$CuO_{(s)} + H_{2(g)} \xrightarrow{\Delta} Cu_{(s)} + H_2O_{(l)}$$

$$ZnO_{(s)} + C_{(s)} \rightarrow Zn_{(s)} + CO_{(g)}$$

$$MnO_{2(s)} + 4HCl_{(aq)} \rightarrow MnCl_{2(aq)} + 2H_2O_{(l)} + Cl_{2(g)}\uparrow$$

Lab file: the three NCERT activities

Activity	Observation	Type
1.9: Fe + $CuSO_4$	Blue → light green; brown coat	Displacement
1.10: $BaCl_2$ + Na_2SO_4	White ppt of $BaSO_4$	Double displacement
1.11: heating Cu	Brown → black (CuO)	Oxidation

Coming up: Day 4

Exothermic and endothermic reactions, rusting, rancidity and the **Chapter 1 Boss Level** test!

★ How we starred (Eduspark PYQ analysis): we counted how often each reaction or topic has come in Chapter 1 board and sample-paper questions. Year tags appear only where the paper year is known.

DAY 04 LEVEL 4

CHEMISTRY | CHAPTER 1 | PART 4 ASTRA SERIES

ENERGY, CORROSION & RANCIDITY

Goal: heat changes, rusting, rancidity and the Boss Level.



by Nitesh Prakash

01 Exo vs Endo | 02 Respiration | 03 Corrosion | 04 Rancidity | 05 Boss Level | +200 XP today | Level 4 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 EXOTHERMIC VS ENDOTHERMIC

DEFINITION

- **Exothermic:** a reaction in which **heat is given out**
- **Endothermic:** a reaction in which **heat is absorbed**

EXOTHERMIC ★★★	ENDOTHERMIC ★★☆
Heat **released**; gets **warm**	Heat **absorbed**; gets **cool**
+ **Heat** on the product side	+ **Heat** with reactants (or Δ)
Burning, respiration, compost	Photosynthesis, $CaCO_3$

Exothermic: heat given out ★★★

$C_{(s)} + O_{2(g)} \rightarrow CO_{2(g)} + \text{Heat}$
burning of coal

$CH_{4(g)} + 2O_{2(g)} \rightarrow CO_{2(g)} + 2H_2O_{(g)} + \text{Heat}$
burning of natural gas (methane)

$CaO_{(s)} + H_2O_{(l)} \rightarrow Ca(OH)_{2(aq)} + \text{Heat}$
quicklime + water: slaked lime, beaker becomes hot

Endothermic: heat absorbed ★★☆

$CaCO_{3(s)} \xrightarrow{\Delta} CaO_{(s)} + CO_{2(g)}$
limestone needs strong heating

$2AgCl_{(s)} \xrightarrow{\text{Sunlight}} 2Ag_{(s)} + Cl_{2(g)}$
energy from light; white → grey

$6CO_{2(g)} + 6H_2O_{(l)} \xrightarrow[\text{Chlorophyll}]{\text{Sunlight}} C_6H_{12}O_{6(aq)} + 6O_{2(g)}$
photosynthesis: plants absorb sunlight energy

EXTRA FROM CLASS

$N_{2(g)} + O_{2(g)} \xrightarrow{\text{very high temp.}} 2NO_{(g)}$
endothermic; happens in lightning and hot car engines



MEME BREAK
*Coal jal raha, stove on: "This is fine." Heat bahar = **exothermic**!*

2 RESPIRATION AND COMPOST

Respiration: an exothermic reaction ★★★

Rice, potatoes and bread contain **carbohydrates**. Digestion breaks them down into **glucose**. In our cells, glucose combines with oxygen and releases energy.

$C_6H_{12}O_{6(aq)} + 6O_{2(aq)} \rightarrow 6CO_{2(aq)} + 6H_2O_{(l)} + \text{Energy}$
glucose + oxygen → carbon dioxide + water + energy

Balance check: C 6 = 6, H 12 = 12, O 18 = 18 ✔



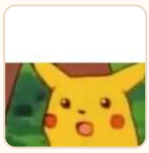
BUJHO *Saans lena hi respiration hai na? Toh heat kahan se aayi?*

PAHELI *Saans lena sirf breathing hai. Respiration cells ke andar glucose ka oxidation hai, jisme energy nikalti hai. Isliye ye exothermic hai!*



Decomposition of vegetable matter into compost ★★★

- Kitchen waste and leaves are broken down by microbes into **compost**
- Heat is **released**, so a compost pit feels **warm** inside: **exothermic**
- Most decomposition reactions need energy (endothermic); compost making is the famous **exception**



MEME BREAK
*Teacher: "Decomposition endothermic hoti hai." Compost pit andar se garam nikla. Bujho: **exothermic decomposition?!***



★★★ *Respiration aur compost dono exothermic: Nitesh Bhaiya ki guarantee, MCQ mein pakka aayega!*

How to spot the type from an equation ★★☆

You see	Type
+ Heat or + Energy on the **right**	**Exothermic**
+ Heat on the **left**; Δ, sunlight or electricity on the arrow	**Endothermic**
Test tube becomes hot	**Exothermic**
Test tube becomes cold	**Endothermic**

3 CORROSION AND RUSTING

DEFINITION

- **Corrosion:** a metal is slowly **eaten away** when attacked by air, moisture, acids or other substances around it
- **Rusting:** corrosion of **iron**; a reddish-brown, flaky coat forms

Rusting of iron ★★★

$$4Fe_{(s)} + 3O_{2(g)} + 2xH_2O_{(l)} \rightarrow 2Fe_2O_3.xH_2O_{(s)}$$

rust = hydrated iron(III) oxide, reddish-brown

Balance check: Fe 4 = 4, O (6 + 2x) = (6 + 2x), H 4x = 4x ✔

- Rusting needs **both air (oxygen) and water (moisture)**

Activity: which nail rusts? ★★★

Tube	Condition	Result
A	Nail in ordinary tap water (air + water)	**Rusts**
B	Boiled water + oil layer on top (no air)	No rust
C	Dry air with anhydrous $CaCl_2$ (no water)	No rust



MEME BREAK
*Iron Man ko bhi **hawa + paani** mil jaaye toh jung lag jaayegi. Suit pe paint ki parat zaroori hai!*

Corrosion of other metals ★★☆

Metal	Coating formed	Due to
Silver	Black: silver sulphide, Ag_2S	H_2S in air
Copper	Green: basic copper carbonate, $CuCO_3.Cu(OH)_2$	Moist CO_2 in air

Preventing rusting ★★★

1. **Painting** gates, grills and bridges
2. **Oiling or greasing** machine parts and cycle chains
3. **Galvanising:** coating iron with a thin layer of **zinc**
4. **Chrome plating** (taps, cycle handles) and tin coating
5. **Alloying:** stainless steel (iron + chromium + nickel) does not rust



MEME BREAK *Paint ya oil ki parat **hawa aur nami** ko iron se door rakhti hai. Parat tooti, rusting shuru!*



BUJHO *Ye iron clips baarish mein bhool gaya, kal tak kuch hoga?*

PAHELI *Hawa + nami = jung! Clips sookhe rakho, ya oil laga do. Paani ke bina rust nahi lagta.*





★★★ ***Rusting ke liye hawa + paani dono chahiye: Nitesh Bhaiya ki guarantee, activity wala question pakka!***

4 RANCIDITY

DEFINITION

- **Rancidity:** fats and oils in food get **oxidised** when left exposed to air for long
- Food develops a **bad smell and taste** and is unsafe to eat

Preventing rancidity ★★★

- ✔ **Antioxidants** added to food containing fats and oils
- ✔ Chips packets **flushed with nitrogen**, an unreactive gas
- ✔ Storing food in **airtight containers**
- ✔ **Refrigeration:** low temperature slows oxidation
- ✔ Keeping food away from light; vacuum packing



BUJHO *Chips packet mein aadhi hawa bhar ke loot rahe hain!*

PAHELI *Hawa nahi, **nitrogen** hai! Oxygen chips ko rancid kar deta, nitrogen unhe crispy aur fresh rakhta hai.*





MEME BREAK
*Mummy ka chhupaya namkeen, 3 mahine baad khola. Smell aur taste dono kharab: **rancidity**!*

Corrosion vs rancidity: both are oxidation

	Corrosion	Rancidity
Happens to	Metals	Fats and oils in food
Caused by	Air, moisture, acids	Oxygen of air
Result	Metal gets weak	Bad smell and taste
Stopped by	Paint, oil, zinc coat	Antioxidants, N_2, airtight

× WRONG	✔ RIGHT
Respiration is endothermic	It releases energy, so it is exothermic
Rust is Fe_2O_3	Rust is hydrated: $Fe_2O_3.xH_2O$
Iron rusts in dry air	It needs air **and** water
Chips packets contain oxygen	They are flushed with nitrogen

5 PRACTICE MCQS

Pehle khud solve karo, answer key page 3 ke end mein hai!



1. Which of the following is an endothermic reaction? ★★★
(a) Burning of coal (b) Respiration
(c) Heating calcium carbonate (d) Quicklime added to water

2. Making compost from vegetable matter is an example of: ★★★
(a) an endothermic reaction (b) an exothermic reaction
(c) a photolytic decomposition (d) a displacement reaction

3. The chemical formula of rust is: ★★★
(a) Fe_2O_3 (b) Fe_3O_4
(c) $Fe_2O_3.xH_2O$ (d) FeO

4. An iron nail in boiled water under a layer of oil does not rust because: ★★☆
(a) oil reacts with iron (b) no air reaches it
(c) the water is hot (d) there is no light

5. The black coating formed on silver articles is: ★★☆
(a) Ag_2O (b) AgCl
(c) Ag_2S (d) $AgNO_3$

6. Packets of potato chips are flushed with: ★★★
(a) oxygen (b) hydrogen
(c) nitrogen (d) chlorine

7. Galvanisation is the process of coating iron with: ★★☆
(a) tin (b) zinc
(c) chromium (d) copper

8. **Assertion (A):** Respiration is an exothermic reaction.
Reason (R): During respiration, glucose is oxidised in our cells and energy is released. ★★★
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.



★★★ *Rust hamesha hydrated iron(III) oxide hota hai! Nitesh Bhaiya ki guarantee, formula wala MCQ yahin se.*

Answer key: practice MCQs

Q	Ans	Why
1	(c)	$CaCO_3$ needs heat to break down.
2	(b)	Compost pit becomes warm: heat released.
3	(c)	Rust is hydrated iron(III) oxide.
4	(b)	Boiling removes air; oil stops fresh air.
5	(c)	Silver reacts with H_2S in air.
6	(c)	Nitrogen is unreactive; no oxidation.
7	(b)	A zinc layer protects the iron.
8	(a)	Energy released, because glucose is oxidised.

Short answer questions

1. Why do we apply paint on iron articles? NCERT
2. Oil and fat containing food items are flushed with nitrogen. Why? NCERT
3. Explain corrosion and rancidity with one example each. NCERT

Model answers: short answer

1 Paint on iron
- Paint keeps **air and moisture** away from iron, so it does not rust

2 Nitrogen in food packets
- Nitrogen is **unreactive**; it pushes out oxygen, so fats are not oxidised and food does not turn **rancid**

3 Corrosion and rancidity
- **Corrosion:** metal eaten away by air, moisture or acids; e.g. rusting of iron
- **Rancidity:** oxidation of fats and oils in food; e.g. stale chips smell bad

6 BOSS LEVEL: CHAPTER 1 TEST

BOSS LEVEL: CHAPTER 1

Days 1–4 | 21 marks | 30 minutes



PAHELI *Poora chapter ek saath! Timer lagao, book band karo, phir answers se milao.*

BUJHO *Har equation balance karke likhunga, boss ko harana hai!*



SECTION A: MCQS (1 MARK EACH)

1. Which of the following is a chemical change? ★★☆
(a) Melting of wax (b) Dissolving salt
(c) Souring of milk (d) Boiling water

2. To balance $Na + H_2O \rightarrow NaOH + H_2$, the coefficients are: ★★★
(a) 1, 1, 1, 1 (b) 2, 2, 2, 1
(c) 2, 1, 2, 1 (d) 1, 2, 1, 2

3. On heating ferrous sulphate crystals, the residue left is: ★★★
(a) green (b) white
(c) reddish-brown (d) blue

4. In electrolysis of water, the volume ratio of gas at the cathode to gas at the anode is: ★★★
(a) 1 : 2 (b) 2 : 1
(c) 1 : 1 (d) 8 : 1

5. $Fe_{(s)} + CuSO_{4(aq)} \rightarrow FeSO_{4(aq)} + Cu_{(s)}$ is a: ★★★
(a) combination reaction (b) decomposition reaction
(c) displacement reaction (d) double displacement

6. The reaction between sodium sulphate and barium chloride solutions is a: ★★★
(a) displacement (b) double displacement
(c) combination (d) decomposition

7. In $CuO + H_2 \xrightarrow{\Delta} Cu + H_2O$, the substance oxidised is: ★★★

(a) CuO (b) H_2
(c) Cu (d) H_2O

8. In $MnO_2 + 4HCl \rightarrow MnCl_2 + 2H_2O + Cl_2$, the oxidising agent is: ★★☆

(a) MnO_2 (b) HCl
(c) $MnCl_2$ (d) Cl_2

9. **Assertion (A):** Silver chloride turns grey when kept in sunlight.
Reason (R): Sunlight decomposes silver chloride into silver and chlorine. ★★★

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

10. **Assertion (A):** Rusting of iron is a slow oxidation reaction.
Reason (R): Iron rusts quickly in dry air. ★★☆

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

Section B and C: written questions

1. What happens when water is added to quicklime? Write the balanced equation and name the type of reaction. **(2)**
2. Lead nitrate is heated in a dry test tube. State one observation, write the balanced equation and name the reaction. **(2)**
3. In $ZnO + C \rightarrow Zn + CO$, identify the substance oxidised and the substance reduced. **(2)**
4. A shiny brown metal X turns black (Y) when heated in air. When hydrogen is passed over hot Y, X is obtained again. (a) Identify X and Y. (b) Write both balanced equations and name each reaction type. (c) What colour change is seen when an iron nail is dipped in the blue solution of a salt of X? **(5)**



★★★ *Boss Level ke ye sawaal = poore chapter ka revision. Nitesh Bhaiya ki guarantee, board mein yahi type aayega!*

Boss Level answers: MCQs

Q	Ans	Why
1	(c)	A new substance (curd) forms.
2	(b)	$2Na + 2H_2O \rightarrow 2NaOH + H_2$
3	(b)	Fe_2O_3 is reddish-brown.
4	(b)	H_2 at cathode is double the O_2.
5	(c)	Fe is more reactive; it displaces Cu.
6	(b)	White $BaSO_4$ ppt forms.
7	(b)	H_2 gains oxygen and becomes H_2O.
8	(a)	MnO_2 gives oxygen; it is reduced.
9	(a)	Photolytic decomposition explains it.
10	(c)	Iron needs air and moisture to rust.

Boss Level answers: written questions

1 Quicklime + water (2)

$CaO_{(s)} + H_2O_{(l)} \rightarrow Ca(OH)_{2(aq)} + \textbf{Heat}$
combination reaction; also exothermic

2 Heating lead nitrate (2)

> **Brown fumes** of NO_2 are seen

$2Pb(NO_3)_{2(s)} \xrightarrow{\Delta} 2PbO_{(s)} + 4NO_{2(g)} + O_{2(g)}$
thermal decomposition

3 ZnO + C (2)

> **Oxidised:** carbon (C gains oxygen → CO)
> **Reduced:** zinc oxide (ZnO loses oxygen → Zn)

4 Metal X and black Y (5)

> **(a)** X = copper (Cu); Y = copper(II) oxide (CuO)

$2Cu_{(s)} + O_{2(g)} \xrightarrow{\Delta} 2CuO_{(s)}$
(b) combination; copper is oxidised

$CuO_{(s)} + H_{2(g)} \xrightarrow{\Delta} Cu_{(s)} + H_2O_{(g)}$
redox: CuO is reduced, H_2 is oxidised

> **(c)** Blue $CuSO_4$ solution turns **light green** ($FeSO_4$); a **brown** coat of Cu forms on the nail

Day 4 checklist: tick before you sleep

- ☐ I can tell exothermic from endothermic and give two examples of each
- ☐ I can write the respiration equation and explain why compost pits are warm
- ☐ I can explain rusting, its equation and four ways to prevent it
- ☐ I can explain rancidity and why chips packets are flushed with nitrogen
- ☐ I cleared the Boss Level and checked every answer with the key

My score: ____ / 5

5/5 = Jalwa hai! 4 se kam? Page 1–2 phir se padho.



LEVEL 4 COMPLETE

+200 XP earned. Chapter 1 cleared. Next: Level 5



BUJHO *Samajh gaya! Respiration exothermic, rust ko hawa + paani chahiye, aur chips ko nitrogen bachata hai!*

PAHELI *Shabaash! Chapter 1 ka boss haar gaya. Kal se acids, bases and salts.*



NITESH SIR SAYS
Boss Level mein jo galat hua, us topic ka Day phir se padho. ***Revision hi asli jalwa hai!***

CHEMISTRY | CHAPTER 1 COMPLETE

CHAPTER 1 SWAHAA!

Chemical Reactions and Equations | Days 1–4



BUJHO *Reaction pehchanna, equation balance karna, redox, corrosion... sab samajh gaya! Chapter 1 Swahaa!*

PAHELI *Shabaash Bujho! Ab Acids, Bases aur Salts ki baari. Kal milte hain Day 5 pe!*

CHAPTER XP EARNED
+500 XP

LEVELS CLEARED
4 levels (Days 1–4)

BOOK PROGRESS
Level 4 / 60 500 XP total

WHAT YOU HAVE MASTERED

- ✔ Chemical change and its 5 signs
- ✔ Combination and decomposition
- ✔ Oxidation, reduction and redox
- ✔ Writing and balancing equations
- ✔ Displacement and double displacement
- ✔ Exo/endothermic, corrosion, rancidity

NEXT CHAPTER Chapter 2: Acids, Bases and Salts (Day 5) *Jalwa jaari rahe!*

DAY 05 · LEVEL 5

CHEMISTRY | CHAPTER 2 | PART 1 · ASTRA SERIES

ACIDS, BASES & SALTS

Goal: test acids and bases, and predict how they react.



by Nitesh Prakash

01 Basics · 02 Indicators · 03 Metals · 04 Carbonates · 05 Neutralisation

+100 XP today | Level 5 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 ACIDS VS BASES

ACIDS	BASES
Sour in taste (lemon, curd, vinegar)	Bitter; soapy to touch; corrosive
Turn **blue litmus red**	Turn **red litmus blue**
HCl, H_2SO_4, HNO_3, acetic acid	NaOH (caustic soda), KOH (caustic potash), $Ca(OH)_2$

2 INDICATORS ★★★ MOST ASKED

DEFINITION

- An **indicator** is a substance that shows, by a change in **colour** or **smell**, whether a solution is **acidic** or **basic** (never taste lab chemicals!)
- Three types: **natural**, **synthetic** and **olfactory**

Colour chart: learn it by heart ★★★

Indicator	In acid	In base
Litmus (natural)	red	blue
Turmeric (natural)	yellow	reddish-brown
Red cabbage (natural)	red	green
China rose (natural)	dark pink	green
Phenolphthalein (synthetic)	colourless	pink
Methyl orange (synthetic)	red	yellow

China rose (hibiscus) petals soaked in warm water give the indicator: dark pink (magenta) in acid, green in base. **Hydrangea**, **petunia** and **geranium** petals also work; litmus comes from **lichen**.



MEME BREAK

*Haldi ka daag sabun se dhoya aur daag **laal-bhoora** ho gaya! Soap is basic, and turmeric turns **reddish-brown** in a base.*



BUJHO *Phenolphthalein acid mein laal hota hai na?*

PAHELI *Nahi! Acid mein **colourless**, base mein **pink**. Acid mein laal hota hai **methyl orange**.*

Olfactory indicators: test by smell ★★☆

- **Olfactory** = related to the sense of smell; the odour changes in an acid or a base
- **Onion**, **vanilla** extract and **clove** oil: the smell **disappears in a base** (NaOH) but stays in an acid (HCl)
- Useful for a visually impaired student to tell an acid from a base

Solved Example 1: Only red litmus given

Q. Three test tubes hold distilled water, an acid and a base. You have only red litmus paper. Identify each. NCERT

1. Dip red litmus in all three. The one that turns it **blue** is the **base**.
2. Dip this blue litmus in the other two. The one that turns it **red** is the **acid**.
3. The tube that changes neither paper is **distilled water**.



★★★ *Indicator colour chart har saal aata hai: Nitesh Bhaiya ki guarantee, isse ratt lo!*

3 ACIDS AND BASES WITH METALS ★★★

GENERAL RULE

- Acid + Metal → **Salt + Hydrogen gas**
- Hydrogen of the acid is **displaced** by the metal

$$Zn_{(s)} + H_2SO_{4(aq)} \rightarrow ZnSO_{4(aq)} + H_{2(g)}\uparrow$$

Zinc granules + dilute sulphuric acid: bubbles of H_2

Activity: The "pop" test for hydrogen ★★★

1. Add dilute H_2SO_4 to zinc granules; pass the gas into **soap solution**.
2. Soap bubbles filled with gas rise up.
3. Bring a **burning candle** near a bubble: it burns with a **pop sound**. The gas is **H_2**.

Same result with dilute HCl, HNO_3 (very dilute) and other reactive metals.

Base + metal: zinc and sodium hydroxide ★★☆

$$2NaOH_{(aq)} + Zn_{(s)} \rightarrow Na_2ZnO_{2(aq)} + H_{2(g)}\uparrow$$

Na_2ZnO_2 = sodium zincate; H_2 again gives the pop test

Not every metal reacts with a base; zinc and aluminium do.

4 ACIDS WITH CARBONATES ★★★ MOST ASKED

GENERAL RULE

- Metal carbonate or hydrogencarbonate + Acid → **Salt + Carbon dioxide + Water**
- Brisk **effervescence** (fizzing) shows CO_2 is given off

$Na_2CO_3(s) + 2HCl(aq) \rightarrow 2NaCl(aq) + H_2O(l) + CO_2(g)\uparrow$
Sodium carbonate + hydrochloric acid

$NaHCO_3(s) + HCl(aq) \rightarrow NaCl(aq) + H_2O(l) + CO_2(g)\uparrow$
Sodium hydrogencarbonate (baking soda) + hydrochloric acid

Activity: Lime water test ★★★

1. Pass the gas through **lime water** [$Ca(OH)_2$ solution]: it turns **milky** due to white $CaCO_3$.
2. Keep passing **excess CO_2**: the milkiness **disappears**, as soluble calcium hydrogencarbonate forms.

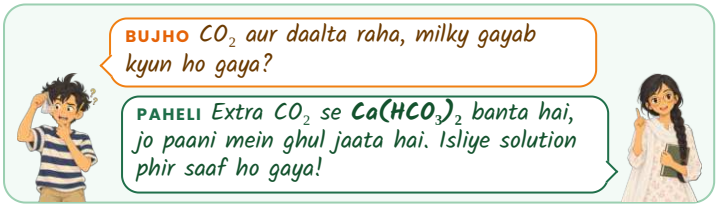
$Ca(OH)_2(aq) + CO_2(g) \rightarrow CaCO_3(s)\downarrow + H_2O(l)$
lime water turns milky

$CaCO_3(s) + H_2O(l) + CO_2(g) \rightarrow Ca(HCO_3)_2(aq)$
excess CO_2: milkiness disappears

Limestone, chalk and marble are all forms of $CaCO_3$, so each fizzes with dilute acid and gives CO_2.



MEME BREAK
*Baking soda + nimbu ka ras = **fizz hi fizz**! Wahi CO_2, wahi acid + hydrogencarbonate reaction.*



Reactions at a glance: 6 patterns ★★★

Reaction	Products
Acid + metal	Salt + $H_2\uparrow$
Base (NaOH) + metal (Zn, Al)	Salt + $H_2\uparrow$
Acid + carbonate or hydrogencarbonate	Salt + H_2O + $CO_2\uparrow$
Acid + base (neutralisation)	Salt + H_2O
Acid + metallic oxide (basic)	Salt + H_2O
Base + non-metallic oxide (acidic)	Salt + H_2O

Exam tips

- ✔ Name the gas **and** its test: H_2 → pop sound; CO_2 → lime water milky.
- ✔ Write state symbols and ↑ for gases in every final equation.
- ✔ For indicators, always give the colour in **both** acid and base.
- ✔ "Effervescence" usually means CO_2 from a carbonate.

5 NEUTRALISATION ★★★

DEFINITION

- The reaction of an **acid** with a **base** to give **salt and water** is called **neutralisation**
- Acid + Base → Salt + Water (heat is given out)

$NaOH(aq) + HCl(aq) \rightarrow NaCl(aq) + H_2O(l)$
Sodium hydroxide + hydrochloric acid → sodium chloride + water

Activity: Watch neutralisation happen ★★☆

1. Take 2 mL dilute NaOH and add 2 drops of **phenolphthalein**: it turns **pink**.
2. Add dilute HCl drop by drop: the pink colour **disappears** (base neutralised).
3. Add a few drops of NaOH again: the **pink returns**.



MEME BREAK
LEFT: NaOH + PHENOLPHTHALEIN
Solution is **pink**, all happy
RIGHT: HCl ADDED DROP BY DROP
Pink fades, solution turns **colourless**
Acid ne base ko neutralise kar diya, pink gayab!

6 OXIDES WITH ACIDS AND BASES

Metallic oxides are basic ★★★

$CuO(s) + 2HCl(aq) \rightarrow CuCl_2(aq) + H_2O(l)$
black CuO dissolves and the solution turns **blue-green**
Metal oxide + Acid → Salt + Water, just like a base.

Non-metallic oxides are acidic ★★☆

$Ca(OH)_2(aq) + CO_2(g) \rightarrow CaCO_3(s)\downarrow + H_2O(l)$
base + non-metal oxide → salt + water
Non-metal oxide + Base → Salt + Water, just like an acid.



MEME BREAK
Acid + base = salt + paani? Toh salt sirf namak nahi: NaCl, KNO_3, $CaCl_2$, $CuSO_4$, sab salts hain!

× WRONG	✔ RIGHT
Phenolphthalein: pink in acid	**Pink only in a base**; colourless in acid
$Na_2CO_3 + HCl \rightarrow NaCl + CO_2$	$Na_2CO_3 + 2HCl \rightarrow 2NaCl + H_2O + CO_2$
Gas from acid + metal is CO_2	It is **H_2**, shown by the pop test

REAL LIFE: WHY NOT IN COPPER OR BRASS?
Curd, pickles and other sour foods contain acids. In **copper or brass** vessels these acids react with the metal and form harmful salts. Store them in glass or steel.

Solved Example 2: Find compound A ★★★

Q. Metal compound A reacts with dilute HCl and gives effervescence. The gas puts out a burning candle. One product is calcium chloride. Write the balanced equation. NCERT

- Gas puts out a flame and fizzes out of an acid → **CO_2**, so A is a carbonate
- Product is $CaCl_2$ → A is **calcium carbonate, $CaCO_3$**

$$CaCO_3(s) + 2HCl(aq) \rightarrow CaCl_2(aq) + H_2O(l) + CO_2(g)\uparrow$$

Ca 1 = 1, C 1 = 1, O 3 = 3, H 2 = 2, Cl 2 = 2 ✔

Day 5 mind map: revise in 2 minutes

INDICATORS ★★★
Litmus, turmeric, red cabbage; phenolphthalein, methyl orange; onion, vanilla, clove

METALS
Acid + metal → salt + H_2; NaOH + Zn → $Na_2ZnO_2 + H_2$

ACIDS & BASES: REACTIONS

CARBONATES
Acid + carbonate → salt + H_2O + CO_2; lime water milky

NEUTRALISATION
Acid + base → salt + water

OXIDES
Metallic oxides are basic (react with acids); non-metallic oxides are acidic (react with bases)

★★★ **PYQ FAVOURITE** Zinc + dilute H_2SO_4 gives **H_2** (CBSE Term-1 SQP 2021); methyl orange is **red** in acid and **yellow** in base (CBSE SQP 2025).

7 PRACTICE MCQS

Pehle khud solve karo, phir answer key dekho. Cheating se XP nahi milta!



1. Dilute sulphuric acid is added to zinc granules and the gas formed is passed through soap solution. The gas filling the soap bubbles is: CBSE Term-1 SQP 2021 ★★★

(a) nitrogen (b) hydrogen
(c) oxygen (d) carbon dioxide

2. Methyl orange is added to dilute HCl and to aqueous NaOH. Its colours are, respectively: CBSE SQP 2025 ★★★

(a) orange, red (b) red, yellow
(c) red, orange (d) yellow, red

3. A turmeric stain on a white shirt turns reddish-brown when soap is scrubbed on it. Why? (I) Soap is acidic. (II) Soap is basic. (III) Turmeric gives a reddish tinge in bases. (IV) Turmeric gives a reddish tinge in acids. CBSE Term-1 SQP 2021 ★★★

(a) (I) and (III) (b) (II) and (III)
(c) (I) and (IV) (d) (II) and (IV)

4. Excess CO_2 is passed through lime water. The milkiness disappears because: CBSE SQP 2025 ★★★

(a) $CaCO_3$ changes into CaO
(b) insoluble $CaCO_3$ changes into soluble $Ca(HCO_3)_2$
(c) $Ca(OH)_2$ is formed again
(d) insoluble $CaCO_3$ changes into insoluble $Ca(HCO_3)_2$

5. Which of these is an olfactory indicator? ★★☆

(a) Litmus (b) Turmeric
(c) Vanilla extract (d) Methyl orange

6. Two drops of phenolphthalein are added to dilute NaOH solution. The solution becomes: ★★★

(a) colourless (b) pink
(c) red (d) yellow

7. Zinc is heated with sodium hydroxide solution. What does it give? ★★☆

(a) ZnO and H_2O (b) Na_2ZnO_2 and H_2
(c) $Zn(OH)_2$ and Na (d) no reaction takes place

8. Black copper(II) oxide is stirred with dilute HCl. What colour does the solution turn? ★★☆

(a) colourless (b) blue-green
(c) pink (d) yellow

9. **Assertion (A):** Carbon dioxide is an acidic oxide.
Reason (R): CO_2 reacts with lime water, which is a base, to give salt and water. ★★★

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

10. **Assertion (A):** Phenolphthalein is a natural indicator.
Reason (R): Phenolphthalein turns pink in a basic solution. ★★☆

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

Short answer questions

1. Which gas is usually given off when an acid reacts with a metal? Give an example with an equation, and state how you will test for this gas. NCERT
2. Write the balanced equations, with states, for: (a) dilute sulphuric acid with zinc granules, (b) dilute HCl with magnesium ribbon, (c) dilute sulphuric acid with sodium hydroxide. NCERT

Case-based question: Sara's titration

Sara adds two drops of phenolphthalein to 2 mL of dilute NaOH: it turns **pink**. She adds dilute H_2SO_4 drop by drop; after **40 drops** it turns colourless. One more drop of NaOH makes it pink again.
Based on CBSE SQP 2025 ★★★

1. Why did the solution first turn pink, and why did it later become colourless?
2. How many drops of the same acid will 4 mL of the same NaOH need?
3. Write the balanced equation and name the type of reaction.

TRY YOURSELF (ANSWERS ON PAGE 4)

Complete and balance:

(i) Mg + HCl →
(ii) KOH + HNO_3 →
(iii) ZnO + H_2SO_4 →
(iv) $CaCO_3$ + HCl →

Answer key: MCQs

Q	Ans	Why
1	(b)	Zn + dil. H_2SO_4 gives H_2; it burns with a pop.
2	(b)	Methyl orange: red in acid, yellow in base.
3	(b)	Soap is basic; turmeric turns reddish-brown in bases.
4	(b)	$Ca(HCO_3)_2$ dissolves in water.
5	(c)	Its smell disappears in a base.
6	(b)	Phenolphthalein is pink in a base.
7	(b)	Sodium zincate and hydrogen gas form.
8	(b)	Copper(II) chloride solution is blue-green.
9	(a)	Non-metal oxide + base → salt + water.
10	(d)	It is synthetic; R is true.

Model answers: short answer

1 Gas from acid + metal: hydrogen

- Hydrogen gas is given off, e.g. zinc with dilute sulphuric acid
- **Test:** bring a burning candle near the gas; it burns with a **pop sound**

$$Zn_{(s)} + H_2SO_{4(aq)} \rightarrow ZnSO_{4(aq)} + H_{2(g)}\uparrow$$

2 Balanced equations

$$Zn_{(s)} + H_2SO_{4(aq)} \rightarrow ZnSO_{4(aq)} + H_{2(g)}\uparrow$$
(a) zinc granules + dilute sulphuric acid

$$Mg_{(s)} + 2HCl_{(aq)} \rightarrow MgCl_{2(aq)} + H_{2(g)}\uparrow$$
(b) magnesium ribbon + dilute hydrochloric acid

$$2NaOH_{(aq)} + H_2SO_{4(aq)} \rightarrow Na_2SO_{4(aq)} + 2H_2O_{(l)}$$
(c) neutralisation: Na 2 = 2, S 1 = 1, O 6 = 6, H 4 = 4 ✔

Answers: Sara's case

- Phenolphthalein is pink in a base (NaOH). The acid neutralises the base, so the solution becomes colourless.
- **80 drops**: double the base needs double the acid.
- It is **neutralisation** (and double displacement); no precipitate forms, as Na_2SO_4 stays dissolved.

$$2NaOH_{(aq)} + H_2SO_{4(aq)} \rightarrow Na_2SO_{4(aq)} + 2H_2O_{(l)}$$

Answers: Try yourself

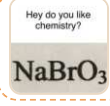
(i) $Mg + 2HCl \rightarrow MgCl_2 + H_2$
(ii) $KOH + HNO_3 \rightarrow KNO_3 + H_2O$
(iii) $ZnO + H_2SO_4 \rightarrow ZnSO_4 + H_2O$
(iv) $CaCO_3 + 2HCl \rightarrow CaCl_2 + H_2O + CO_2$



BUJHO *Toh har metal acid se H_2 deta hai?*

PAHELI *Nahi! **Copper** aur **silver** jaise kam reactive metals dilute acid se H_2 nahi dete. Chapter 3 mein detail!*





★★★ *Zn + acid, carbonate + acid, lime water, neutralisation: ye chaar pakka aayenge, skip mat karna! Nitesh Bhaiya ki guarantee!*

Day 5 checklist: tick before you sleep

- ☐ I can give the colours of litmus, turmeric, red cabbage, phenolphthalein and methyl orange in acid and base
- ☐ I can name the olfactory indicators and say how they work
- ☐ I can write acid + metal and NaOH + Zn equations, with the pop test
- ☐ I can write acid + carbonate equations and explain the lime water test
- ☐ I solved all 10 MCQs, including neutralisation and oxide questions

My score: ____ / 5

5/5 = Jalwa hai! 4 se kam? Dimaag nahi chal raha? Page 1–2 phir se padho.



LEVEL 5 COMPLETE +100 XP earned. Next: Level 6



BUJHO *Samajh gaya! Metal se H_2, carbonate se CO_2, aur acid + base = salt + paani!*

PAHELI *High five! Kal Level 6: pH scale.*

Day 5 hit-list: Eduspark PYQ analysis

Topic / reaction	Rating
Indicator colours (litmus, phenolphthalein, methyl orange, turmeric)	★★★
Zn + dilute H_2SO_4: H_2 and the pop test	★★★
Lime water test; excess CO_2 clears milkiness	★★★
Acid + carbonate or hydrogencarbonate	★★★
Neutralisation with phenolphthalein	★★☆
Olfactory indicators; NaOH + Zn	★★☆

CLASS QUESTION

Q. Are all acids sour and all bases bitter?
Ans. Generally acids taste sour and bases bitter, but taste is not a reliable or safe test. **Coffee** tastes bitter yet is slightly **acidic** (pH about 5), so bitter does not always mean basic. Always use an indicator; never taste chemicals in the lab.

Which acid gives which salt? HCl → chlorides (NaCl, $CaCl_2$); H_2SO_4 → sulphates ($ZnSO_4$, Na_2SO_4); HNO_3 → nitrates (KNO_3)



NITESH SIR SAYS
*Indicator ka colour table aaj hi yaad karo. **Kal pH ka jalwa!***

★ Stars (Eduspark PYQ analysis): how often a topic came in Chapter 2 board and sample papers; year tags only where the year is known.

DAY 06 · LEVEL 6

CHEMISTRY | CHAPTER 2 | PART 2 · ASTRA SERIES

IONS, STRENGTH & THE pH SCALE

Goal: ions in water, dilution, strength and the pH scale.



by Nitesh Prakash

01 Ions in Water | 02 Conduction | 03 Dilution | 04 Strength | 05 pH Scale

+100 XP today | Level 6 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 ACIDS & BASES IN WATER ★★★

DEFINITION

- **Acid:** gives **$H^+(aq)$ ions** when dissolved in water
- **Base:** gives **$OH^-(aq)$ ions** when dissolved in water
- H^+ cannot exist alone; it joins water to form **hydronium ion, H_3O^+**

$$HCl(aq) + H_2O(l) \rightarrow H_3O^+(aq) + Cl^-(aq)$$

HCl gives H^+, which is carried as H_3O^+

$$H^+ + H_2O(l) \rightarrow H_3O^+(aq)$$

so $H^+(aq)$ and $H_3O^+(aq)$ mean the same thing in answers

ACID IN WATER	→ together they make water	BASE IN WATER
H^+ / H_3O^+ ions blue litmus → red		OH^- ions red litmus → blue

$$H^+(aq) + OH^-(aq) \rightarrow H_2O(l)$$

neutralisation, in one line (Day 5)

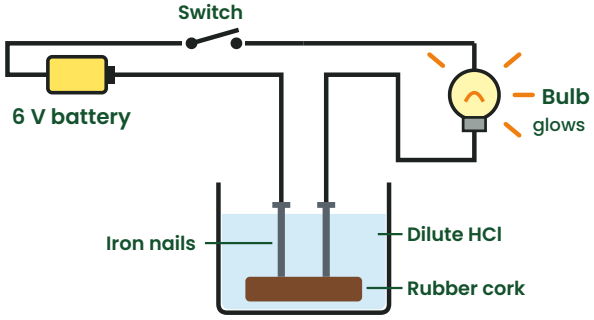


MEME BREAK

*Glucose ($C_6H_{12}O_6$) and alcohol (C_2H_5OH): "Hum mein bhi H hai, hum acid kyun nahi?" Kyunki paani mein **hydrogen ions nahi dete**.*

2 CONDUCTION EXPERIMENT ★★☆

NCERT activity: do acids conduct?



- Fix two iron nails on a rubber cork in a beaker; connect them to a 6 V battery through a bulb and a switch
- Pour **dilute HCl**, switch on: the **bulb glows**
- Repeat with dilute H_2SO_4 (glows), then glucose and alcohol solutions (bulb **does not glow**)

Result and conclusion

Solution	Bulb	Reason
Dilute HCl, H_2SO_4	✔ glows	free H^+ ions carry current
Glucose, alcohol	× no glow	no ions formed in water

Conclusion: acids give $H^+(aq)$ ions in water and these ions carry the current. Bases conduct due to OH^- ions.



BUJHO *Glucose mein bhi toh H hai, phir bulb kyun nahi jala?*

PAHELI *Kyunki glucose paani mein H^+ ions nahi banata. Ions nahi, toh current nahi!*



3 ACIDS NEED WATER ★★★

NCERT activity: acids need water to act

$$NaCl(s) + H_2SO_4(l) \rightarrow NaHSO_4(s) + HCl(g)\uparrow$$

salt + conc. H_2SO_4; $CaCl_2$ guard tube keeps the gas dry

DRY LITMUS	MOIST LITMUS
× Dry blue litmus: **no change**	✔ Moist blue litmus turns **red**
No water, so HCl does not ionise; no H^+ ions	HCl dissolves in the water and gives H_3O^+ ions



MEME BREAK

*Dry HCl gas litmus ko red kar dega? **Bilkul nahi!** Acidic behaviour sirf **aqueous solution** mein dikhta hai, kyunki paani ke bina hydrogen ions bante hi nahi.*

4 BASES AND ALKALIS ★★☆

DEFINITION

- A base that **dissolves in water** is called an **alkali**: NaOH, KOH, $Ca(OH)_2$, NH_4OH
- All alkalis are bases, but **all bases are not alkalis**: $Cu(OH)_2$, $Fe(OH)_3$ do not dissolve

$$NaOH(s) \xrightarrow{H_2O} Na^+(aq) + OH^-(aq)$$

alkalis feel soapy, taste bitter and are corrosive

5 DILUTION OF ACIDS & BASES ★★★ MOST ASKED

DEFINITION

- **Dilution:** mixing an acid or base with water to **decrease the concentration** of ions per unit volume
- Mixing a concentrated acid or base with water is **highly exothermic** (a lot of heat is released)

Safety rule: acid into water, never water into acid

- Add the **acid slowly to water** with **constant stirring**; the large amount of water absorbs the heat
- Water added to concentrated acid: heat is released suddenly, the mixture **splashes out** and causes burns
- Local overheating can even **crack the glass** container

× WRONG	✓ RIGHT
Water poured into conc. acid	Acid added slowly to water
Fast mixing, no stirring	Slow mixing with stirring



MEME BREAK
*Conc. acid mein paani daal diya? Garam mixture uchhal ke haath par: **ouch!** Yaad rakho, **acid ko paani mein**, dheere se, hilate hue.*

What dilution does to the ions

Diluting...	Ion per unit volume	pH moves
an acid	$[H_3O^+]$ **decreases**	up, towards 7
a base	$[OH^-]$ **decreases**	down, towards 7

Dilution never crosses 7: a diluted acid stays acidic, a diluted base stays basic.



BUJHO *Jaldi hai, conc. acid mein hi paani daal doon?*

PAHELI *Bilkul nahi! Heat se mixture uchhal ke bahar aayega. Hamesha acid ko paani mein, hilate hue.*



★★★ *"Acid to water, why?" 3-mark sawaal har saal ka hai: Nitesh Bhaiya ki guarantee!*

Exam tips

- ✔ For acids in water write "H^+(aq) or H_3O^+(aq)", not a bare H^+.
- ✔ Bulb glows = ions present. Glucose and alcohol have H but give no ions.
- ✔ Dilution answer keywords: "highly exothermic", "acid to water", "slowly with stirring".

6 STRONG & WEAK ACIDS, BASES ★★★ MOST ASKED

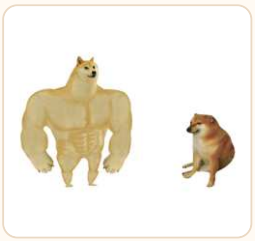
STRONG	WEAK
Ionises **completely** in water; gives **many** H^+ (or OH^-) ions	Ionises **partially**; gives **fewer** H^+ (or OH^-) ions
Acids: HCl, H_2SO_4, HNO_3	**Acids:** CH_3COOH, H_2CO_3, citric acid
Bases: NaOH, KOH	**Bases:** NH_4OH, $Mg(OH)_2$

$HCl(aq) \rightarrow H^+(aq) + Cl^-(aq)$
strong: every HCl molecule breaks into ions

$CH_3COOH(aq) \rightleftharpoons CH_3COO^-(aq) + H^+(aq)$
weak: only a few molecules ionise (⇌ = partial, reversible)



MEME BREAK
*Acetic acid thoda sa ionise; HCl ka har molecule toot jaata hai. **100% ionisation = strong acid.***



MEME BREAK
LEFT: STRONG (HYDROCHLORIC ACID)
lower pH, bright bulb, fast fizz with Mg
RIGHT: WEAK (ACETIC ACID)
same concentration: higher pH, dim bulb
Same concentration, alag taaqat: H^+ ions ka khel hai!

Strength vs concentration: don't mix them up

	Strength	Concentration
Meaning	degree of ionisation	amount of acid in a volume
Words	strong / weak	concentrated / dilute
On dilution	does not change	decreases

So: dilute HCl is still a **strong** acid; concentrated acetic acid is still a **weak** acid.

CLASS QUESTION

Q. Why is HF (hydrofluoric acid) not stored in glass bottles?
Ans. HF reacts with silica (SiO_2) in glass and eats it away, so it is kept in **plastic (polythene)** or wax-lined bottles.

$SiO_2(s) + 4HF(aq) \rightarrow SiF_4(g) + 2H_2O(l)$

7 THE pH SCALE ★★★ MOST ASKED

DEFINITION

- **pH scale:** measures the **H^+ ion concentration** of a solution, from **0** (very acidic) to **14** (very basic)
- "p" comes from **potenz**, German for power
- **Higher the H^+ concentration, lower the pH**

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14

0: STRONGLY ACIDIC — 7: NEUTRAL — 14: STRONGLY BASIC

← acidic: H^+ increases, pH falls | basic: OH^- increases, pH rises →

Read the scale ★★★

pH	Nature	Ions in excess
< 7	acidic (lower pH, more acidic)	H^+ more than OH^-
= 7	neutral (pure water)	$H^+ = OH^-$
> 7	basic (higher pH, more basic)	OH^- more than H^+

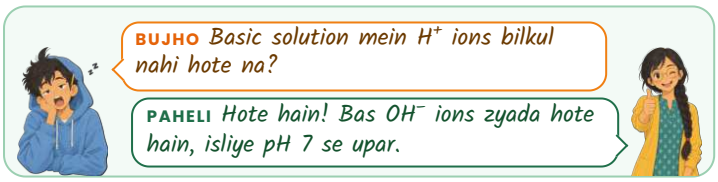
pH of common substances (approx.)

Substance	pH	Substance	pH
Gastric juice	1.2	Pure water	7
Lemon juice	2.2	Blood	7.4
Vinegar	about 3	Milk of magnesia	10
Milk	about 6.5	NaOH solution	14

Universal indicator

- A **mixture of several indicators**; shows a different colour at each H^+ concentration, so it gives the **pH value**
- Litmus only says acid or base; universal indicator also says **how acidic or basic**

pH	Colour	pH	Colour
0–2	red	7	green
3–4	orange	8–10	blue
5–6	yellow	11–14	violet



Solved Example: hydronium ions CBSE SQP 2024 ★★★

Q. Solutions A, B, C have pH 6, 8 and 9.5. Which has the most hydronium ions? Arrange them in increasing order of H^+ ion concentration.

- Lowest pH = most H^+, so **A (pH 6)** has the maximum hydronium ions
- Increasing $[H^+]$: **C (9.5) < B (8) < A (6)**

8 PRACTICE MCQS

Kyun nahi ho rahi padhai? Phone side mein, pehle khud solve karo! Answer key page 4 par hai.



1. Which of the following substances, when dissolved in an equal volume of water, will have the highest pH value? CBSE SQP 2025 ★★★

(a) Sulphuric acid (b) Acetic acid
(c) Magnesium hydroxide (d) Sodium hydroxide

2. Two nails dipped in a solution are connected to a battery and a bulb. The bulb does not glow. The solution is likely to be: CBSE SQP 2024 ★★★

(a) HNO_3 (b) C_2H_5OH
(c) H_2SO_4 (d) CO_2 in water

3. HCl gas from NaCl and conc. H_2SO_4 changes the colour of moist blue litmus paper. This is due to the presence of: CBSE SQP 2022 ★★☆

(a) an acid (b) a base
(c) H^+(aq) in the solution (d) litmus

4. Which option correctly matches the solution, its pH paper colour, pH and nature? CBSE 2023 ★★★

(a) Lemon juice: orange, 3, basic
(b) Milk of magnesia: blue, 10, basic
(c) Gastric juice: red, 6, acidic
(d) Pure water: yellow, 7, neutral

5. Solution P has pH 6 and Q has pH 8. So: ★★★

(a) P has more H^+ ions and is acidic
(b) Q has more H^+ ions and is acidic
(c) P is basic and Q is acidic
(d) Both are neutral

6. The safe way to dilute concentrated sulphuric acid is to: ★★★

(a) add water to the acid quickly
(b) add the acid to water slowly with stirring
(c) add both together at once
(d) heat the acid, then add water

7. Which of the following is a weak acid? ★★☆

(a) HCl (b) HNO_3
(c) CH_3COOH (d) H_2SO_4

8. Adding an acid to water involves: (i) ionisation (ii) neutralisation (iii) dilution (iv) salt formation ★★☆

(a) (i) and (ii) (b) (i) and (iii)
(c) (ii) and (iii) (d) (ii) and (iv)

9. A few drops of universal indicator turn a solution green. Its pH and nature are: ★★☆

(a) pH 1, strongly acidic
(b) pH 7, neutral
(c) pH 10, basic
(d) pH 14, strongly basic

10. **Assertion (A):** Dry HCl gas does not change the colour of dry blue litmus paper.
Reason (R): HCl produces H^+ ions only in the presence of water. ★★★

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

11. **Assertion (A):** A concentrated acid is diluted by adding it slowly to water with constant stirring.
Reason (R): Mixing a concentrated acid with water is a highly exothermic process. ★★★

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

Answer key: MCQs

Q	Ans	Why
1	(d)	NaOH: strong base, completely ionised; most OH^-.
2	(b)	Ethanol gives no ions, so no current.
3	(c)	HCl dissolves in the moisture and gives H^+(aq).
4	(b)	Milk of magnesia: blue, pH about 10, basic.
5	(a)	Lower pH means more H^+; pH 6 is acidic.
6	(b)	Heat is absorbed by the large amount of water.
7	(c)	Acetic acid ionises only partially.
8	(b)	The acid ionises and its ions get diluted.
9	(b)	Green means neutral, pH 7.
10	(a)	No water, no H^+ ions; R explains A.
11	(a)	Highly exothermic, so add acid slowly.

Short and long answer questions

1. Why does an aqueous solution of an acid conduct electricity, while a glucose solution does not? (2 marks) NCERT
2. While diluting an acid, why is it recommended that the acid be added to water and not water to the acid? (2 marks) NCERT
3. How is the concentration of H_3O^+ ions affected when an acid solution is diluted? (1 mark) NCERT
4. (a) Describe an activity to show that HCl is acidic only in water. (b) Differentiate strong and weak acids, with one example each. (5 marks)

Model answers

1 Acid conducts, glucose does not

- Acid in water gives **H^+(aq) ions**; moving ions carry current
- Glucose does not ionise in water: **no ions, no current**

2 Acid to water

- Dilution is **highly exothermic**; water into acid makes the mixture **splash** and burn, and glass may crack
- Acid added slowly with stirring: the water absorbs the heat

3 On dilution

H_3O^+ ions per unit volume **decrease**, so pH rises towards 7.

4 (a) Dry vs moist litmus

- Add conc. H_2SO_4 to common salt; pass the gas through a $CaCl_2$ guard tube
- Dry blue litmus: no change; moist blue litmus turns red, so H^+ ions form only in water

$$NaCl_{(s)} + H_2SO_{4(l)} \rightarrow NaHSO_{4(s)} + HCl_{(g)}\uparrow$$

(b) Strong acid ionises completely, e.g. HCl; weak acid ionises partially, e.g. CH_3COOH.

× COMMON MISTAKE	✓ CORRECT IDEA
Dilute HCl is a weak acid	Dilute HCl is still strong
Bases have no H^+ ions	They have fewer H^+ than OH^-
pH 9 is more acidic than pH 4	Lower pH = more acidic

Day 6 checklist: tick before you sleep

- [ ] I can explain hydrogen, hydronium and hydroxide ions with equations
- [ ] I can describe the conduction and dry HCl activities
- [ ] I know why acid is always added to water
- [ ] I can tell strength from concentration and read the pH scale
- [ ] I solved all 11 MCQs and checked them with the key

My score: ____ / 5

5/5 = Jalwa hai, ab chain ki neend lo! 4 se kam? Page 1–3 phir se padho.



LEVEL 6 COMPLETE

+100 XP earned. Next: Level 7



BUJHO *Samajh gaya! H^+ zyada toh pH kam, aur acid hamesha paani mein!*

PAHELI *Shabaash! Kal Level 7: pH in everyday life.*

Day 6 mind map: revise in 2 minutes

IONS
Acid gives H^+(aq) or H_3O^+; base gives OH^-(aq) ions

ONLY IN WATER
Bulb glows with acids; dry HCl does not affect dry litmus

ACIDS & BASES IN WATER

DILUTION ★★★
Exothermic; acid into water, slowly, stirring

STRENGTH
Complete vs partial ionisation; not the same as concentration

pH SCALE ★★★
0–14; 7 neutral; below 7 acidic, above 7 basic; more H^+ = lower pH; universal indicator gives the value

Coming up: Day 7

pH in everyday life: your stomach, your teeth, the soil, a bee sting, acid rain. Plus the family of salts and their pH.



NITESH SIR SAYS
*pH ka khel samajh gaye toh Chapter 2 aadha jeet liya. **Kal milte hai Day 7 pe!***

★ How we starred (Eduspark PYQ analysis): we counted how often each topic has come in Chapter 2 board and sample-paper questions. Year tags appear only where the paper year is known.

DAY 07 · LEVEL 7

CHEMISTRY | CHAPTER 2 | PART 3 · ASTRA SERIES

pH IN DAILY LIFE & SALTS

Goal: apply pH to daily life and predict the pH of salts.



by Nitesh Prakash

01 Body & Rain | 02 Digestion | 03 Soil & Teeth | 04 Natural Acids | 05 Salts & pH | +100 XP today | Level 7 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 pH IN OUR BODY AND RAIN

KEY IDEA

- Living things survive only in a **narrow pH range**
- Our body works well between **pH 7.0 and 7.8**; a change upsets body processes
- Rain water with **pH below 5.6** is **acid rain**

Quick pH numbers to remember

Where	pH	What it means
Gastric juice (stomach)	about 2	HCl helps digest food
Mouth, tooth decay	< 5.5	enamel starts to corrode
Acid rain	< 5.6	harms fish and plants
Fresh milk	about 6	slightly acidic
Human body	7.0-7.8	blood is about 7.4
Toothpaste	about 8	mild base, fights decay

Acid rain: why rivers suffer ★★☆

- SO_2 and NO_2 from burning fuels dissolve in rain and make acids
- Acid rain flows into rivers and **lowers their pH**; aquatic life finds it hard to survive
- It also corrodes marble ($CaCO_3$) monuments

2 pH IN THE DIGESTIVE SYSTEM

- Stomach makes **hydrochloric acid (HCl)**; it helps digest food without harming the stomach
- **Indigestion:** too much acid causes pain and irritation
- **Antacids** are **mild bases** that neutralise the extra acid

$$Mg(OH)_{2(s)} + 2HCl_{(aq)} \rightarrow MgCl_{2(aq)} + 2H_2O_{(l)}$$

Milk of magnesia: Mg 1 = 1, O 2 = 2, H 4 = 4, Cl 2 = 2 ✔

$$NaHCO_{3(s)} + HCl_{(aq)} \rightarrow NaCl_{(aq)} + H_2O_{(l)} + CO_{2(g)}\uparrow$$

Baking soda is also used as an antacid



MEME BREAK

*Pet mein jalan? "Kya tapleek hai aapko?" Tapleek = extra HCl. Ilaaj = antacid, jo karta hai **neutralisation**!*



BUJHO *Pet mein jalan hai. NaOH pee loon? Base hi to hai!*

PAHELI *Bilkul nahi! NaOH strong aur corrosive hai. **Mild base** lo: milk of magnesia ya baking soda.*



3 pH OF SOIL

- Most plants grow best when soil pH is **close to 7**
- **Too acidic soil:** add a base: quicklime (CaO), slaked lime ($Ca(OH)_2$) or chalk ($CaCO_3$)
- **Too basic soil:** add decaying organic matter (manure, compost), which is acidic



BUJHO *Mitti ka pH 4.5 aaya. Thoda nimbu daal doon?*

PAHELI *Ulta ho jayega! Acidic mitti ko **base** chahiye, jaise quicklime, slaked lime ya chalk.*



4 TOOTH DECAY ★★★ MOST ASKED

1. Bacteria in the mouth break down **sugar and food particles** left after eating and produce **acids**.
2. The pH of the mouth falls; decay starts when it goes **below 5.5**.
3. Enamel is **calcium hydroxyapatite** (a form of calcium phosphate), the hardest substance in the body. It does not dissolve in water but is corroded by acid.
4. **Prevention:** rinse the mouth after eating; brush with toothpaste, which is basic and neutralises the acid.



★★★ *Tooth decay = pH 5.5 se neeche. Ye number pakka aayega, Nitesh Bhaiya ki guarantee!*

5 SELF-DEFENCE BY ANIMALS AND PLANTS

STING	RELIEF
Bee or ant sting: injects an acid (ant: methanoic acid)	Rub a mild base: **baking soda**
Nettle leaf: stinging hairs inject **methanoic acid**	Rub a **dock leaf** (it contains a base)
Wasp sting: injects a basic liquid	Rub a mild acid: **vinegar**



BUJHO *Jungle mein ek paudha chhua. Haath mein jalan, aankh mein aansu!*

PAHELI *Nettle hai! Uske baal methanoic acid daalte hain. Paas wale* ***dock ke patte*** *ragdo, unme base hota hai.*

6 NATURALLY OCCURRING ACIDS ★★★

Natural source	Acid present
Vinegar	**Acetic acid (ethanoic acid)**
Orange, lemon	**Citric acid**
Tamarind	**Tartaric acid**
Tomato	**Oxalic acid**
Sour milk (curd)	**Lactic acid**
Ant sting, nettle sting	**Methanoic acid (formic acid)**

These **organic acids** are weak acids, safe in food. Mineral acids (HCl, H_2SO_4, HNO_3) are strong.

7 SALTS AND FAMILIES OF SALTS

DEFINITION

- A **salt** forms when an acid neutralises a base: Acid + Base → Salt + Water
- Its **positive ion** comes from the base, its **negative ion** from the acid
- Salts with the **same positive or same negative ion** form a **family**

NCERT activity: salts, parent acids and bases ★★☆

Salt	Formula	Acid	Base
Potassium sulphate	**K_2SO_4**	H_2SO_4	KOH
Sodium sulphate	**Na_2SO_4**	H_2SO_4	NaOH
Calcium sulphate	**$CaSO_4$**	H_2SO_4	$Ca(OH)_2$
Magnesium sulphate	**$MgSO_4$**	H_2SO_4	$Mg(OH)_2$
Copper sulphate	**$CuSO_4$**	H_2SO_4	$Cu(OH)_2$
Sodium chloride	**NaCl**	HCl	NaOH
Sodium nitrate	**$NaNO_3$**	HNO_3	NaOH
Sodium carbonate	**Na_2CO_3**	H_2CO_3	NaOH
Ammonium chloride	**NH_4Cl**	HCl	NH_4OH

Sulphate family: K_2SO_4, Na_2SO_4, $CaSO_4$, $MgSO_4$, $CuSO_4$. **Sodium family:** Na_2SO_4, NaCl, $NaNO_3$, Na_2CO_3. **Chloride family:** NaCl, NH_4Cl.



MEME BREAK

SO_4^{2-} to Na, K, Ca, Mg, Cu: "Humare liye kaam karoge?" Sab haan bole, ban gayi ***sulphate family****!*

× WRONG	✓ RIGHT
Every salt is neutral	pH depends on the parent acid and base
"Salt" means only NaCl	NaCl is common salt; $CuSO_4$, Na_2CO_3 are salts too
$NaHCO_3$ has H, so it is acidic	Weak acid + strong base: basic

8 pH OF SALTS ★★★ MOST ASKED

NEUTRAL pH = 7	ACIDIC pH < 7	BASIC pH > 7
Strong acid + strong base	Strong acid + weak base	Weak acid + strong base
NaCl, KNO_3, Na_2SO_4, K_2SO_4	NH_4Cl, $CuSO_4$, $ZnSO_4$, $AlCl_3$	Na_2CO_3, CH_3COONa, $NaHCO_3$

KNOW THE PARENTS

Strong acids: HCl, H_2SO_4, HNO_3. **Weak acids:** CH_3COOH, H_2CO_3. **Strong bases:** NaOH, KOH. **Weak bases:** NH_4OH and insoluble hydroxides such as $Cu(OH)_2$, $Zn(OH)_2$.



★★★ *Salt ka swabhav = jo parent strong, uska. Dono strong? Neutral! Nitesh Bhaiya ki guarantee, ye rule har saal aata hai.*

Solved example: predict the pH

Q. Sodium acetate, CH_3COONa: acidic, basic or neutral?

$$CH_3COOH(aq) + NaOH(aq) \rightarrow CH_3COONa(aq) + H_2O(l)$$

Weak acid + strong base → **basic salt**, pH more than 7 ✓

NCERT activity: test the pH of these salts ★★★

Salt	Acid	Base	Nature
NaCl	HCl (S)	NaOH (S)	Neutral
KNO_3	HNO_3 (S)	KOH (S)	Neutral
$AlCl_3$	HCl (S)	$Al(OH)_3$ (W)	Acidic
$ZnSO_4$	H_2SO_4 (S)	$Zn(OH)_2$ (W)	Acidic
$CuSO_4$	H_2SO_4 (S)	$Cu(OH)_2$ (W)	Acidic
CH_3COONa	CH_3COOH (W)	NaOH (S)	Basic
Na_2CO_3	H_2CO_3 (W)	NaOH (S)	Basic
$NaHCO_3$	H_2CO_3 (W)	NaOH (S)	Basic

S = strong, W = weak. Dissolve each salt in distilled water and test with pH paper.

Real life: the milkman and baking soda ★★☆

- Fresh milk has pH about 6; as it sours, **lactic acid** forms and the pH **falls**
- A milkman adds a little **baking soda**: pH shifts to slightly alkaline
- So the milk takes **longer to set into curd**; the base must first neutralise the acid formed

Exam tips

- ✓ Always write the **number**: 7.0–7.8 (body), 5.5 (teeth), 5.6 (acid rain).
- ✓ Antacid answers need the word **mild base** and the neutralisation equation.
- ✓ For a salt, name **both parents** and say which is strong before giving the pH.
- ✓ Tamarind = tartaric, tomato = oxalic: the most confused pair in MCQs.

Day 7 mind map: revise in 2 minutes

BODY & RAIN
Body pH 7.0–7.8; acid rain below 5.6 harms aquatic life

DIGESTION
Extra HCl → antacid (mild base): $Mg(OH)_2$, $NaHCO_3$

pH IN DAILY LIFE & SALTS

SOIL & TEETH
Acidic soil + lime; decay below pH 5.5

STINGS
Methanoic acid → baking soda, dock leaf

SALTS ★★★
Family = same positive or negative ion; pH follows the stronger parent: SA + SB neutral, SA + WB acidic, WA + SB basic

TRY YOURSELF: ACIDIC, BASIC OR NEUTRAL?

(i) K_2SO_4 (ii) NH_4NO_3 (iii) CH_3COOK
(iv) $(NH_4)_2SO_4$ (v) KCl (vi) K_2CO_3
Hint: name the parent acid and base first, then mark which one is strong.

9 PRACTICE MCQS

Pehle khud solve karo, answer key page 4 par hai. Cheating = apne aap se dhokha!



1. Tooth decay starts when the pH of the mouth falls below: ★★★
(a) 7.0 (b) 6.5
(c) 5.5 (d) 8.0

2. Tamarind juice turns blue litmus red because it contains: CBSE 2025 ★★★
(a) acetic acid (b) methanoic acid
(c) oxalic acid (d) tartaric acid

3. Aqueous solutions: A = potassium nitrate, B = ammonium chloride, C = sodium carbonate. The ascending order of their pH is: CBSE 2025 ★★★
(a) A < B < C (b) B < C < A
(c) C < A < B (d) B < A < C

4. Sonia tests solutions of sodium acetate, sodium chloride and ammonium chloride and gets pH values 4.6, 7.0 and 8.9. Which order matches pH 4.6, 7.0, 8.9? CBSE Practice Set 2023 ★★☆
(a) sodium acetate, sodium chloride, ammonium chloride
(b) sodium chloride, ammonium chloride, sodium acetate
(c) ammonium chloride, sodium acetate, sodium chloride
(d) ammonium chloride, sodium chloride, sodium acetate

5. The pH of a soil sample is 5. Which of these should be added to make it suitable for crops? (I) calcium chloride (II) calcium hydroxide (III) calcium sulphate (IV) calcium oxide ★★☆
(a) (I) and (II) (b) (II) and (IV)
(c) (III) and (IV) (d) only (III)

6. Which pairs belong to the same family of salts? (I) NaCl and CH_3COONa (II) $CaSO_4$ and $MgSO_4$ (III) Na_2CO_3 and $NaHCO_3$ (IV) NaCl and $MgSO_4$ ★★☆
(a) (I) and (II) only (b) (II) and (III) only
(c) (I), (II) and (III) (d) (I), (II) and (IV)

7. The aqueous solution of which salt turns red litmus blue? ★★☆
(a) NH_4Cl (b) NaCl
(c) Na_2CO_3 (d) $CuSO_4$

8. "Milk of magnesia", a common antacid, is: ★★★
(a) magnesium hydroxide (b) sodium hydroxide
(c) calcium oxide (d) magnesium sulphate

9. **Assertion (A):** Aquatic animals find it difficult to survive in a river into which acid rain flows.
Reason (R): Acid rain lowers the pH of the river water. ★★★
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

10. **Assertion (A):** An aqueous solution of sodium carbonate is basic in nature.
Reason (R): Sodium carbonate is formed from a strong acid and a weak base. ★★★
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

Short answer questions (PYQ)

1. Comment on: (A) a bee sting is treated with baking soda paste, but a wasp sting with dilute vinegar; (B) farmers treat soil with quicklime when tilling; (C) marble sculptures are protected by treating them with certain chemicals. CBSE SQP 2024
2. (A) How does pH help us when (i) the stomach makes too much acid during indigestion, (ii) we are stung by a honey bee or nettle leaves? (B) Fresh milk has pH 6. When it turns into curd, will its pH rise or fall? Why? CBSE 2022
3. Write the name and formula of (i) an acidic salt and (ii) a basic salt. Name the parent acid and parent base of each. CBSE 2024

Case-based question (4 marks)

A farmer finds his wheat growing poorly; his soil has **pH 4.5**. In a pond nearby, fed by rain water, fish are dying. The rain water tests **pH 4.8**.

1. Is this rain acid rain? Give a reason. (1)
2. Name two substances the farmer can add to his soil, and the type of reaction that takes place. (2)
3. Calcium carbonate is added to the pond. What happens to the pH of its water? (1)

Long answer question (5 marks)

1. What is a family of salts? Give one example each of a sodium family and a sulphate family. (2)
2. Explain why an aqueous solution of ammonium chloride is acidic, while that of sodium acetate is basic. Write the equation for the formation of each salt. (2)
3. Why is a strong base like sodium hydroxide never used as an antacid? (1)

Answer key: MCQs

Q	Ans	Why
1	(c)	Decay starts below pH 5.5.
2	(d)	Tamarind has tartaric acid.
3	(d)	NH_4Cl acidic < KNO_3 neutral < Na_2CO_3 basic.
4	(d)	NH_4Cl 4.6, NaCl 7.0, CH_3COONa 8.9.
5	(b)	$Ca(OH)_2$ and CaO are bases; the others are neutral salts.
6	(c)	(I), (III) sodium family; (II) sulphate family.
7	(c)	Na_2CO_3 is a basic salt.
8	(a)	Milk of magnesia = $Mg(OH)_2$.
9	(a)	Lower pH is the reason fish suffer.
10	(c)	Na_2CO_3 comes from a weak acid (H_2CO_3) and a strong base (NaOH).

Model answers

1 Comment on the statements

- **(A)** Bee sting is acidic, so a mild base (baking soda) neutralises it; wasp sting is basic, so a mild acid (vinegar) neutralises it.
- **(B)** Soil is often too acidic; quicklime is a base and neutralises the extra acid.
- **(C)** Acid rain corrodes marble ($CaCO_3$); a protective treatment keeps the acid away.

$$CaCO_{3(s)} + H_2SO_{4(aq)} \rightarrow CaSO_{4(s)} + H_2O_{(l)} + CO_{2(g)}\uparrow$$

Ca 1 = 1, S 1 = 1, C 1 = 1, H 2 = 2, O 7 = 7 ✔

2 pH and indigestion, stings, milk

- **(i)** An antacid (mild base) such as milk of magnesia neutralises the extra HCl.
- **(ii)** Rub baking soda (or a dock leaf for nettle) to neutralise the acid.
- **(B)** pH **falls**: bacteria turn lactose into lactic acid, so curd is more acidic.

3 Acidic and basic salt

- **Acidic:** ammonium chloride, NH_4Cl, from HCl (strong) and NH_4OH (weak)
- **Basic:** sodium carbonate, Na_2CO_3, from H_2CO_3 (weak) and NaOH (strong)

4 Case-based

- **(a)** Yes: its pH (4.8) is below 5.6.
- **(b)** Quicklime (CaO) and slaked lime ($Ca(OH)_2$), or chalk; neutralisation.
- **(c)** The pH rises towards 7 as $CaCO_3$ neutralises the acid.

5 Long answer

- **Family:** salts with the same positive or negative ion. Sodium family: NaCl, Na_2SO_4. Sulphate family: K_2SO_4, $CuSO_4$.
- NH_4Cl: strong acid + weak base, so acidic. CH_3COONa: weak acid + strong base, so basic.

$$NH_4OH_{(aq)} + HCl_{(aq)} \rightarrow NH_4Cl_{(aq)} + H_2O_{(l)}$$

$$CH_3COOH_{(aq)} + NaOH_{(aq)} \rightarrow CH_3COONa_{(aq)} + H_2O_{(l)}$$

- NaOH is strong and corrosive; it would damage the mouth, food pipe and stomach lining, so only mild bases are used.

Try-yourself answers

(i) neutral, (ii) acidic, (iii) basic, (iv) acidic, (v) neutral, (vi) basic.
Rule: the stronger parent decides.

Day 7 hit-list: Eduspark PYQ analysis

Topic	Rating
Tooth decay below pH 5.5	★★★
Antacids: $Mg(OH)_2$, $NaHCO_3$	★★★
pH of salts (parent acid and base)	★★★
Natural acids: tamarind, tomato, curd	★★★
Acid rain and aquatic life	★★☆
Soil treated with lime	★★☆
Bee, wasp and nettle stings	★★☆
Families of salts	★☆☆

Day 7 checklist: tick before you sleep

- ☐ I can give the key pH numbers: body 7.0–7.8, teeth 5.5, acid rain 5.6
- ☐ I can explain antacids and write the $Mg(OH)_2$ + HCl equation
- ☐ I can match natural sources with their acids
- ☐ I can name a salt's family and its parent acid and base
- ☐ I solved all 10 MCQs and checked them with the key

My score: ____ / 5

5/5 = Jalwa hai! 4 se kam? Page 1–2 phir se padho.



LEVEL 7 COMPLETE +100 XP earned. Next: Level 8



BUJHO *Samajh gaya! Salt ka swabhav strong parent se aata hai, aur daant 5.5 se neeche pH pe kharab hote hain!*

PAHELI *Shabaash! Kal Level 8: common salt se bane chemicals.*

pH FIXERS AROUND YOU

Bases: antacid, toothpaste, baking soda, quicklime, chalk, dock leaf.
Acids: vinegar (wasp sting), manure or compost (basic soil).

Coming up: Day 8

Chemicals from common salt: NaOH from the chlor-alkali process, bleaching powder, baking soda, washing soda and Plaster of Paris.

★ How we starred (Eduspark PYQ analysis): we counted how often each topic has come in Chapter 2 board and sample-paper questions. Year tags appear only where the paper year is known.

DAY 08 · LEVEL 8

CHEMISTRY | CHAPTER 2 | PART 4 · ASTRA SERIES

CHEMICALS FROM COMMON SALT

Goal: prepare, write and use five chemicals made from NaCl.



by Nitesh Prakash

01 Chlor-alkali | 02 Bleaching Powder | 03 Sodas | 04 Hydrates | 05 PoP

+125 XP today | Level 8 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

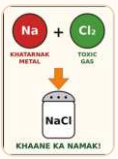
1 COMMON SALT: THE RAW MATERIAL

COMMON SALT (NaCl)

- A **neutral salt** of strong acid HCl and strong base NaOH
- **Sea water:** salt is obtained by **evaporation**; also from salt lakes: Sambhar (Rajasthan), Great Salt Lake (USA)
- **Rock salt** (*sendha namak*): beds of large crystals, often **brown** due to impurities; mined like coal at Khewra (Pakistan) and Mandi (Himachal Pradesh)

$NaOH_{(aq)} + HCl_{(aq)} \rightarrow NaCl_{(aq)} + H_2O_{(l)}$

neutralisation: Na 1 = 1, Cl 1 = 1, H 2 = 2, O 1 = 1 ✔



MEME BREAK

*Ek **khatarnak metal** (Na) + ek **toxic gas** (Cl_2) = khaane ka namak! Aur isi se 5 naye chemicals banenge.*

Family tree of common salt

From NaCl, by…	We get
Electrolysis of brine	NaOH, Cl_2, H_2
Cl_2 + dry slaked lime	Bleaching powder
Brine + NH_3 + CO_2	Baking soda, $NaHCO_3$
Heat $NaHCO_3$, then recrystallise	Washing soda

PoP is made from gypsum, not NaCl; NCERT teaches it in this section.

Real life: salt and freedom

- Used in food, to preserve pickles and meat, and in industry
- 1930: Gandhiji's **Dandi March** made salt from sea water

2 CHLOR-ALKALI PROCESS ★★★ MOST ASKED

DEFINITION

- **Electricity** is passed through **brine** (aqueous NaCl); it decomposes to give NaOH, Cl_2 and H_2
- Name: **chlor** for chlorine + **alkali** for sodium hydroxide

$2NaCl_{(aq)} + 2H_2O_{(l)} \xrightarrow{\text{Electricity}} 2NaOH_{(aq)} + Cl_{2(g)}\uparrow + H_{2(g)}\uparrow$

Na 2 = 2, Cl 2 = 2, H 4 = 4, O 2 = 2 ✔

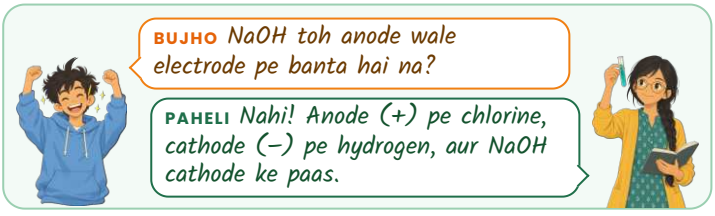
Which product, where?

Where?	What forms
Anode (+)	Cl_2 gas
Cathode (−)	H_2 gas
Near the cathode	NaOH solution

Brine = a concentrated solution of NaCl in water.

Uses of the products ★★☆

Product	Important uses
NaOH	de-greasing metals, soaps and detergents, paper making, artificial fibres
Cl_2	water treatment, swimming pools, PVC, disinfectants, CFCs, pesticides
H_2	fuels, margarine (hydrogenation of oils), ammonia for fertilisers
HCl	from H_2 + Cl_2: cleaning steel, ammonium chloride, medicines, cosmetics
Bleach	from NaOH + Cl_2: household bleaches, bleaching fabrics





★★★ *Chlor-alkali ki equation aur teeno products ke uses pakka aayenge, Nitesh Bhaiya ki guarantee!*

3 BLEACHING POWDER ★★★ MOST ASKED

PREPARATION

- **Chlorine gas** is passed over **dry slaked lime**, $Ca(OH)_2$
- Cl_2 itself comes from the chlor-alkali process
- NCERT writes bleaching powder as **$CaOCl_2$** (calcium oxychloride); its real composition is quite complex

$Ca(OH)_{2(s)} + Cl_{2(g)} \rightarrow CaOCl_{2(s)} + H_2O_{(l)}$

Ca 1 = 1, O 2 = 2, H 2 = 2, Cl 2 = 2 ✔

Uses of bleaching powder

1. **Bleaching** cotton and linen (textile industry), wood pulp (paper factories) and washed clothes (laundry)
2. **Oxidising agent** in many chemical industries
3. **Disinfecting** drinking water, making it free from germs



FUN FACT

Bleaching powder ki tez smell = chlorine! Bujho ko chheenk aa gayi, aur paani ke germs ka khel khatam.

4 BAKING SODA ★★★ MOST ASKED

PREPARATION

- Chemical name: **sodium hydrogencarbonate**; made from brine with **NH_3 and CO_2**
- A **mild, non-corrosive base**

$$NaCl_{(aq)} + H_2O_{(l)} + CO_{2(g)} + NH_{3(g)} \rightarrow NH_4Cl_{(aq)} + NaHCO_{3(s)}$$

Na 1 = 1, Cl 1 = 1, H 5 = 5, O 3 = 3, C 1 = 1, N 1 = 1 ✔

On heating (while cooking) ★★★

$$2NaHCO_{3(s)} \xrightarrow{\text{Heat}} Na_2CO_{3(s)} + H_2O_{(l)} + CO_{2(g)}\uparrow$$

Na 2 = 2, H 2 = 2, C 2 = 2, O 6 = 6 ✔

The CO_2 released makes food soft; the Na_2CO_3 left behind tastes bitter.

Baking powder: soda + edible acid ★★★

- Acid used: **tartaric acid**
- With water or heat: $NaHCO_3 + H^+ \rightarrow CO_2 + H_2O$ + sodium salt of the acid
- CO_2 bubbles make bread and cake **soft and spongy**
- The acid **neutralises** the bitter Na_2CO_3, so no bitter taste



MEME BREAK

*Baking soda + tartaric acid: **perfectly balanced**, as all things should be. Cake bhi phoola, swaad bhi kadwa nahi!*

Uses of baking soda

1. To make **baking powder** (cakes, bread, dhokla)
2. In **antacids**: being basic, it neutralises extra acid in the stomach
3. In **soda-acid fire extinguishers**



BUJHO *Cake mein baking soda daala, phir bhi kadwa kyun?*

PAHELI *Garam hokar soda, sodium carbonate chhodta hai, wahi kadwa hai. Baking powder lo: tartaric acid use neutralise kar deta hai!*



5 WASHING SODA ★★☆

Preparation in three steps

1. Make baking soda from brine (equation above).
2. Heat it to get **soda ash**, anhydrous Na_2CO_3.
3. Recrystallise soda ash from water:

$$Na_2CO_{3(s)} + 10H_2O_{(l)} \rightarrow Na_2CO_3{\cdot}10H_2O_{(s)}$$

sodium carbonate decahydrate: H 20 = 20, O 13 = 13 ✔

Uses of washing soda ★★★

1. **Glass, soap and paper** industries
2. Making sodium compounds such as **borax**
3. **Cleaning agent** for domestic purposes
4. Removing **permanent hardness** of water

Baking soda vs washing soda ★★★

	Baking soda	Washing soda
Chemical name	sodium hydrogencarbonate	sodium carbonate
Formula	$NaHCO_3$	$Na_2CO_3{\cdot}10H_2O$
Nature	mild base	stronger base; harsh on skin
Main use	baking powder, antacid	cleaning, hard water, glass

6 WATER OF CRYSTALLISATION ★★★ MOST ASKED

DEFINITION

- The **fixed number of water molecules** present in **one formula unit** of a salt
- Crystals look dry, but this water is part of the crystal

Salt	Formula	H_2O
Copper sulphate (blue vitriol)	$CuSO_4{\cdot}5H_2O$	5
Washing soda	$Na_2CO_3{\cdot}10H_2O$	10
Ferrous sulphate (green vitriol)	$FeSO_4{\cdot}7H_2O$	7
Gypsum	$CaSO_4{\cdot}2H_2O$	2

Activity: heating copper sulphate crystals ★★★

- Heat blue $CuSO_4$ crystals in a dry boiling tube: **water droplets** appear and the salt turns **white**

$$CuSO_4{\cdot}5H_2O_{(s)} \xrightarrow{\text{Heat}} CuSO_{4(s)} + 5H_2O_{(g)}$$

blue → white (anhydrous)

- Add 2–3 drops of water to the white powder: the **blue colour returns**

$$CuSO_{4(s)} + 5H_2O_{(l)} \rightarrow CuSO_4{\cdot}5H_2O_{(s)}$$

white → blue: water of crystallisation regained



MEME BREAK

*$CuSO_4$ garam hone ke baad: paani gaya, neela rang gaya, poora **look hi badal gaya**. Do boond paani daalo, wapas neela hero!*



BUJHO *Crystals bilkul sukhe the, toh tube mein paani kahan se aaya?*

PAHELI *Wahi water of crystallisation! Garam karte hi nikal jaata hai, isliye neela $CuSO_4$ safed ho jaata hai.*



7 PLASTER OF PARIS (PoP) ★★★ MOST ASKED

PREPARATION

- Heat **gypsum** at **373 K**: it loses water and becomes calcium sulphate **hemihydrate**
- Formula **$CaSO_4 \cdot \frac{1}{2}H_2O$**: two formula units of $CaSO_4$ share one water molecule
- **Heat with care:** above 373 K all water is lost, giving anhydrous $CaSO_4$, called **dead-burnt plaster**, which does not set with water

$$CaSO_4 \cdot 2H_2O(s) \xrightarrow{373\ K} CaSO_4 \cdot \tfrac{1}{2}H_2O(s) + 1\tfrac{1}{2}H_2O(g)$$

gypsum → Plaster of Paris

Setting of PoP ★★★

- Mixed with water, PoP changes back into **gypsum**, a hard solid mass

$$CaSO_4 \cdot \tfrac{1}{2}H_2O(s) + 1\tfrac{1}{2}H_2O(l) \rightarrow CaSO_4 \cdot 2H_2O(s)$$

Ca 1 = 1, S 1 = 1, O 6 = 6, H 4 = 4 ✓

So PoP is stored in a **moisture-proof** container: moisture would make it set.

Uses of Plaster of Paris

1. Plaster for supporting **fractured bones** in the right position
2. Making **toys** and materials for **decoration**
3. Making surfaces **smooth** (walls, ceilings)



MEME BREAK

*PoP + paani = gypsum. Statue ki tarah **set**! Isliye doctor tooti haddi pe yahi lagate hai: ab hilna mana hai.*

Formula table: learn by heart ★★★

Common name	Formula	Made from
Bleaching powder	$CaOCl_2$	Cl_2 + dry $Ca(OH)_2$
Baking soda	$NaHCO_3$	brine + NH_3 + CO_2
Washing soda	$Na_2CO_3 \cdot 10H_2O$	soda ash + $10H_2O$
Plaster of Paris	$CaSO_4 \cdot \frac{1}{2}H_2O$	gypsum at 373 K
Caustic soda	NaOH	electrolysis of brine



★★★ *PoP, bleaching powder aur copper sulphate ki equations har saal aati hai, Nitesh Bhaiya ki guarantee!*

× WRONG	✓ RIGHT
PoP = $CaSO_4 \cdot H_2O$	$CaSO_4 \cdot \frac{1}{2}H_2O$ (hemihydrate)
Washing soda = $NaHCO_3$	$Na_2CO_3 \cdot 10H_2O$
Cl_2 over lime water	Cl_2 over **dry** slaked lime
H_2 at the anode	H_2 at the cathode (−)

Exam tips

- ✔ Gypsum is heated at exactly **373 K**; write it over the arrow
- ✔ Hydrates: write the dot and the water count, e.g. $CuSO_4 \cdot 5H_2O$
- ✔ "Uses" questions: give the use with the reason (basic, oxidising, sets hard)
- ✔ Colours: $CuSO_4 \cdot 5H_2O$ blue, anhydrous $CuSO_4$ white

8 PRACTICE MCQS

Pehle khud solve karo, phir answer key se milao. Bujho ki tarah guess mat karna!



1. A mild, non-corrosive basic salt is: CBSE SQP 2023 ★★★
(a) $Ca(OH)_2$ (b) NaCl
(c) NaOH (d) $NaHCO_3$

2. A chemical compound used in glass, soap and paper industries is: CBSE 2024 ★★★
(a) washing soda (b) baking soda
(c) bleaching powder (d) common salt

3. Which compound has the maximum number of water molecules of crystallisation in one formula unit? CBSE 2024 ★★★
(a) $FeSO_4$ (b) $CuSO_4$
(c) $CaSO_4$ (d) Na_2CO_3

4. In the chlor-alkali process, the gas formed at the anode is: ★★★
(a) hydrogen (b) chlorine
(c) oxygen (d) carbon dioxide

5. Bleaching powder is made by passing chlorine over: ★★☆
(a) quicklime (b) dry slaked lime
(c) limestone (d) gypsum

6. The chemical formula of Plaster of Paris is: ★★★
(a) $CaSO_4 \cdot 2H_2O$ (b) $CaSO_4 \cdot \frac{1}{2}H_2O$
(c) $CaSO_4 \cdot H_2O$ (d) $CaSO_4$

7. On strong heating, blue copper sulphate crystals turn: ★★★
(a) green (b) black
(c) white (d) red

8. Baking powder is a mixture of baking soda and: ★★☆
(a) tartaric acid (b) hydrochloric acid
(c) washing soda (d) common salt

9. **Assertion (A):** Fresh milk to which baking soda is added takes a longer time to set as curd.
Reason (R): Baking soda decreases the pH value of fresh milk to below 6. CBSE Term-1 SQP 2021 ★★☆

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

10. **Assertion (A):** Plaster of Paris is stored in a moisture-proof container.
Reason (R): Plaster of Paris reacts with water and sets into a hard mass of gypsum. ★★☆

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

9 SHORT ANSWER QUESTIONS

1. A compound X prepared from gypsum hardens when mixed with the proper quantity of water. Identify X and write its formula. State the difference in chemical composition between baking soda and baking powder. CBSE 2023 (3 marks)

2. Chlorine gas is prepared by the electrolysis of brine. Write the chemical equation. Name the other products and give one use of each. CBSE SQP 2023 (3 marks)

3. Write balanced equations for: (a) heating blue copper sulphate crystals; (b) heating sodium hydrogencarbonate during cooking. CBSE 2023 (2 marks)

4. Write the equation for the preparation of baking soda. Why is it also an ingredient of antacids? CBSE Compartment 2025 (2 marks)

Answer key: MCQs

Q	Ans	Why
1	(d)	$NaHCO_3$ is a mild, non-corrosive base.
2	(a)	Washing soda: glass, soap, paper.
3	(d)	$Na_2CO_3 \cdot 10H_2O$ has 10 water molecules.
4	(b)	Cl_2 at the anode; H_2 at the cathode.
5	(b)	Cl_2 + dry $Ca(OH)_2 \rightarrow CaOCl_2 + H_2O$
6	(b)	Hemihydrate: $CaSO_4 \cdot \frac{1}{2}H_2O$.
7	(c)	It loses water: anhydrous $CuSO_4$ is white.
8	(a)	A mild edible acid: tartaric acid.
9	(c)	Baking soda is basic; it raises the pH.
10	(a)	PoP sets with moisture, so keep it dry.

Coming up: Day 9

The Chapter 2 **Boss Level**: one full test on indicators, pH, salts and today's chemicals. Revise the formula table tonight!



NITESH SIR SAYS
*Formula table roz ek baar likho. **Kal Boss Level, taiyaar raho!***

Model answers

1 X = Plaster of Paris
- Formula $CaSO_4 \cdot \frac{1}{2}H_2O$; made by heating gypsum at 373 K
- **Baking soda:** only $NaHCO_3$. **Baking powder:** $NaHCO_3$ + a mild edible acid such as tartaric acid

2 Electrolysis of brine

$$2NaCl_{(aq)} + 2H_2O_{(l)} \xrightarrow{\text{Electricity}} 2NaOH_{(aq)} + Cl_{2(g)} + H_{2(g)}$$

- **NaOH:** soaps and detergents
- **H_2:** fuel; ammonia for fertilisers
- **Cl_2:** water treatment; bleaching powder

3 Two heating reactions

$$CuSO_4 \cdot 5H_2O_{(s)} \xrightarrow{\text{Heat}} CuSO_{4(s)} + 5H_2O_{(g)}$$

(a) blue crystals → white powder

$$2NaHCO_{3(s)} \xrightarrow{\text{Heat}} Na_2CO_{3(s)} + H_2O_{(l)} + CO_{2(g)} \uparrow$$

(b) CO_2 makes the food soft

4 Baking soda and antacids

$$NaCl_{(aq)} + H_2O_{(l)} + CO_{2(g)} + NH_{3(g)} \rightarrow NH_4Cl_{(aq)} + NaHCO_{3(s)}$$

- It is a mild base, so it **neutralises excess acid** in the stomach and gives relief

Day 8 checklist: tick before you sleep

- ☐ I can write the chlor-alkali equation and name the product at each electrode
- ☐ I can prepare bleaching powder, baking soda and washing soda with equations
- ☐ I can tell baking soda, baking powder and washing soda apart
- ☐ I can explain water of crystallisation and the PoP equations
- ☐ I solved all 10 MCQs and checked them with the key

My score: ____ / 5

5/5 = Jalwa hai! 4 se kam? Page 2–3 phir se padho.



LEVEL 8 COMPLETE +125 XP earned. Next: Level 9



★ How we starred (Eduspark PYQ analysis): we counted how often each topic has come in Chapter 2 board and sample-paper questions. Year tags appear only where the paper year is known.

DAY 09 LEVEL 9

CHEMISTRY | CHAPTER 2 | BOSS LEVEL

ACIDS, BASES & SALTS

Goal: revise Chapter 2, then clear the Boss test.



by Nitesh Prakash

01 Revision Map | 02 14 MCQs | 03 Case Study | 04 Written Answers | 05 Answer Key | +200 XP Boss Level | Level 9 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 CHAPTER 2 IN ONE MAP

Quick revision map: the whole chapter

INDICATORS
Natural: litmus, turmeric, red cabbage; synthetic: phenolphthalein, methyl orange; olfactory: onion, vanilla, clove

REACTIONS ★★★
Acid + metal → salt + H_2; acid + carbonate → salt + H_2O + CO_2; acid + base → salt + water

ACIDS, BASES & SALTS

IONS IN WATER
In water, acids give hydrogen ions, H^+ (as H_3O^+); bases give OH^-; soluble bases are alkalis

pH SCALE ★★★
0–14; below 7 acidic, 7 neutral, above 7 basic; lower pH means more H^+ ions

STRENGTH AND DILUTION
Strong acids (HCl, H_2SO_4, HNO_3) ionise fully; weak (CH_3COOH, H_2CO_3) partly. Dilution is exothermic: add acid to water slowly, never water to acid.

2 INDICATOR COLOURS

Indicator	In acid	In base
Litmus	Blue → red	Red → blue
Phenolphthalein	Colourless	Pink
Methyl orange	Red	Yellow
Turmeric	Stays yellow	Reddish-brown
Red cabbage	Red/pink	Green
Onion, vanilla	Smell stays	Smell is lost

3 REACTIONS OF ACIDS AND BASES

$Zn_{(s)} + 2HCl_{(aq)} \rightarrow ZnCl_{2(aq)} + H_{2(g)}\uparrow$

$2NaOH_{(aq)} + Zn_{(s)} \rightarrow Na_2ZnO_{2(aq)} + H_{2(g)}\uparrow$
Base + metal (sodium zincate)

$Na_2CO_{3(s)} + 2HCl_{(aq)} \rightarrow 2NaCl_{(aq)} + H_2O_{(l)} + CO_{2(g)}\uparrow$

$Ca(OH)_{2(aq)} + CO_{2(g)} \rightarrow CaCO_{3(s)}\downarrow + H_2O_{(l)}$
Lime water turns milky

$CaCO_{3(s)} + H_2O_{(l)} + CO_{2(g)} \rightarrow Ca(HCO_3)_{2(aq)}$
Excess CO_2: milkiness disappears

$NaOH_{(aq)} + HCl_{(aq)} \rightarrow NaCl_{(aq)} + H_2O_{(l)}$

$CuO_{(s)} + 2HCl_{(aq)} \rightarrow CuCl_{2(aq)} + H_2O_{(l)}$
Metal oxide is basic: blue-green solution

4 CHEMICALS FROM COMMON SALT

$2NaCl_{(aq)} + 2H_2O_{(l)} \xrightarrow{\text{Electricity}} 2NaOH_{(aq)} + Cl_{2(g)}\uparrow + H_{2(g)}\uparrow$
Chlor-alkali: Cl_2 at anode, H_2 at cathode

$Ca(OH)_{2(s)} + Cl_{2(g)} \rightarrow CaOCl_{2(s)} + H_2O_{(l)}$
Bleaching powder (dry slaked lime)

$NaCl_{(aq)} + H_2O_{(l)} + CO_{2(g)} + NH_{3(g)} \rightarrow NH_4Cl_{(aq)} + NaHCO_{3(s)}$
Baking soda

$2NaHCO_{3(s)} \xrightarrow{\text{Heat}} Na_2CO_{3(s)} + H_2O_{(l)} + CO_{2(g)}\uparrow$
Heating baking soda

$Na_2CO_{3(s)} + 10H_2O_{(l)} \rightarrow Na_2CO_3{\cdot}10H_2O_{(s)}$
Washing soda (recrystallisation)

$CuSO_4{\cdot}5H_2O_{(s)} \xrightarrow{\text{Heat}} CuSO_{4(s)} + 5H_2O_{(g)}$
Blue crystals turn white

$CaSO_4{\cdot}2H_2O_{(s)} \xrightarrow{373\ K} CaSO_4{\cdot}\frac{1}{2}H_2O_{(s)} + 1\frac{1}{2}H_2O_{(g)}$
Gypsum → Plaster of Paris

$CaSO_4{\cdot}\frac{1}{2}H_2O_{(s)} + 1\frac{1}{2}H_2O_{(l)} \rightarrow CaSO_4{\cdot}2H_2O_{(s)}$
PoP sets into hard gypsum

5 pH SCALE AND SALTS



Lemon juice 2.2, water 7, blood 7.4, milk of magnesia 10, NaOH 14

- Human body works in pH **7.0–7.8**; rain below **5.6** is acid rain
- Tooth enamel corrodes below pH **5.5**; brush after meals
- Ant sting, nettle = **methanoic acid**; bee sting: rub baking soda
- Excess stomach acid: antacid like milk of magnesia, $Mg(OH)_2$

Made from	Salt pH	Examples
Strong acid + strong base	7 (neutral)	NaCl, Na_2SO_4, KNO_3
Strong acid + weak base	below 7 (acidic)	NH_4Cl, $(NH_4)_2SO_4$
Weak acid + strong base	above 7 (basic)	Na_2CO_3, CH_3COONa



★★★ *Chlor-alkali, bleaching powder, baking soda aur PoP ke equations baar baar poochhe jaate hain: Nitesh Bhaiya ki guarantee!*



BUJHO *Boss Level kal hai aur poora chapter abhi bacha hai!*

PAHELI *Tension nahi! Pehle yeh map padho, phir test khud solve karo. Practice ki galti exam mein nahi hogi!*



6 BOSS LEVEL TEST: CHAPTER 2

INSTRUCTIONS

Time: 60 minutes. **Maximum marks:** 40. Section A: Q1–14, MCQs, 1 mark each. Section B: Q15, case-based, 4 marks. Section C: Q16–20, short answers, 2 or 3 marks. Section D: Q21–22, long answers, 5 marks each.

Boss Level mode ON! Phone side mein rakho, timer lagao aur key tab tak mat dekho.



SECTION A: MULTIPLE CHOICE — 14 × 1 = 14

1. Which of the following is a mild, non-corrosive basic salt? CBSE SQP 2023 ★★★
(a) $Ca(OH)_2$ (b) NaCl
(c) NaOH (d) $NaHCO_3$

2. Methyl orange is added to dilute hydrochloric acid and to aqueous sodium hydroxide. Its colours in the two solutions are: CBSE SQP 2025 ★★★
(a) orange; red (b) red; yellow
(c) red; orange (d) yellow; red

3. A compound used in the glass, soap and paper industries is: CBSE 2024 ★★☆
(a) washing soda (b) baking soda
(c) bleaching powder (d) common salt

4. Solution A turns phenolphthalein pink. When solution B is added to A, the pink colour disappears. Which statement is true? CBSE SQP 2024 ★★★
(a) A is strongly basic and B is a weak base.
(b) A is strongly acidic and B is a weak acid.
(c) A has pH greater than 7 and B has pH less than 7.
(d) A has pH less than 7 and B has pH greater than 7.

5. When excess carbon dioxide is passed through lime water, the milkiness disappears because: CBSE SQP 2025 ★★☆
(a) soluble $CaCO_3$ changes to soluble $Ca(HCO_3)_2$
(b) insoluble $CaCO_3$ changes to soluble $Ca(HCO_3)_2$
(c) soluble $CaCO_3$ changes to insoluble $Ca(HCO_3)_2$
(d) insoluble $CaCO_3$ changes to insoluble $Ca(HCO_3)_2$

6. Identify the compound A in the reaction $NaCl + NH_3 + H_2O + CO_2 \rightarrow A + NH_4Cl$ CBSE Compartment 2024 ★★★
(a) Na_2CO_3 (b) $(NH_4)_2CO_3$
(c) $NaHCO_3$ (d) NH_4HCO_3

7. Aqueous solutions: A = potassium nitrate, B = ammonium chloride, C = sodium carbonate. Their pH in ascending order is: CBSE 2025 ★★☆
(a) A < B < C (b) B < C < A
(c) C < A < B (d) B < A < C

8. Tamarind juice turns blue litmus red. The acid present in it is: CBSE 2025 ★★☆
(a) acetic acid (b) methanoic acid
(c) oxalic acid (d) tartaric acid

9. A curry stain on a white shirt turns reddish-brown when soap is scrubbed on it and yellow again after washing with plenty of water. The reasons are: (I) soap is acidic; (II) soap is basic; (III) turmeric turns reddish in bases; (IV) turmeric turns reddish in acids. CBSE Term-1 SQP 2021 ★★☆
(a) (I) and (II) (b) (II) and (III)
(c) (I) and (IV) (d) (II) and (IV)

10. In a conductivity test, the bulb does not glow at all. The liquid in the beaker is most likely: ★★☆
(a) dilute HNO_3 (b) C_2H_5OH (alcohol)
(c) dilute H_2SO_4 (d) dilute HCl

11. The parent acid and parent base of calcium carbonate are: CBSE Term-1 SQP 2021 ★★☆
(a) HCl and NaOH (b) H_2CO_3 and $Ca(OH)_2$
(c) H_2SO_4 and $Ca(OH)_2$ (d) HNO_3 and $Ca(OH)_2$

12. To dilute a concentrated acid safely, you should add: ★★★
(a) water to the acid quickly
(b) the acid to water slowly, with stirring
(c) water to the acid slowly
(d) both together at once

13. **Assertion (A):** Fresh milk to which baking soda is added takes a longer time to set as curd.
Reason (R): Baking soda decreases the pH of fresh milk to below 6. CBSE Term-1 SQP 2021 ★★★
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

14. **Assertion (A):** Metal oxides are acidic in nature.
Reason (R): Calcium hydroxide reacts with carbon dioxide to form a salt and water. ★★☆
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

SECTION B: CASE-BASED — 1 × 4 = 4

READ THE PASSAGE

Electricity passed through **brine** (concentrated aqueous NaCl) gives sodium hydroxide, so the process is called the **chlor-alkali process**. Chlorine forms at the anode, hydrogen at the cathode, and NaOH solution near the cathode. Sodium hydroxide formed here is a strong alkali. All three products are useful; hydrogen, for example, is a fuel and is used to make ammonia for fertilisers.

15. Answer these, based on the passage: ★★★
(a) Write the balanced equation for this process. [1]
(b) Name the gas formed at the anode; give one use. [1]
(c) Chlorine reacts with dry slaked lime to form a compound Z. Name Z, write the equation for its formation and give one use of Z. [2]

OR

(c) Hydrogen and chlorine from this process combine to form a compound. Name it and give one use. Also give two uses of sodium hydroxide. [2]

SECTION C: SHORT ANSWERS — 12 marks

16. Give reasons: CBSE 2020 ★★★ [2]
(a) Sodium hydrogencarbonate is used as an antacid.
(b) Blue copper sulphate crystals turn white on strong heating.

17. Answer both parts: CBSE 2023 ★★★ [2]
(a) Compound X, made from gypsum, hardens when mixed with the right amount of water. Identify X and write down its chemical formula.
(b) How does baking powder differ from baking soda in chemical composition?

18. The pH of rain water collected from cities A and B is 6.1 and 5.3. Which city has more acidic rain water? Justify. State, with reason, what would happen to aquatic life in a pond into which the rain water of city B flows. CBSE Compartment 2024 ★★☆ [3]

19. Answer both parts: CBSE 2022 ★★★ [3]
(a) What would you use to counteract the change in pH in: (i) too much acid in the stomach during indigestion; (ii) a honey-bee or nettle sting?
(b) Fresh milk has a pH of 6. When it turns into curd, will its pH increase or decrease? Why?

20. Three solutions A, B and C have pH values 6, 8 and 9.5. CBSE SQP 2024 ★★☆ [2]
(a) In which solution are the most hydronium ions present?
(b) Arrange A, B and C in increasing order of H^+ ions.



Short answer mein equation aur reason dono likho, tabhi poore marks milenge!

Time plan: 60 minutes

1. Section A (14 MCQs): **15 min**
2. Section B (case study): **8 min**
3. Section C (short answers): **17 min**
4. Section D (long answers): **15 min**
5. Final check of equations and states: **5 min**

Common mistakes to avoid

× WRONG	✓ RIGHT
Higher pH = more acidic	Lower pH = more acidic
Water poured into acid	Acid added to water, slowly
PoP = $CaSO_4 \cdot 2H_2O$	PoP = $CaSO_4 \cdot \frac{1}{2}H_2O$
Cl_2 + lime water	Cl_2 + dry slaked lime
Every salt is neutral	NH_4Cl acidic, Na_2CO_3 basic
Baking soda = Na_2CO_3	Baking soda = $NaHCO_3$

SECTION D: LONG ANSWERS — 2 × 5 = 10

21. Acids and their salts: ★★★
(a) Write balanced chemical equations for the reaction of an acid with (i) a metal, (ii) a base and (iii) a metal carbonate. Name the main product in each case. CBSE 2025 [3]
(b) Name the acid and the base from which (i) sodium chloride and (ii) ammonium sulphate are obtained. State whether each salt is acidic, basic or neutral, with reason. CBSE Compartment 2024 [2]

22. Salts with water of crystallisation: ★★★
(a) What is water of crystallisation? Write the formulae of crystalline copper sulphate and washing soda. CBSE Compartment 2025 [1]
(b) How is Plaster of Paris obtained from gypsum? Write the equation. Why is it stored in a moisture-proof container? [2]
(c) Write the two equations for making washing soda from baking soda. Give one use of washing soda. [2]

7 ANSWER KEY AND MODEL ANSWERS

Section A: MCQ answers

Q	Ans	Why
1	(d)	$NaHCO_3$ is a mild base, so it is non-corrosive.
2	(b)	Red in acid, yellow in base.
3	(a)	Washing soda, $Na_2CO_3 \cdot 10H_2O$.
4	(c)	Pink means basic; B neutralises A, so B is acidic.
5	(b)	$Ca(HCO_3)_2$ is soluble in water.
6	(c)	$NaHCO_3$ forms; NH_4Cl is the by-product.
7	(d)	NH_4Cl acidic, KNO_3 neutral, Na_2CO_3 basic.
8	(d)	Tamarind contains tartaric acid.
9	(b)	Soap is basic; turmeric turns reddish-brown in bases.
10	(b)	Alcohol gives no ions in water, so no current flows.
11	(b)	$H_2CO_3 + Ca(OH)_2 \rightarrow CaCO_3 + 2H_2O$
12	(b)	Dilution is highly exothermic; acid may splash.
13	(c)	A is true; baking soda raises the pH of milk. R is false.
14	(d)	Metal oxides are basic, so A is false; R is true.

15 Case: chlor-alkali process

$2NaCl_{(aq)} + 2H_2O_{(l)} \xrightarrow{\text{Electricity}} 2NaOH_{(aq)} + Cl_{2(g)}\uparrow + H_{2(g)}\uparrow$
(a) Chlor-alkali process

- (b) **Chlorine** at the anode; used in water treatment, PVC, disinfectants
- (c) Z = **bleaching powder**, $CaOCl_2$; used to bleach cotton and linen and to disinfect drinking water

$Ca(OH)_{2(s)} + Cl_{2(g)} \rightarrow CaOCl_{2(s)} + H_2O_{(l)}$

- OR (c): $H_2 + Cl_2$ give **HCl** (cleaning steel, medicines). NaOH: soaps and detergents, paper making, degreasing metals

BOARD MARKING TIPS

- Balanced equation with state symbols = full marks
- Write the product's name along with its formula

16 Antacid; copper sulphate turns white

- (a) $NaHCO_3$ is a mild, non-corrosive base; it neutralises extra HCl in the stomach
- (b) Heating removes the water of crystallisation; anhydrous $CuSO_4$ is white

$$CuSO_4{\cdot}5H_2O_{(s)} \xrightarrow{\text{Heat}} CuSO_{4(s)} + 5H_2O_{(g)}$$

17 Plaster of Paris; baking powder

- (a) X = **Plaster of Paris**, $CaSO_4{\cdot}\frac{1}{2}H_2O$
- (b) Baking soda is only $NaHCO_3$; baking powder is $NaHCO_3$ + a mild edible acid such as tartaric acid

18 Acid rain and aquatic life

- City **B** (pH 5.3): lower pH means more H^+ ions, so more acidic
- Rain below pH 5.6 is acid rain; it lowers the pond's pH
- Aquatic animals survive only in a narrow pH range, so many may die

19 Counteracting pH changes

- (a)(i) An antacid such as milk of magnesia, $Mg(OH)_2$, or baking soda
- (a)(ii) Bee sting: baking soda paste; nettle (methanoic acid): rub a dock leaf
- (b) pH **decreases**: lactic acid forms as milk sets into curd

20 Hydronium ions and pH

- (a) **A** (pH 6): lowest pH, most H_3O^+ ions
- (b) C (9.5) < B (8) < A (6)

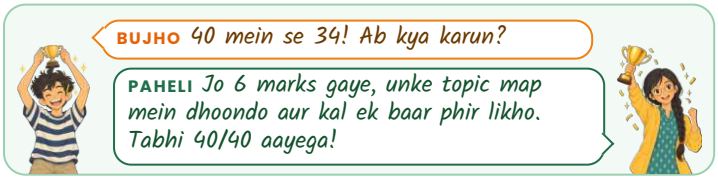
21 Reactions of acids; parent acid and base

$$Zn_{(s)} + 2HCl_{(aq)} \rightarrow ZnCl_{2(aq)} + H_{2(g)}\uparrow$$
Main product: zinc chloride (salt) and hydrogen gas

$$NaOH_{(aq)} + HCl_{(aq)} \rightarrow NaCl_{(aq)} + H_2O_{(l)}$$
Main product: sodium chloride (salt) and water

$$Na_2CO_{3(s)} + 2HCl_{(aq)} \rightarrow 2NaCl_{(aq)} + H_2O_{(l)} + CO_{2(g)}\uparrow$$
Main product: sodium chloride, water and carbon dioxide

- (b)(i) NaCl: HCl + NaOH; **neutral** (strong acid + strong base)
- (b)(ii) $(NH_4)_2SO_4$: H_2SO_4 + NH_4OH; **acidic** (strong acid + weak base)





MEME BREAK
Score dekh ke khush ho gaye, par galat answers dobara nahi likhe? Boss abhi zinda hai!

NITESH SIR SAYS
Galat answers ek "mistake notebook" mein likho aur exam se pehle wahi padho. ***Chapter 2 cleared, jalwa hai!***



★ How we starred (Eduspark PYQ analysis): we counted how often each topic has come in Chapter 2 board and sample-paper questions. Year tags appear only where the paper year is known.

22 Hydrates, Plaster of Paris, washing soda

- (a) A fixed number of water molecules in one formula unit of a salt crystal: $CuSO_4{\cdot}5H_2O$, $Na_2CO_3{\cdot}10H_2O$
- (b) Heat gypsum at 373 K:

$$CaSO_4{\cdot}2H_2O_{(s)} \xrightarrow{373\ K} CaSO_4{\cdot}\tfrac{1}{2}H_2O_{(s)} + 1\tfrac{1}{2}H_2O_{(g)}$$

- Moisture turns it back into hard gypsum, so keep it dry:

$$CaSO_4{\cdot}\tfrac{1}{2}H_2O_{(s)} + 1\tfrac{1}{2}H_2O_{(l)} \rightarrow CaSO_4{\cdot}2H_2O_{(s)}$$

- (c) Heat baking soda, then recrystallise sodium carbonate:

$$2NaHCO_{3(s)} \xrightarrow{\text{Heat}} Na_2CO_{3(s)} + H_2O_{(l)} + CO_{2(g)}\uparrow$$

$$Na_2CO_{3(s)} + 10H_2O_{(l)} \rightarrow Na_2CO_3{\cdot}10H_2O_{(s)}$$

- Use: glass, soap and paper industries; removing permanent hardness of water

Day 9 checklist: tick before you sleep

- [ ] I revised the one-page map and every key equation
- [ ] I attempted all 14 MCQs without looking at the key
- [ ] I solved the case study and all five short answers
- [ ] I wrote both long answers with balanced equations
- [ ] I checked my answers with the key and marked my score

My score: ____ / 5 | Test score: ____ / 40

5/5 = Jalwa hai! 4 se kam? Page 1 ka map phir se padho.



Revise these five first

1. Chlor-alkali equation, electrode products and their uses
2. Making of bleaching powder, baking soda, washing soda and PoP
3. pH of salts: parent acid and base decide the nature
4. Indicator colours: litmus, phenolphthalein, methyl orange
5. Acid + metal, carbonate, base and metal oxide equations

LEVEL 9 COMPLETE

+200 XP earned. Chapter 2 cleared. Next: Level 10



BUJHO *Boss Level clear! Chapter 2 poora samajh gaya!*

PAHELI *Shabaash! Kal milte hain Level 10 pe.*

CHEMISTRY | CHAPTER 2 COMPLETE

CHAPTER 2 SWAHAA!

Acids, Bases and Salts | Days 5–9



BUJHO *Litmus se lekar Plaster of Paris tak, pH scale bhi ab dost hai. Chapter 2 Swahaa!*

PAHELI *High five, Bujho! Ab chalte hain Metals aur Non-metals ki duniya mein.*

CHAPTER XP EARNED
+625 XP

LEVELS CLEARED
5 levels (Days 5–9)

BOOK PROGRESS
Level 9 / 60 1125 XP total

WHAT YOU HAVE MASTERED

- ✔ Indicators and reactions of acids and bases
- ✔ pH scale and universal indicator
- ✔ Family of salts and pH of salts
- ✔ H^+ and OH^- ions, strength, dilution
- ✔ pH in everyday life
- ✔ Chemicals from common salt, Plaster of Paris

NEXT CHAPTER Chapter 3: Metals and Non-metals (Day 10) *Aadhi jung jeet li!*

DAY 10 · LEVEL 10

CHEMISTRY | CHAPTER 3 | PART 1 · ASTRA SERIES

METALS & THEIR REACTIONS

Goal: properties, reactions and the reactivity series.



by Nitesh Prakash

01 Properties | 02 Exceptions | 03 Oxygen & Water | 04 Acids & Salts | 05 Reactivity Series | +100 XP today | Level 10 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 METALS VS NON-METALS: PHYSICAL PROPERTIES

METALS	NON-METALS
✔ **Lustrous** (shiny surface)	× Dull, no shine
Malleable and **ductile**	**Brittle** when solid
Good conductors of heat and electricity	Poor conductors (insulators)
Sonorous: ring when struck	Not sonorous
Hard; high melting and boiling points	Soft solids or gases; low melting points
Solids at room temperature	Solid, liquid (bromine) or gas

METALLOIDS: properties of both metals and non-metals
B, Si, Ge, As, Sb, Te | Si and Ge are semiconductors

DEFINITION

- **Malleability:** can be beaten into **thin sheets**. Gold and silver are the most malleable.
- **Ductility:** can be drawn into **thin wires**. Gold is the most ductile metal (not platinum): 1 g gives about 2 km of wire.
- **Sonorous:** rings when struck, so school bells are metal.

2 THE FAMOUS EXCEPTIONS ★★★ MOST ASKED

Exceptions board examiners love

- ✔ **Mercury** is the only liquid metal at room temperature.
- ✔ **Gallium** and **caesium** melt in your palm (low melting points).
- ✔ **Sodium** and **potassium** are so soft they cut with a knife; low density too.
- ✔ **Lead** and **mercury** are poor conductors of heat.
- ✔ **Silver** is the best conductor, then **copper**.
- ✔ **Iodine** is a shiny non-metal, and **bromine** is a liquid non-metal at room temperature.
- ✔ **Graphite** (carbon) conducts electricity; **diamond** is the hardest natural substance with a very high melting point.



★★★ *Exceptions se 1 mark pakka aata hai: Nitesh Bhaiya ki guarantee, list ratt lo!*



MEME BREAK

*Iron Man ka suit metal ka kyun? Strong, malleable aur tough. Asli hero **Fe** hai, bhai!*



BUJHO *Non-metal current nahi chalate, toh pencil ki nib bhi safe hai na?*

PAHELI *Nahi Bujho! Nib graphite ki hai: non-metal, phir bhi current chalata hai. Yahi exceptions exam mein sabse zyada aate hain!*



3 METALS + OXYGEN

METAL + OXYGEN (burn or heat in air) → combination → **METAL OXIDE** (mostly basic in nature)

$$2Cu_{(s)} + O_{2(g)} \xrightarrow{\Delta} 2CuO_{(s)}$$

copper (reddish-brown) → copper(II) oxide (black)

$$4Al_{(s)} + 3O_{2(g)} \rightarrow 2Al_2O_{3(s)}$$

Al 4 = 4, O 6 = 6 ✔

$$4Na_{(s)} + O_{2(g)} \rightarrow 2Na_2O_{(s)}$$

Na 4 = 4, O 2 = 2 ✔

Metal	With oxygen
K, Na	React so fast they catch fire in open air, so they are **kept in kerosene**
Mg, Al, Zn, Pb	A thin **oxide layer** forms and stops further oxidation
Mg	Burns with a dazzling white flame
Fe	Does not burn as a block; iron filings burn brightly in a flame
Cu	Does not burn; a black layer of CuO forms on heating
Ag, Au	No reaction even at high temperature

Did you know?

Phosphorus, a non-metal, catches fire in air, so it is stored **under water**. Sodium and potassium are stored under **kerosene** instead, because they react with water.

Amphoteric oxides: both acid and base nature ★★★

- Oxides that react with **both acids and bases** to give salt and water: **Al_2O_3**, **ZnO**

$$Al_2O_3(s) + 6HCl(aq) \rightarrow 2AlCl_3(aq) + 3H_2O(l)$$
acts as a base: Al 2 = 2, O 3 = 3, H 6 = 6, Cl 6 = 6 ✔

$$Al_2O_3(s) + 2NaOH(aq) \rightarrow 2NaAlO_2(aq) + H_2O(l)$$
acts as an acid; $NaAlO_2$ = sodium aluminate

Most metal oxides do not dissolve in water. Those that do give **alkalis**: $Na_2O(s) + H_2O(l) \rightarrow 2NaOH(aq)$

★★★ **PYQ FAVOURITE** The metal used in the thermit process burns in air to give an **amphoteric oxide**: the metal is **Al** and the oxide is **Al_2O_3** (CBSE 2023).

4 METALS + WATER ★★★

- Metal + water → **metal oxide + H_2**; a soluble oxide then gives **metal hydroxide**

Metal	Reacts with	What you see
K, Na	cold water	Violent; so much heat that the H_2 catches fire
Ca	cold water	Less violent; Ca **floats** as H_2 bubbles stick to it
Mg	hot water	Mg also **floats** on H_2 bubbles
Al, Zn, Fe	steam only	Metal oxide + H_2
Pb, Cu, Ag, Au	none	No reaction with water at all

$$2Na(s) + 2H_2O(l) \rightarrow 2NaOH(aq) + H_2(g)\uparrow + \textbf{Heat}$$
Na 2 = 2, H 4 = 4, O 2 = 2 ✔

$$Ca(s) + 2H_2O(l) \rightarrow Ca(OH)_2(aq) + H_2(g)\uparrow$$

$$Mg(s) + 2H_2O(l) \xrightarrow{\text{hot water}} Mg(OH)_2(aq) + H_2(g)\uparrow$$

$$2Al(s) + 3H_2O(g) \rightarrow Al_2O_3(s) + 3H_2(g)\uparrow$$
Al 2 = 2, O 3 = 3, H 6 = 6 ✔

$$3Fe(s) + 4H_2O(g) \rightarrow Fe_3O_4(s) + 4H_2(g)\uparrow$$
Fe 3 = 3, O 4 = 4, H 8 = 8 ✔ (Fe_3O_4, not Fe_2O_3!)



BUJHO *Calcium toh metal hai, phir paani pe tairta kyun hai?*

PAHELI *H_2 ke bubbles uski surface pe chipak jaate hain, wahi upar uthaate hain. Garam paani mein Mg bhi aise hi tairta hai!*



5 METALS + DILUTE ACIDS

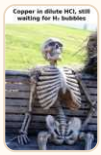
- Metal + dilute acid → **salt + hydrogen gas** (only metals **above H** in the series)

$$Mg(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + H_2(g)\uparrow$$

$$Fe(s) + 2HCl(aq) \rightarrow FeCl_2(aq) + H_2(g)\uparrow$$

$$2Al(s) + 6HCl(aq) \rightarrow 2AlCl_3(aq) + 3H_2(g)\uparrow$$
Al 2 = 2, H 6 = 6, Cl 6 = 6 ✔

Rate of bubbling: **Mg > Al > Zn > Fe**. Copper shows **no reaction**.



MEME BREAK
Copper, dilute HCl ko dekh ke: "Main H se neeche hoon, mera kuch nahi bigad sakte!"

Nitric acid exception and aqua regia ★★★

- **HNO_3** is a strong **oxidising agent**: it oxidises the H_2 formed to water and is itself reduced to N_2O, NO or NO_2. So **H_2 is not evolved**.
- Exception: **Mg** and **Mn** give H_2 with **very dilute** HNO_3.
- **Aqua regia** ("royal water"): freshly mixed conc. HCl and conc. HNO_3 in a **3 : 1** ratio. It dissolves **gold** and **platinum**.

EXTRA FROM CLASS NOCl + nascent Cl attack Au and Pt

$$HNO_3 + 3HCl \rightarrow NOCl + Cl_2 + 2H_2O$$
H 4 = 4, N 1 = 1, O 3 = 3, Cl 3 = 3 ✔ (NOCl = nitrosyl chloride)



★★★ *HNO_3 exception aur aqua regia ka 3 : 1: Nitesh Bhaiya ki guarantee, board mein aata hai!*

6 METALS + SALT SOLUTIONS ★★★

DISPLACEMENT

- A **more reactive** metal pushes a **less reactive** metal out of its salt solution
- Metal A + salt of B → salt of A + B (only if A is above B)

$$Fe(s) + CuSO_4(aq) \rightarrow FeSO_4(aq) + Cu(s)$$
blue solution turns light green; reddish-brown Cu coats the iron nail

$$Zn(s) + CuSO_4(aq) \rightarrow ZnSO_4(aq) + Cu(s)$$
blue colour fades; reddish-brown Cu deposits on zinc

$$Cu(s) + 2AgNO_3(aq) \rightarrow Cu(NO_3)_2(aq) + 2Ag(s)$$
solution turns blue; shiny silver deposits on copper

No reaction: $Cu(s) + FeSO_4(aq) \rightarrow$ no reaction since Cu is below Fe.



MEME BREAK
Zyada reactive metal aaya, copper ko solution se bahar! Mg aur Fe dono yahi karte hain.

7 REACTIVITY SERIES ★★★ MOST ASKED

Metals arranged in **decreasing order of reactivity**. Hydrogen is the reference point.

K *Kal*	**Na** *Naani*	**Ca** *Cake*	**Mg** *Mangwayi*	**Al** *Aaj*	**Zn** *Zara*	**Fe** *Feeka*
Pb *Pada*	**H** *Hum*	**Cu** *Chupke*	**Hg** *Hi*	**Ag** *Aadha*	**Au** *Aur*	

MOST REACTIVE — H: reference — LEAST REACTIVE

Trick yaad karo: "Kal Naani Cake Mangwayi, Aaj Zara Feeka Pada; Hum Chupke Hi Aadha Aur kha gaye!"



MEME BREAK

LEFT: POTASSIUM (TOP OF THE SERIES)
Cold water mein daalo, aag laga deta hai!
RIGHT: GOLD (BOTTOM OF THE SERIES)
Paani, hawa, acid: kuch farak nahi padta.
Upar wala jitna active, neeche wala utna noble!

Use the series to predict

- **Above H**: displaces H_2 from dilute acids. **Below H** (Cu, Hg, Ag, Au): no H_2.
- **Higher metal** displaces a lower metal from its salt solution.
- Bottom metals (Ag, Au) are found **free in nature** and stay shiny for years.

TRY YOURSELF: REACTION OR NO REACTION?

Series dekho, phir likho: reaction hoga ya nahi? Answer page 4 par.



(i) $Zn + FeSO_4$
(ii) $Ag + CuSO_4$
(iii) $Fe + ZnSO_4$
(iv) Mg + dil. HCl
(v) Cu + dil. H_2SO_4
(vi) $Pb + CuCl_2$

Solved Example: Identify the metal

Q. Metal X reacts with cold water, gives hydroxide Y, and floats on the water. Name X and Y and give the equation.

- Reacts with **cold** water but not violently, and **floats**, so X is **calcium**
- The H_2 bubbles stick to Ca and lift it; Y = **calcium hydroxide**

$$Ca_{(s)} + 2H_2O_{(l)} \rightarrow Ca(OH)_{2(aq)} + H_{2(g)}\uparrow$$

Ca 1 = 1, O 2 = 2, H 4 = 4 ✔

8 EXAM TIPS: COMMON MISTAKES

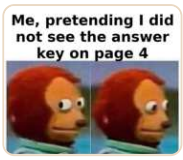
× WRONG	✔ RIGHT
$Cu + 2HCl \rightarrow CuCl_2 + H_2$	No reaction: Cu is below H
Zn + dil. HNO_3 gives H_2	HNO_3 oxidises H_2; nitrogen oxides form
$3Fe + 4H_2O \rightarrow Fe_2O_3 + \ldots$	Iron + steam gives Fe_3O_4
Al_2O_3 is a basic oxide	Al_2O_3 and ZnO are amphoteric
Sodium is stored in water	Stored in kerosene

Short answer questions

1. Give reasons: (A) Copper, not steel, is used to make hot water tanks. (B) Sodium is kept immersed in kerosene. (C) Aluminium is highly reactive, yet it is used to make cooking utensils. NCERT
2. Complete and balance: (A) $Al_2O_3 + HCl \rightarrow$ (B) $K_2O + H_2O \rightarrow$ (C) $Fe + H_2O$ (steam) $\rightarrow$ CBSE 2020
3. A man offered to polish dull gold bangles with a special solution. They sparkled like new, but their weight dropped sharply. Which solution did he use and why? NCERT

9 PRACTICE MCQS

Copy-paste nahi chalega! Khud solve karo, answer key page 4 par hai.



1. Metal X is used in the thermit process. When X is burnt in air, it gives an amphoteric oxide Y. X and Y are: CBSE 2023 ★★★
(a) Fe and Fe_2O_3 (b) Al and Al_2O_3
(c) Fe and Fe_3O_4 (d) Zn and ZnO

2. The arrangement of copper, tin, lead and mercury according to the reactivity series is: Delhi Gov. SQP 2024 ★★★
(a) Sn > Pb > Cu > Hg (b) Pb > Cu > Hg > Sn
(c) Cu > Hg > Sn > Pb (d) Hg > Sn > Pb > Cu

3. Which statements are correct? (I) Al_2O_3 reacts with both dil. HCl and NaOH(aq). (II) CaO reacts with both dil. HCl and NaOH(aq). (III) ZnO reacts with both dil. HCl and NaOH(aq). (IV) SO_2 reacts with neither. CBSE SQP 2025 ★★★
(a) (I) and (II) (b) (I) and (III)
(c) (II) and (IV) (d) (III) and (IV)

4. Dilute HCl is added to Al, Cu, Fe, Mg and Zn separately. The correct observation is: CBSE Compartment 2025 ★★☆
(a) most vigorous with Al; no reaction with Fe
(b) most vigorous with Mg; no reaction with Fe
(c) most vigorous with Mg; no reaction with Cu
(d) most vigorous with Zn; no reaction with Cu

5. Which metal is stored under kerosene? ★★☆
(a) Magnesium (b) Sodium
(c) Aluminium (d) Zinc

6. Which metal reacts with steam but not with cold or hot water? ★★☆
(a) Calcium (b) Magnesium
(c) Iron (d) Potassium

7. The non-metal that is a liquid at room temperature is: ★★★
(a) mercury (b) bromine
(c) iodine (d) gallium

8. Aqua regia is a freshly prepared mixture of: ★★☆
(a) conc. HCl and conc. HNO_3 in 1 : 3
(b) conc. HCl and conc. HNO_3 in 3 : 1
(c) conc. H_2SO_4 and conc. HCl in 3 : 1
(d) conc. HNO_3 and conc. H_2SO_4 in 1 : 1

Q9 and Q10: choose (a), (b), (c) or (d) as given below.

(a) Both (A) and (R) are true, and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true, and (R) is not the correct explanation of (A).
(c) (A) is true, but (R) is false.
(d) (A) is false, but (R) is true.

9. **Assertion (A):** A piece of zinc gets a reddish-brown coating when kept in copper sulphate solution for some time. **Reason (R):** Copper is a more reactive metal than zinc. CBSE 2024 ★★★

10. **Assertion (A):** Hydrogen gas is not evolved when zinc reacts with dilute nitric acid. **Reason (R):** Nitric acid is a strong oxidising agent; it oxidises the hydrogen formed to water. ★★☆

Answer key: MCQs

Q	Ans	Why
1	(b)	Al burns to Al_2O_3, an amphoteric oxide.
2	(a)	Sn and Pb are above H; Cu, then Hg below.
3	(b)	CaO is basic; SO_2 is acidic and reacts with NaOH.
4	(c)	Mg is most reactive; Cu is below H.
5	(b)	Na catches fire in air.
6	(c)	Fe reacts only with steam: gives Fe_3O_4.
7	(b)	Mercury is a metal; bromine is the liquid non-metal.
8	(b)	3 parts conc. HCl : 1 part conc. HNO_3.
9	(c)	Zn is more reactive and displaces Cu.
10	(a)	HNO_3 oxidises H_2 to H_2O.

Model answers: short answer

1 Reasons

- (A) Copper does not react with hot water or steam; iron in steel slowly reacts and rusts.
- (B) Sodium reacts so vigorously with air and moisture that it catches fire; kerosene keeps them away.
- (C) A thin, tough layer of Al_2O_3 protects it from further reaction; it is also light and a good conductor of heat.

2 Balanced equations

$$Al_2O_3(s) + 6HCl(aq) \rightarrow 2AlCl_3(aq) + 3H_2O(l)$$

$$K_2O(s) + H_2O(l) \rightarrow 2KOH(aq)$$

$$3Fe(s) + 4H_2O(g) \rightarrow Fe_3O_4(s) + 4H_2(g)\uparrow$$

3 The bangle mystery

- He used **aqua regia** (conc. HCl : conc. HNO_3 = 3 : 1).
- It dissolved the dull outer layer of gold, so the bangles shone, but the dissolved gold was lost and the weight fell.

Try-yourself answers

(i) Yes: $ZnSO_4$ + Fe
(ii) No: Ag is below Cu
(iii) No: Fe is below Zn
(iv) Yes: $MgCl_2$ + H_2
(v) No: Cu is below H
(vi) Yes: $PbCl_2$ + Cu

Coming up: Day 11

How metals and non-metals react: why sodium gives away an electron and chlorine grabs it, and how **NaCl** and **$MgCl_2$** are formed.

Real life: Metals around you

- **Aluminium foil** wraps food: malleable, light, does not rust.
- **Copper wires** at home: ductile, and the best conductor after silver.
- **Gold and silver** jewellery stays shiny: they sit at the bottom of the series.

★ How we starred (Eduspark PYQ analysis): we counted how often each topic has come in Chapter 3 board and sample-paper questions. Year tags appear only where the paper year is known.

Day 10 mind map: revise in 2 minutes

PROPERTIES
Lustrous, malleable, ductile, sonorous and good conductors (Hg is a liquid)

OXYGEN
Metal oxides are basic; Al_2O_3 and ZnO amphoteric

METALS & THEIR REACTIONS

WATER
K, Na, Ca cold; Mg hot; Al, Zn, Fe steam; Cu, Ag, Au none

ACIDS
Above H gives salt + H_2; no H_2 with HNO_3

SERIES ★★★
K > Na > Ca > Mg > Al > Zn > Fe > Pb > H > Cu > Hg > Ag > Au; a higher metal displaces a lower one from its salt solution

Day 10 hit-list: Eduspark PYQ analysis

Day 10 topic	Rating
Reactivity series and displacement	★★★
Amphoteric oxides (Al_2O_3, ZnO)	★★★
Metals with water (Na, Ca, Mg, Fe)	★★★
Exceptions in physical properties	★★☆
HNO_3 exception and aqua regia	★★☆

Day 10 checklist: tick before you sleep

- ☐ I can compare metals and non-metals and list the exceptions
- ☐ I can write metal + oxygen and metal + water equations
- ☐ I can explain amphoteric oxides with two equations
- ☐ I can use the reactivity series to predict displacement
- ☐ I solved all 10 MCQs and checked them with the key

My score: ____ / 5

5/5 = Jalwa hai! 4 se kam? Page 1–2 phir se padho.



LEVEL 10 COMPLETE +100 XP earned. Next: Level 11



BUJHO *Samajh gaya! Upar wala metal neeche wale ko bahar nikaalta hai, aur H se neeche wale metal acid se H_2 nahi dete!*

PAHELI *Shabaash! Kal Level 11 mein hum ionic bonds aur electron-dot banana seekhenge.*



NITESH SIR SAYS
Reactivity series roz ek baar bolo, 7 din mein dimaag mein chhap jayegi. ***Kal milte hai Day 11 pe!***

DAY 11 · LEVEL 11

CHEMISTRY | CHAPTER 3 | PART 2 · ASTRA SERIES

IONIC BONDS & COMPOUNDS

Goal: show electron transfer and explain ionic properties.



by Nitesh Prakash

01 Octet Rule | 02 Ions | 03 Electron-dot | 04 Ionic Properties | 05 Non-metals | +100 XP today | Level 11 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 WHY DO ATOMS REACT?

OCTET RULE

- Noble gases (He 2; Ne 2, 8; Ar 2, 8, 8) have a **full outer shell**, so they hardly react
- Other atoms react to get **8 electrons** in the outermost shell (an octet; 2 for the first shell, a duplet)
- Metals **lose** their few valence electrons; non-metals **gain** electrons

Valence electrons decide the ion

Atom	Z	Shells	Does what	Ion
Na	11	2, 8, 1	loses 1 e^-	Na^+ (2, 8)
Mg	12	2, 8, 2	loses 2 e^-	Mg^{2+} (2, 8)
Al	13	2, 8, 3	loses 3 e^-	Al^{3+} (2, 8)
O	8	2, 6	gains 2 e^-	O^{2-} (2, 8)
Cl	17	2, 8, 7	gains 1 e^-	Cl^- (2, 8, 8)



MEME BREAK

*Sodium ne apna **ikloota valence electron** chlorine ko de diya. Sodium: "Ouch, mera dil!" Par ab sodium ion ka octet complete hai.*

2 CATION AND ANION

CATION (+)	ANION (-)
Formed when a **metal** atom **loses** electrons	Formed when a **non-metal** atom **gains** electrons
Electrons fewer than protons: positive charge	Electrons more than protons: negative charge
$Na \rightarrow Na^+ + e^-$ $Mg \rightarrow Mg^{2+} + 2e^-$	$Cl + e^- \rightarrow Cl^-$ $O + 2e^- \rightarrow O^{2-}$



BUJHO *Sodium ne electron de diya, toh sodium ion mein ek proton bhi kam ho gaya?*

PAHELI *Nahi Bujho! Nucleus ke 11 protons wahi rehte hain. Electrons 10 bache, isliye charge +1 hai.*



3 ELECTRON-DOT: NaCl ★★★

IONIC (ELECTROVALENT) BOND

- Formed when **electrons are transferred** from a metal atom to a non-metal atom
- Opposite ions are held by **strong electrostatic attraction**
- Such compounds are **ionic** or **electrovalent** compounds



× metal electron | ● non-metal electron | curved arrow: electron transferred

$Na \rightarrow Na^+ + e^-$ $\quad Cl + e^- \rightarrow Cl^-$

Na^+ like neon (2, 8); Cl^- like argon (2, 8, 8)

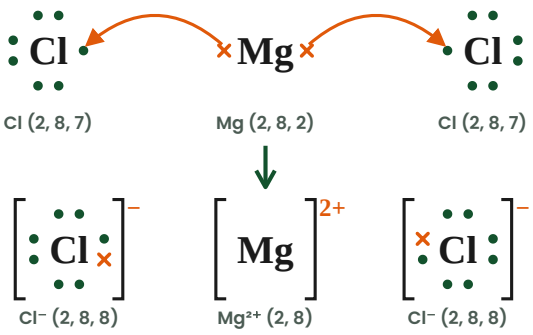
$2Na_{(s)} + Cl_{2(g)} \rightarrow 2NaCl_{(s)}$

Overall: Na 2 = 2, Cl 2 = 2 ✓



★★★ *Cl ko 8wa electron mila, full sukoon! Ye diagram har saal aata hai: Nitesh Bhaiya ki guarantee!*

4 ELECTRON-DOT: $MgCl_2$ ★★★



$Mg \rightarrow Mg^{2+} + 2e^-$ $\quad 2Cl + 2e^- \rightarrow 2Cl^-$

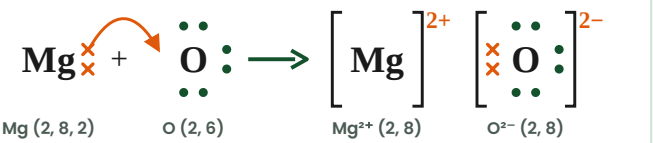
Mg gives two electrons, each Cl takes one: $Mg^{2+}(Cl^-)_2 = MgCl_2$

LINK TO VALENCY

Valency = electrons lost or gained: Na 1, Mg 2, Al 3, Cl 1, O 2. Total charge is zero: in Al_2O_3, $2 \times (+3) + 3 \times (-2) = 0$.

Solved Example 1: magnesium oxide ★★☆

Q. Show the formation of magnesium oxide by the transfer of electrons. NCERT



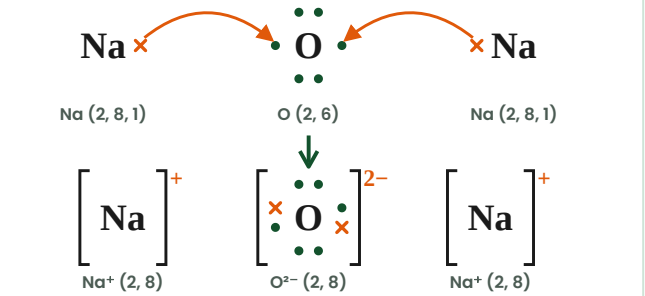
Mg gives **both** valence electrons to one O atom. Ions: Mg^{2+} and O^{2-}.

$$2Mg_{(s)} + O_{2(g)} \rightarrow 2MgO_{(s)}$$

Mg 2 = 2, O 2 = 2 ✔

Solved Example 2: sodium oxide ★★☆

Q. Show the formation of sodium oxide by the transfer of electrons. NCERT



Each Na gives **one** electron; O needs two, so **two** Na atoms are used. Ions: $2Na^{+}$ and O^{2-}.

$$4Na_{(s)} + O_{2(g)} \rightarrow 2Na_2O_{(s)}$$

Na 4 = 4, O 2 = 2 ✔

5 COMMON MISTAKES

× WRONG	✔ RIGHT
Both Mg electrons go to one Cl	One electron to each of two Cl atoms
A shared pair drawn between Na and Cl	A curved arrow shows the transfer
Ions drawn with no brackets or charge	Square brackets and the charge on each ion
Mg^{+} or MgCl in the answer	Mg^{2+} and the formula $MgCl_2$



MEME BREAK
*Electron-dot ekdum perfect banaya... phir **brackets aur charge** likhna bhool gaye. Marks gaye!*

TRY YOURSELF: SHOW THE TRANSFER

1. Potassium (19) and chlorine (17): draw the electron-dot formation of potassium chloride and write both ions with their shells.
2. Calcium (20) and oxygen (8): show how calcium oxide forms. How many electrons move, and which ions form?

6 PROPERTIES OF IONIC COMPOUNDS ★★★

Four properties with reasons

Property	Ionic compounds	Reason
Physical nature	Solids; hard but **brittle**	Strong attraction between + and − ions; break into pieces under pressure
Melting and boiling points	**High**	A lot of energy is needed to overcome the strong forces between ions
Solubility	Soluble in **water**; insoluble in kerosene, petrol	Water is a polar solvent and pulls the ions apart
Conduction	Only when **molten** or **in water**	Ions must be free to move

Compound	Melting point (K)	Boiling point (K)
NaCl	1074	1686
$MgCl_2$	981	1685
CaO	2850	3120

Solved Example 3: spot the ionic compound ★★☆

Q. Which of these are ionic compounds?

Substance	Melting point	Solid conducts?	Molten conducts?
A	1074 K	No	Yes
B	1235 K	Yes	Yes
C	387 K	No	No
D	2850 K	No	Yes

Ans. **A and D**: high melting point, and they conduct only when molten. B conducts as a solid (a **metal**); C melts low and never conducts.

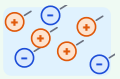
Do they conduct electricity? ★★★

SOLID

× no current
ions fixed in a rigid lattice

MOLTEN

✔ conducts
heat frees the ions

IN WATER

✔ conducts
ions move in solution



MEME BREAK
*Solid NaCl: "Main conduct karu? Mere ions toh **fixed** hain!" Pighlao ya paani mein ghol do, tab karega.*



BUJHO *Toh NaCl ek molecule hai na, ek Na aur ek Cl wala?*

PAHELI *Nahi! Ionic compound mein alag molecule nahi hote; sodium aur chloride ions ka giant crystal hota hai. NaCl sirf 1 : 1 ratio batata hai.*



7 PHYSICAL PROPERTIES OF NON-METALS

Rule and exception ★★☆

Property	Non-metals (general)	Exception
State	Solids or gases	**Bromine** is a liquid
Lustre	Dull	**Iodine** is lustrous
Hardness	Soft, brittle solids	**Diamond** is the hardest natural substance
Conduction	Poor conductors of heat and electricity	**Graphite** conducts electricity
Melting point	Generally low	**Diamond** has a very high melting point

Metal exceptions: mercury is a liquid; gallium and caesium melt on the palm; sodium and potassium are soft enough to cut with a knife; lead and mercury are poor conductors of heat.

Day 11 mind map: revise in 2 minutes

OCTET
8 electrons in the outer shell is stable (noble gas)

IONS
Metal loses e^-: cation; non-metal gains e^-: anion

IONIC BONDS & COMPOUNDS

ELECTRON-DOT ★★★
NaCl, $MgCl_2$, MgO, Na_2O with arrows, brackets, charges

PROPERTIES ★★★
Hard, brittle, high melting point, water-soluble, conduct when molten or dissolved

NON-METALS
Dull, brittle, poor conductors; exceptions: bromine, iodine, graphite, diamond



NUDGE
*Kerosene ne NaCl se kaha "ghul jao". NaCl ne seedha mana kar diya! Ionic compounds kerosene, petrol mein **insoluble** hote hain.*

Short answer questions

1. Write the electronic configurations of X (atomic number 12) and Y (atomic number 17). Show the formation of the compound formed when X and Y combine, and write its name and formula. What happens when an electric current is passed through its aqueous solution? Name the ion that moves to (i) the cathode, (ii) the anode.
CBSE Compartment 2024 (5 marks)

2. A farmer dissolved a copper sulphate fertiliser in water and sprayed it on his copper-deficient field; it worked. Later he mixed the same fertiliser in kerosene and it did not work. Give the reason. (2 marks)

3. (a) Why do ionic compounds have high melting points?
NCERT (b) Why does solid sodium chloride not conduct electricity, while its solution in water does? (2 marks)

8 PRACTICE MCQS

Sab milke solve karenge? Nahi! Pehle khud solve karo, answer key page 4 par hai.



1. In which form do electrovalent compounds conduct electricity?
CBSE Competency 2025 ★★★
(a) Only in the solid form
(b) In the solid form and in aqueous solution
(c) In aqueous solution and in the molten form
(d) In the solid, molten and aqueous forms

2. Which ions are present in sodium oxide? CBSE SQP 2023 ★★★
(a) Na^+ and O^- (b) $2Na^+$ and O^{2-}
(c) Na^{2+} and O^{2-} (d) Na^+ and O_2^{2-}

3. The electronic configurations of X, Y and Z are X: 2, 8, 7; Y: 2, 8, 2; Z: 2, 8. Which statement is correct? CBSE SQP 2024 ★★☆
(a) Y and Z are metals.
(b) Y and X are non-metals.
(c) X is a non-metal and Y is a metal.
(d) Y is a non-metal and Z is a metal.

4. The cation and anion present in aluminium oxide are: ★★☆
(a) Al^{2+} and O^{2-} (b) Al^{3+} and O^{2-}
(c) Al^{3+} and O^{3-} (d) Al^+ and O^-

5. How many electrons does one magnesium atom transfer when $MgCl_2$ is formed? ★★★
(a) one (b) two
(c) three (d) four

6. The electronic configuration of the Mg^{2+} ion is: ★★☆
(a) 2, 8, 2 (b) 2, 8
(c) 2, 8, 8 (d) 2, 6

7. Which of these is NOT a property of ionic compounds? ★★★
(a) High melting point (b) Soluble in water
(c) Conduct electricity in the solid state (d) Hard but brittle

8. Which non-metal is lustrous? ★★☆
(a) Sulphur (b) Iodine
(c) Phosphorus (d) Bromine

9. **Assertion (A):** The compound $MgCl_2$ does not conduct electricity in the solid state.
Reason (R): Movement of ions is not possible in the solid state of an ionic compound. ★★★
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

10. **Assertion (A):** Ionic compounds conduct electricity in the solid state.
Reason (R): Ionic compounds are made up of oppositely charged ions. ★★☆
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

Answer key: MCQs

Q	Ans	Why
1	(c)	Ions move only when molten or dissolved.
2	(b)	Two Na each lose 1 e^-; O gains 2.
3	(c)	7 valence e^-: non-metal; 2: metal.
4	(b)	Al loses 3 e^-; O gains 2 e^-.
5	(b)	Mg (2, 8, 2) gives one e^- to each Cl.
6	(b)	Mg^{2+} has 10 electrons: 2, 8.
7	(c)	Solid: ions fixed, so no conduction.
8	(b)	Iodine is a lustrous non-metal.
9	(a)	R correctly explains A.
10	(d)	A is false (no conduction as solid); R is true.

Model answers: short answer

1 X = magnesium, Y = chlorine

- X (12): 2, 8, 2 = Mg; Y (17): 2, 8, 7 = Cl
- Mg gives one electron to each of two Cl atoms (diagram in Section 4)
- Compound: **magnesium chloride, $MgCl_2$**
- In water its ions are free: the solution conducts and **electrolysis** takes place
- (i) Cathode: **Mg^{2+}** (cation); (ii) anode: **Cl^-** (anion)

$$Mg_{(s)} + Cl_{2(g)} \rightarrow MgCl_{2(s)}$$

Mg 1 = 1, Cl 2 = 2 ✔

2 Copper sulphate in water vs kerosene

- $CuSO_4$ is an **ionic compound**: soluble in water, so Cu^{2+} ions reach the plants
- Ionic compounds are **insoluble in kerosene**, so no ions were released and the spray failed

3 Melting point and conduction

- (a) Strong electrostatic attraction between oppositely charged ions needs a lot of energy to break
- (b) In solid NaCl the ions are held in a rigid lattice; in water they are free to move and carry charge

Try-yourself answers

- **(i)** K (2, 8, 8, 1) gives one electron to Cl (2, 8, 7): $[K]^+$ (2, 8, 8) and $[Cl]^-$ (2, 8, 8); formula KCl.
- **(ii)** Ca (2, 8, 8, 2) gives two electrons to one O (2, 6): $[Ca]^{2+}$ (2, 8, 8) and $[O]^{2-}$ (2, 8); formula CaO.

Day 11 hit-list: Eduspark PYQ analysis

Topic	Rating
Electron-dot formation of NaCl, $MgCl_2$	★★★
Conduction: molten or aqueous, not solid	★★★
Why ionic compounds have high melting points	★★★
Ions in Na_2O, MgO, Al_2O_3	★★☆
Solubility: water vs kerosene	★★☆
Non-metal exceptions (iodine, graphite, diamond)	★★☆

Exam tips: electron-dot answers

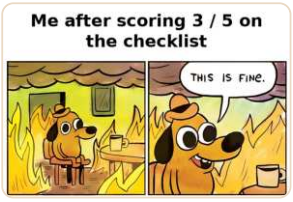
- ✔ Show only **valence** electrons: dots for one atom, crosses for the other.
- ✔ Draw a **curved arrow** from each transferred electron to the atom that receives it.
- ✔ Write ions in **square brackets** with the charge; add shell configurations below.
- ✔ Conduction answer: "molten or aqueous, not solid" **plus the reason** (ions free to move).
- ✔ Balance charges for the formula: Mg^{2+} needs two Cl^-; O^{2-} needs two Na^+.

Day 11 checklist: tick before you sleep

- ☐ I can explain the octet rule and why metals form cations
- ☐ I can draw the electron-dot formation of NaCl and $MgCl_2$
- ☐ I can show the formation of MgO and Na_2O and name the ions
- ☐ I can state four properties of ionic compounds with reasons
- ☐ I solved all 10 MCQs and checked them with the key

My score: ____ / 5

5/5 = Jalwa hai! 4 se kam? Page 1–2 phir se padho.



LEVEL 11 COMPLETE +100 XP earned. Next: Level 12



BUJHO *Samajh gaya! Metal electron deta hai, non-metal leta hai, aur ulte charge wale ions chipak jaate hain. Yahi ionic bond hai!*

PAHELI *Shabaash! Kal Level 12: ores se metal nikaalna.*

Coming up: Day 12

Occurrence and extraction of metals: ores and gangue, roasting vs calcination, the thermit reaction that welds railway tracks, and electrolytic refining of copper.

IONIC COMPOUNDS AT HOME

Common salt (NaCl), quicklime (CaO) and baking soda ($NaHCO_3$).



NITESH SIR SAYS

Electron-dot diagram teen baar haath se banao, exam mein haath khud chalega. ***Kal milte hai Day 12 pe!***

★ How we starred (Eduspark PYQ analysis): we counted how often each topic has come in Chapter 3 board and sample-paper questions. Year tags appear only where the paper year is known.

DAY 12 · LEVEL 12

CHEMISTRY | CHAPTER 3 | PART 3 · ASTRA SERIES

EXTRACTION OF METALS

Goal: trace a metal from its ore to the pure metal.



by Nitesh Prakash

01 Ores & Gangue | 02 Extraction Flow | 03 Calcination & Roasting | 04 Reduction | 05 Refining | +100 XP today | Level 12 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 MINERALS, ORES & GANGUE

DEFINITIONS

- **Mineral:** an element or compound that **occurs naturally** in the Earth's crust
- **Ore:** a mineral from which a metal can be extracted **profitably**. All ores are minerals, but all minerals are not ores.
- **Gangue:** unwanted impurities in the ore, such as soil, sand and rock
- **Metallurgy:** the steps of extracting a metal from its ore and refining it

Common ores to know ★★☆

Metal	Ores (formula)
Mercury	cinnabar HgS
Copper	copper glance Cu_2S, copper pyrites $CuFeS_2$, cuprite Cu_2O
Zinc	zinc blende ZnS, calamine $ZnCO_3$, zincite ZnO
Iron	haematite Fe_2O_3, magnetite Fe_3O_4, siderite $FeCO_3$, iron pyrites FeS_2
Lead	galena PbS
Aluminium	bauxite $Al_2O_3 \cdot 2H_2O$ (hydrated alumina)
Sodium	rock salt NaCl

Where metals occur: it depends on reactivity

- **Low reactivity** (Au, Ag, Pt, Cu): found **free** in nature; Cu and Ag also as sulphide or oxide ores
- **Middle** (Zn, Fe, Pb): mostly as **oxides, sulphides or carbonates**
- **High reactivity** (K, Na, Ca, Mg, Al): **never free**, always as compounds

Fact file: oxygen is the most abundant element in the earth's crust (about 46%), silicon is second, and **aluminium** is third and the **most abundant metal** (about 8%). **Astatine** is the rarest naturally occurring element. Many ores are oxides because oxygen is so reactive and abundant.



MEME BREAK

Gold ka swag: "Main free state mein milta hoon!" Least reactive metal, isliye kisi se react hi nahi karta.

2 EXTRACTION FLOW CHART

ORE → Enrichment: gangue removed

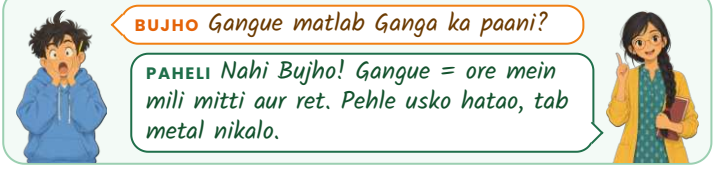
HIGH REACTIVITY (K, Na, Ca, Mg, Al)	MIDDLE (Zn, Fe, Pb)	LOW REACTIVITY (Hg, Cu)
Electrolysis of the molten ore	**Calcination** (carbonate) or **roasting** (sulphide)	Sulphide ore of the metal
Pure metal at the cathode	Oxide of the metal	**Roasting:** heat in air; the oxide is reduced by heat alone
	Reduction with carbon or a more reactive metal	**Metal, then refining**
	Metal, then refining	

EXAM FAVOURITE

Learn the three branches by reactivity and draw neat arrows.

Step 1: Enrichment of ore

- Removing gangue from the ore **before** extraction
- Uses the difference in **physical or chemical properties** of ore and gangue
- **Hydraulic washing** (gravity separation): ore powder is washed in a stream of water; heavy ore particles settle, light gangue is washed away
- Others: magnetic separation, froth flotation



Why convert ores into oxides first?

Sulphide and carbonate ores are converted to **metal oxides** because it is much easier to obtain a metal from its oxide than from its sulphide or carbonate. NCERT Exemplar

3 LOW REACTIVITY: HEAT ALONE WORKS

Mercury from cinnabar (HgS) ★★★

$$2HgS_{(s)} + 3O_{2(g)} \xrightarrow{\Delta} 2HgO_{(s)} + 2SO_{2(g)}$$
Step 1, roasting: Hg 2 = 2, S 2 = 2, O 6 = 6 ✔

$$2HgO_{(s)} \xrightarrow{\Delta} 2Hg_{(l)} + O_{2(g)}$$
Step 2, further heating reduces the oxide to liquid mercury

Copper from copper glance (Cu_2S) ★★☆

$$2Cu_2S_{(s)} + 3O_{2(g)} \xrightarrow{\Delta} 2Cu_2O_{(s)} + 2SO_{2(g)}$$
Cu 4 = 4, S 2 = 2, O 6 = 6 ✔

$$2Cu_2O_{(s)} + Cu_2S_{(s)} \xrightarrow{\Delta} 6Cu_{(s)} + SO_{2(g)}$$
Cu 6 = 6, O 2 = 2, S 1 = 1 ✔



SPIDER-SENSE
*Question mein "liquid metal from its sulphide ore" dikha? Turant socho: **mercury aur cinnabar**!*

4 CALCINATION VS ROASTING

CALCINATION	ROASTING
Carbonate ores	**Sulphide** ores
Strong heating in **limited air** (or no air)	Strong heating in **excess air**
Gives metal oxide + **CO_2**	Gives metal oxide + **SO_2**
Example: calamine, $ZnCO_3$	Example: zinc blende, ZnS

$$ZnCO_{3(s)} \xrightarrow[\text{limited air}]{\Delta} ZnO_{(s)} + CO_{2(g)}$$
Calcination: Zn 1 = 1, C 1 = 1, O 3 = 3 ✔

$$2ZnS_{(s)} + 3O_{2(g)} \xrightarrow[\text{excess air}]{\Delta} 2ZnO_{(s)} + 2SO_{2(g)}$$
Roasting: Zn 2 = 2, S 2 = 2, O 6 = 6 ✔



★★★ *Trick: Carbonate → Calcination, Sulphide → roaSting. Nitesh Bhaiya ki guarantee, ye table pakka aayegi!*

× WRONG	✔ RIGHT
Calcination and roasting are reduction steps	Both only change the ore into an oxide; reduction comes next
Roasting releases CO_2	Roasting releases SO_2; calcination releases CO_2

Real life: a crack in the railway track

A train stops because of a crack in the track. Engineers fill the gap with **thermit**: a mixture of iron(III) oxide and aluminium powder is ignited, molten iron flows into the crack and joins the rails on cooling.

5 REDUCTION: OXIDE TO METAL

(a) Reduction with carbon (coke) ★★★

$$ZnO_{(s)} + C_{(s)} \xrightarrow{\Delta} Zn_{(s)} + CO_{(g)}$$
Carbon removes oxygen: it is the reducing agent

Coke is the cheap carbon used in industry. ZnO is **reduced** (loses oxygen) and C is **oxidised** (gains oxygen).

(b) Thermit reaction: aluminium as reducer ★★★

$$Fe_2O_{3(s)} + 2Al_{(s)} \rightarrow 2Fe_{(l)} + Al_2O_{3(s)} + \textbf{Heat}$$
Fe 2 = 2, Al 2 = 2, O 3 = 3 ✔ (iron comes out molten)

$$3MnO_{2(s)} + 4Al_{(s)} \rightarrow 3Mn_{(l)} + 2Al_2O_{3(s)} + \textbf{Heat}$$
Mn 3 = 3, Al 4 = 4, O 6 = 6 ✔

- So much heat is released that the iron formed is **molten**
- Used to **weld railway tracks** and cracked machine parts
- Al is the reducing agent; Fe_2O_3 is the oxidising agent

EXTRA KNOWLEDGE (HIGH LEVEL) not in current CBSE syllabus

Blast furnace: haematite + coke + limestone give **pig iron**.

$C + O_2 \rightarrow CO_2$ $\quad$ $CO_2 + C \rightarrow 2CO$

$Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$ (CO is the reducer)

$CaCO_3 \rightarrow CaO + CO_2$ $\quad$ $CaO + SiO_2 \rightarrow CaSiO_3$ (slag)



MEME BREAK
Thermit mein Al ne Fe_2O_3 se oxygen chheena, aur itni garmi nikli ki iron pighal gaya. Railway track jud gaya!



BUJHO *Carbon se toh sab reduce ho jayega na? Sodium bhi?*

PAHELI *Nahi! Na, Mg, Al ko oxygen carbon se zyada pyaara hai. Inhe electrolysis se nikalte hain.*



6 ELECTROLYTIC REDUCTION

Highly reactive metals: Na, Mg, Ca, Al ★★★

- These metals have **more affinity for oxygen than carbon**, so carbon cannot reduce their oxides
- They are obtained by **electrolysis of their molten chlorides** (Al from molten aluminium oxide)
- Metal is deposited at the **cathode** (−); chlorine is released at the **anode** (+)

$$2NaCl_{(l)} \xrightarrow{\text{Electric current}} 2Na_{(l)} + Cl_{2(g)}$$
Electrolysis of molten sodium chloride: Na 2 = 2, Cl 2 = 2 ✔

CATHODE $Na^+ + e^- \rightarrow Na$

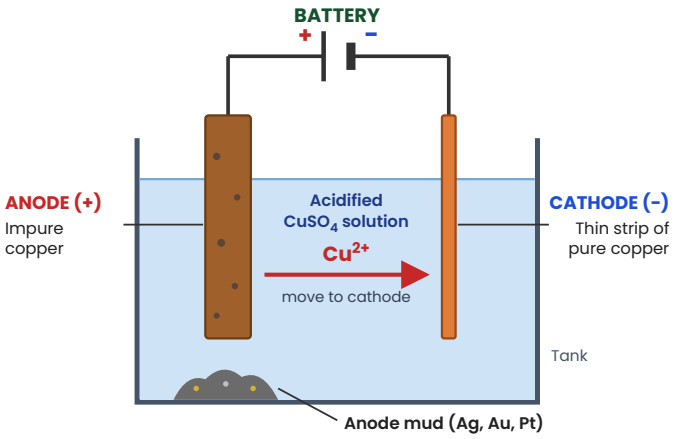
ANODE $2Cl^- \rightarrow Cl_2 + 2e^-$



OFFICIAL NOTICE
*K, Na, Ca, Mg, Al ke oxides par **carbon ki entry band**! Inke liye sirf electrolysis chalega.*

7 ELECTROLYTIC REFINING OF COPPER

Labelled diagram: draw this in the exam ★★★



How it works

ANODE $Cu_{(s)} \rightarrow Cu^{2+}_{(aq)} + 2e^-$ (impure Cu dissolves)

CATHODE $Cu^{2+}_{(aq)} + 2e^- \rightarrow Cu_{(s)}$ (pure Cu deposits)

- Anode: thick block of **impure** metal, joined to the **positive** terminal
- Cathode: thin strip of **pure** metal, joined to the **negative** terminal
- Electrolyte: **acidified copper sulphate** solution (a soluble salt of the same metal)
- Soluble impurities stay in the solution; **insoluble** ones settle below the anode as **anode mud**
- Also refined this way: Zn, Sn, Ni, Ag, Au

DID YOU KNOW?

The anode mud from copper refining is not waste: it contains **silver and gold**, which are recovered from it.

Which method for which metal? ★★☆

Metal	Position	Method of extraction
K, Na, Ca, Mg, Al	top	electrolysis of molten compound
Zn, Fe, Pb	middle	roasting or calcination, then reduction with carbon
Hg, Cu	low	roasting; oxide reduced by heating alone
Au, Ag, Pt	bottom	found free; only purified

Exam tips: common mistakes

- ✔ Method depends on **reactivity**: high = electrolysis, middle = carbon reduction, low = heating alone.
- ✔ Write **Δ** over the arrow for roasting, calcination and HgO decomposition.
- ✔ In refining, **impure metal is always the anode**; never swap the electrodes in the diagram.
- ✔ Label all five parts of the refining diagram: anode, cathode, electrolyte, anode mud, battery.

8 PRACTICE MCQS

Solve all 10 first, then check with the answer key on page 4.

1. Cinnabar is an ore of: ★★★

(a) zinc (b) mercury
(c) copper (d) iron

2. A metal X is used in the thermit process. When X is burnt in air, it gives an amphoteric oxide Y. X and Y are: CBSE 2023 ★★★

(a) Fe and Fe_2O_3 (b) Al and Al_2O_3
(c) Fe and Fe_3O_4 (d) Cu and CuO

3. Heating a carbonate ore strongly in limited air is called: ★★★

(a) roasting (b) calcination
(c) smelting (d) refining

4. Which gas is given off when zinc blende is roasted? ★★☆

(a) CO_2 (b) H_2
(c) SO_2 (d) O_2

5. Electrolysis of its molten chloride gives which metal? ★★★

(a) Iron (b) Copper
(c) Zinc (d) Sodium

6. In the electrolytic refining of copper, the anode is made of: ★★★

(a) a thin strip of pure copper (b) a block of impure copper
(c) graphite (d) zinc

7. The electrolyte used in refining copper is: ★★☆

(a) acidified copper sulphate solution
(b) sodium chloride solution
(c) dilute nitric acid
(d) zinc sulphate solution

8. Which metal is usually found in the free state in nature? ★☆☆

(a) Sodium (b) Gold
(c) Zinc (d) Aluminium

9. ★★★ **Assertion (A):** Zinc oxide can be reduced to zinc metal by heating with carbon.
Reason (R): Carbon is less reactive than zinc.
CBSE Practice Set 2023

(a) Both (A) and (R) are true and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
(c) (A) is true, but (R) is false.
(d) (A) is false, but (R) is true.

10. ★★☆ **Assertion (A):** Sodium is obtained by electrolysis of molten sodium chloride and not by heating its oxide with carbon.
Reason (R): Sodium has a greater affinity for oxygen than carbon.

(a) Both (A) and (R) are true and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
(c) (A) is true, but (R) is false.
(d) (A) is false, but (R) is true.

QUICK CHECK

Calcination gives off _____; roasting gives off _____.
Answers: CO_2, SO_2

Answer key: MCQs

Q	Ans	Why
1	(b)	Cinnabar = HgS, the ore of mercury.
2	(b)	Al reduces Fe_2O_3; Al_2O_3 is amphoteric.
3	(b)	Carbonate + limited air = calcination.
4	(c)	Sulphide ore gives SO_2 on roasting.
5	(d)	Na is too reactive for carbon reduction.
6	(b)	Impure metal is always the anode.
7	(a)	A soluble salt of the same metal.
8	(b)	Least reactive, so it stays free.
9	(c)	A true; R false: carbon removes O from ZnO.
10	(a)	Na holds O more strongly than C can.

Short and long answer questions

1. A reddish-brown metal X, when heated in air, gives a black compound Y, which on heating in hydrogen gives X back. X is refined by electrolysis. Identify X and Y and draw a labelled diagram of the refining of X. **CBSE SQP 2023**
2. A metal that is a liquid at room temperature is obtained by heating its sulphide in air. Name the metal and its ore, and write the reactions. **Delhi SQP 2025**
3. An element X is stored in kerosene and cannot be extracted from its ore by a reducing agent. (A) Can X be stored in water? (B) Identify X and write how it is extracted. **CBSE SQP 2025**
4. Differentiate between calcination and roasting with one balanced equation each.

Model answers

1 X = copper (Cu), Y = copper(II) oxide (CuO)

$$2Cu_{(s)} + O_{2(g)} \xrightarrow{\Delta} 2CuO_{(s)}$$

$$CuO_{(s)} + H_{2(g)} \xrightarrow{\Delta} Cu_{(s)} + H_2O_{(l)}$$

Cu 1 = 1, O 1 = 1, H 2 = 2 ✔

Diagram: draw the one in Section 7.

2 Mercury (Hg); ore: cinnabar (HgS)

$2HgS + 3O_2 \rightarrow 2HgO + 2SO_2$, then $2HgO \rightarrow 2Hg + O_2$ (both on heating; see Section 3).

3 X = sodium (potassium is also accepted)

- (A) No. It reacts violently with water and catches fire.
- (B) Electrolysis of molten NaCl: $Na^+ + e^- \rightarrow Na$ at the cathode; $2Cl^- \rightarrow Cl_2 + 2e^-$ at the anode.

4 Calcination vs roasting

- **Calcination:** carbonate ore heated strongly in limited air; CO_2 released
- **Roasting:** sulphide ore heated strongly in excess air; SO_2 released

$$ZnCO_{3(s)} \xrightarrow{\Delta} ZnO_{(s)} + CO_{2(g)}$$

$$2ZnS_{(s)} + 3O_{2(g)} \xrightarrow{\Delta} 2ZnO_{(s)} + 2SO_{2(g)}$$

Day 12 mind map: revise in 2 minutes

ORE
Mineral that gives metal profitably; gangue removed first

TO OXIDE
Carbonate: calcination; sulphide: roasting

EXTRACTION OF METALS

REDUCTION
Heat alone (Hg, Cu); carbon (Zn); Al (thermit)

ELECTROLYSIS ★★★
Na, Mg, Ca, Al from molten compounds

REFINING
Anode impure Cu, cathode pure Cu, acidified $CuSO_4$; anode mud below the anode

Day 12 checklist: tick before you sleep

- ☐ I can define mineral, ore and gangue with an example
- ☐ I can draw the extraction flow chart for low, middle and high reactivity metals
- ☐ I can compare calcination and roasting with balanced equations
- ☐ I can write the thermit reaction and explain electrolytic reduction
- ☐ I can draw and label the electrolytic refining of copper

My score: ____ / 5

5/5 = Jalwa hai! 4 se kam? Page 1–3 phir se padho.



LEVEL 12 COMPLETE +100 XP earned. Next: Level 13



BUJHO *Samajh gaya! Ore saaf karo, oxide banao, reduce karo, phir refine karo!*

PAHELI *Shabaash! Kal Level 13: corrosion aur alloys.*

Day 12 hit-list: Eduspark PYQ analysis

Topic	Rating
Refining of copper (labelled diagram)	★★★
Calcination vs roasting with equations	★★★
Thermit reaction and its use	★★★
Hg from cinnabar; Cu from Cu_2S	★★☆
Why Na, Mg, Al need electrolysis	★★☆



NITESH SIR SAYS
*Diagram ek baar khud banao, exam mein haath apne aap chalega. **Practice hi jalwa hai!***

★ How we starred (Eduspark PYQ analysis): we counted how often each topic has come in Chapter 3 board and sample-paper questions. Year tags appear only where the paper year is known.

DAY 13 · LEVEL 13

CHEMISTRY | CHAPTER 3 | PART 4 · ASTRA SERIES

CORROSION & ALLOYS

Goal: explain corrosion, prevent rusting, know key alloys.



by Nitesh Prakash

01 Corrosion · 02 Rusting Test · 03 Prevention · 04 Alloys · 05 Gold Carats

+125 XP today | Level 13 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 WHAT IS CORROSION?

DEFINITION

- A metal is **slowly eaten up** by air, moisture, acids or other substances around it
- The metal gets **oxidised** and goes back to a stable compound, like its ore
- Bridges, ships, railings and car bodies get weak; huge loss every year

Three famous examples ★★★

Metal	Attacked by	Coating formed
Iron	moist air	**Rust**: reddish-brown, flaky; hydrated iron(III) oxide, $Fe_2O_3 \cdot xH_2O$
Silver	sulphur compounds (H_2S) in air	**Black** coat of silver sulphide, Ag_2S (tarnishing)
Copper	moist CO_2 in air	**Green** coat of basic copper carbonate, $CuCO_3 \cdot Cu(OH)_2$

Equations behind the coats ★★☆

$4Ag(s) + 2H_2S(g) + O_2(g) \rightarrow 2Ag_2S(s) + 2H_2O(l)$

Silver tarnish: Ag 4 = 4, H 4 = 4, S 2 = 2, O 2 = 2 ✓

$2Cu(s) + H_2O(l) + CO_2(g) + O_2(g) \rightarrow CuCO_3 \cdot Cu(OH)_2(s)$

Green coat on copper: Cu 2 = 2, C 1 = 1, H 2 = 2, O 5 = 5 ✓



MEME BREAK

*Mummy ki chandi ki payal kaali ho gayi! Hawa ki H_2S ne silver ko **black Ag_2S** bana diya.*



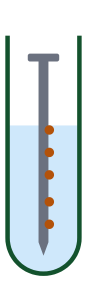
BUJHO *Aluminium bhi oxygen se react karta hai, toh usme rust kyun nahi lagta?*

PAHELI *Rust sirf iron ka hota hai! Aluminium par Al_2O_3 ki patli layer ban jaati hai jo aage corrosion rok deti hai.*

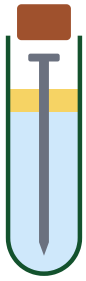


2 RUSTING: AIR + WATER ★★★ MOST ASKED

Activity: Conditions for rusting (NCERT)



A: nails in tap water, tube open
air ✓ water ✓
Rust forms



B: boiled distilled water, oil layer on top, corked
air × water ✓
No rust



C: anhydrous $CaCl_2$, corked
air ✓ water ×
No rust

- **Boiled** water: dissolved air is driven out; **oil layer** stops air dissolving again
- **Anhydrous $CaCl_2$** absorbs moisture, so the air in tube C stays dry
- Rust only in A: iron rusts only when **both air (oxygen) and water** are present



★★★ *Air ya paani? Dono chahiye! Teen test tube wala experiment pakka aayega, Nitesh Bhaiya ki guarantee!*

$4Fe(s) + 3O_2(g) + 2xH_2O(l) \rightarrow 2Fe_2O_3 \cdot xH_2O(s)$

Rust: hydrated iron(III) oxide. Fe 4 = 4, O 6 + 2x = 6 + 2x, H 4x = 4x ✓

RUSTING SPEEDS UP WHEN

- ✓ Air is humid, as in the rainy season and near the sea
- ✓ Water has **salt** in it (sea water): ships and coastal railings rust fast
- ✓ Air carries acidic gases (pollution) or the iron is scratched

Rust is **porous and flaky**: it falls off and exposes fresh iron, so rusting goes on and on.



MEME BREAK

Baarish mein cycle bahar chhod di? Rust: "Mere liye toh party hai!" Paani + hawa = rust ka buffet.

3 PREVENTION OF CORROSION

Six ways to stop rusting ★★★

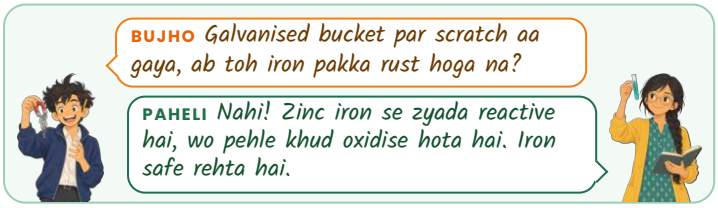
Method	How it protects	Used on
Painting	coat of paint keeps out air and moisture	railings, bridges, car bodies
Oiling, greasing	thin film of oil or grease blocks air and water	machine parts, tools, cycle chains
Galvanising ★★★	thin coat of **zinc** on iron or steel	buckets, roof sheets, water pipes
Chrome plating	chromium layer by electroplating; shiny and rust-proof	taps, cycle handlebars
Anodising ★★☆	thicker protective Al_2O_3 layer on aluminium	cookware, window frames
Alloying	mix iron with other metals, e.g. stainless steel	utensils, surgical tools

Galvanising: protected even when scratched ★★★

- Zinc first acts as a **barrier** between iron and moist air
- If the zinc coat breaks, iron still does not rust: zinc is **more reactive than iron** and gets oxidised first
- **Sacrificial protection:** iron is joined to or coated with a **more reactive metal (Zn or Mg)** that corrodes first, e.g. galvanising, Mg blocks on ship hulls and buried pipelines

Anodising of aluminium

1. Clean aluminium article is made the **anode**; dilute sulphuric acid is the electrolyte.
2. Oxygen released at the anode reacts with aluminium and **thickens the Al_2O_3 layer**.
3. The thick oxide layer resists corrosion and can be dyed in attractive colours.



× WRONG	✓ RIGHT
Rusting is caused by water alone	Rusting needs both air and water
Aluminium and copper "rust"	Only iron rusts; other metals corrode
Galvanising coats iron with tin	Galvanising coats iron with zinc



MEME BREAK
Cycle ki chain ko kabhi oil nahi diya, ab rust aur awaaz dono free mein! Oiling karo, chain bachao.

4 ALLOYS ★★★ MOST ASKED

DEFINITION

- An alloy is a **homogeneous mixture** of two or more metals, or a metal and a non-metal
- Made by melting the main metal and dissolving the others in it, then cooling
- **Amalgam**: an alloy in which one of the metals is **mercury**

Common alloys: composition and uses

Alloy	Made of	Special property and use
Steel	Fe + C (small amount)	hard, strong: buildings, bridges, rails
Stainless steel ★★★	Fe + Ni + Cr (+ a little C)	does not rust: utensils, surgical tools
Brass ★★★	Cu + Zn	harder than copper: utensils, taps, musical instruments
Bronze ★★★	Cu + Sn	strong, resists corrosion: statues, medals, bells
Solder ★★☆	Pb + Sn	**low melting point**: welding electrical wires
Amalgam ★☆☆	Hg + another metal	soft paste that hardens: dental fillings

Alloy vs pure metal ★★☆

- ✔ **Harder and stronger**: pure iron is soft and stretches when hot; a little carbon makes it hard steel
- ✔ **More resistant to corrosion** than the pure metal
- ✔ **Lower electrical conductivity** than the pure metal
- ✔ **Lower melting point** than its constituents in some cases, e.g. solder

COMMON MISTAKE

Steel is **not** a metal-only alloy: it contains carbon, a non-metal. Brass, bronze and amalgam contain only metals. CBSE 2020

5 GOLD: 24 VS 22 CARAT ★★☆

24 CARAT	22 CARAT
Pure gold (24 parts out of 24)	22 parts gold + **2 parts copper or silver**
Very soft; bends easily, not good for ornaments	Harder and more durable: used for jewellery

Solved example: Purity of gold

Q. What percentage of gold is present in 22 carat and in 18 carat gold?

- 22 carat: 22/24 × 100 = **91.67%** gold
- 18 carat: 18/24 × 100 = **75%** gold



MEME BREAK

TOP: 24 CARAT, PURE GOLD
Bahut soft, payal turant mud jaayegi
BOTTOM: 22 CARAT
22 gold + 2 copper or silver: strong aur classy
Thoda copper milao, jewellery ko tuxedo pehnao!

Exam tips

- ✔ Draw and label all three test tubes; write what each one proves.
- ✔ Write rust as **hydrated iron(III) oxide**; the formula alone can lose marks.
- ✔ "Why is a galvanised article safe when scratched?" Answer with reactivity: zinc is above iron.
- ✔ Name the constituents in the right order: brass Cu + Zn, bronze Cu + Sn, solder Pb + Sn.
- ✔ Silver turns **black**, copper turns **green**, iron turns **reddish-brown**: never mix them up.

Day 13 mind map: revise in 2 minutes

CORROSION
Iron: rust; silver: black Ag_2S; copper: green coat

RUSTING ★★★
Needs air + water; rust only in tube A

CORROSION & ALLOYS

PREVENTION
Paint, oil, zinc, chromium, anodising, alloying

ALLOYS ★★★
Steel, stainless steel, brass, bronze, solder, amalgam

GOLD
24 carat = pure and soft; 22 carat = 22 gold + 2 Cu or Ag

TRY YOURSELF: NAME THE BEST METHOD

(i) bicycle chain
(ii) iron roof sheets
(iii) car body
(iv) shiny water tap
(v) aluminium pressure cooker
(vi) surgical scissors

Short and long answer questions

1 Copper domes of old buildings slowly turn green. (A) Why do the domes appear green though copper is reddish-brown? (B) Should the copper domes be replaced by iron domes? Justify. (C) Why did ancient architects use copper sheets for domes? CBSE SQP 2025

2 Iron nails coated with zinc do not rust, even when the coat is damaged. (A) Name this process and give two other methods to prevent rusting. (B) Where will rusting be the most: an iron bucket electroplated with zinc, iron hinges on a gate, or a painted iron fence? Give a reason. CBSE SQP 2022

3 Describe an activity with a labelled diagram to show that both air and water are needed for the rusting of iron. Why is boiled water and a layer of oil used in one test tube? (5 marks)

6 PRACTICE MCQS

Padhai tabhi hogi jab MCQs khud solve karoge! Answer key page 4 par hai.



1. Rust is chemically: ★★★
(a) Fe_3O_4 (b) hydrated iron(III) oxide
(c) iron(II) sulphate (d) iron(II) oxide

2. In the three test tube experiment, rust appears only in the tube with: ★★★
(a) boiled water + oil (b) anhydrous $CaCl_2$
(c) tap water, open (d) all three tubes

3. Oil is poured over the boiled water in test tube B to: ★★☆
(a) add moisture to the air
(b) stop air from dissolving in the water
(c) dry the air in the tube
(d) speed up rusting

4. Silver articles turn black in air because of a coating of: ★★★
(a) Ag_2O (b) Ag_2S
(c) AgCl (d) $AgNO_3$

5. The green coating formed on copper in moist air is: ★★☆
(a) copper(II) oxide (b) copper sulphate
(c) basic copper carbonate (d) copper nitrate

6. Galvanisation is the process of coating iron with a thin layer of: ★★★
(a) tin (b) chromium
(c) zinc (d) aluminium

7. Which alloy contains a non-metal as one of its constituents? CBSE 2020 ★★★
(a) Brass (b) Bronze
(c) Amalgam (d) Steel

8. In stainless steel, iron is mixed with: NCERT Exemplar ★★★
(a) Ni and Cr (b) Cu and Cr
(c) Ni and Cu (d) Cu and Au

9. **Assertion (A):** A galvanised iron article stays protected from rusting even if the zinc coating is broken.
Reason (R): Zinc is more reactive than iron. ★★★
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

10. **Assertion (A):** Aluminium does not corrode easily in air.
Reason (R): Aluminium is less reactive than iron. ★★☆
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

QUICK RECALL: ONE-MARK QUESTIONS

(a) Gas in air that blackens silver?
(b) Metal used for galvanising?
(c) Constituents of bronze?
(d) Why is solder used for wires?

Answer key: MCQs

Q	Ans	Why
1	(b)	Rust = $Fe_2O_3 \cdot xH_2O$, hydrated iron(III) oxide.
2	(c)	Only tube A has both air and water.
3	(b)	Oil seals the water from air.
4	(b)	Silver reacts with H_2S to give black Ag_2S.
5	(c)	Green coat = basic copper carbonate.
6	(c)	Galvanising = zinc coating.
7	(d)	Steel contains carbon, a non-metal.
8	(a)	Stainless steel = Fe + Ni + Cr.
9	(a)	Zinc corrodes first and protects iron.
10	(c)	A true; Al is more reactive than Fe, but its Al_2O_3 layer protects it.

Try-yourself and quick recall answers

(i) oiling or greasing
(ii) galvanising
(iii) painting
(iv) chrome plating
(v) anodising
(vi) alloying (stainless steel)
(a) hydrogen sulphide, H_2S
(b) zinc
(c) copper and tin
(d) its melting point is low

Model answers: short and long answer

1 Copper domes (CBSE SQP 2025)

- (A) Copper reacts with moist CO_2 in air to form a **green coat of basic copper carbonate**
- (B) No. Iron would rust; flaky rust falls off, exposes fresh iron and the dome weakens. The copper coat protects the metal below
- (C) Copper is highly **malleable**: thin sheets can be shaped into a dome, and it does not rust

2 Zinc-coated nails (CBSE SQP 2022)

- (A) **Galvanisation**. Other methods: painting, oiling or greasing, chrome plating, alloying
- (B) **Iron hinges on a gate**: bare iron in contact with both air and moisture; the bucket and fence are coated

3 Activity: air and water both needed

- Take three test tubes with clean iron nails: A with tap water (open), B with boiled distilled water + oil layer (corked), C with anhydrous $CaCl_2$ (corked); draw the diagram from page 1
- After a few days, rust only in A. B has no air (boiling removes air, oil stops it re-entering); C has no moisture ($CaCl_2$ absorbs it)
- Conclusion: rusting needs **both air and water**

Day 13 hit-list: Eduspark PYQ analysis

Topic	Rating
Rusting experiment (air + water)	★★★
Galvanisation and why it works	★★★
Alloys: constituents and uses	★★★
Silver black, copper green	★★★
Methods of preventing rusting	★★☆
22 carat gold, anodising	★★☆

Day 13 checklist: tick before you sleep

- [ ] I can define corrosion and name the coats on iron, silver and copper
- [ ] I can draw the three test tube experiment and explain each tube
- [ ] I can list six ways to prevent rusting and explain galvanisation
- [ ] I know the constituents and uses of six alloys, and 22 carat gold
- [ ] I solved all 10 MCQs and checked them with the key

My score: ____ / 5

5/5 = Jalwa hai! 4 se kam? Page 1–2 phir se padho.



LEVEL 13 COMPLETE +125 XP earned. Next: Level 14



BUJHO *Samajh gaya! Rust ke liye hawa aur paani dono chahiye, aur zinc khud kurbaan hokar iron ko bachata hai!*

PAHELI *Shabaash! Kal Level 14: Chapter 3 Boss Level.*

Real life: why kitchens love these

- Pressure cookers and kadhais: **anodised aluminium**, hard and non-rusting
- Steel plates and spoons: **stainless steel** never rusts
- Pooja thali and temple bells: **brass and bronze** stay bright for years

Did you know? The iron pillar of Delhi

The iron pillar near the Qutub Minar was made by Indian iron workers more than **1600 years ago** and has hardly rusted. Ancient Indians had a special way of working iron that resists rusting.

Coming up: Day 14

Chapter 3 **Boss Level**: one test on the whole chapter, from reactivity series to alloys. Revise Days 10–13 tonight!



NITESH SIR SAYS

*Iron ko paint chahiye, tumhe revision. **Roz thoda padho, rust nahi lagega!***

★ How we starred (Eduspark PYQ analysis): we counted how often each topic has come in Chapter 3 board and sample-paper questions. Year tags appear only where the paper year is known.

DAY 14 · LEVEL 14

CHEMISTRY | CHAPTER 3 | BOSS LEVEL

CHAPTER 3 BOSS LEVEL

Goal: test all of Metals and Non-metals in 75 minutes.



by Nitesh Prakash

01 Revision Map · 02 MCQs · 03 Short Answers · 04 Long Answers · 05 Answer Key · +200 XP today | Level 14 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 CHAPTER 3 REVISION MAP

Chapter 3 mind map: revise in 5 minutes

PROPERTIES
Metals: lustrous, malleable, ductile, good conductors. Exceptions: Hg is a liquid; Na, K are soft; iodine is lustrous; graphite conducts

REACTIONS
Metal + O_2 → basic oxide (Al_2O_3, ZnO amphoteric); + water → hydroxide or oxide + H_2; + dilute acid → salt + H_2 (not with HNO_3)

REACTIVITY SERIES ★★★
K > Na > Ca > Mg > Al > Zn > Fe > Pb > [H] > Cu > Hg > Ag > Au; a more reactive metal displaces a less reactive one from its salt solution

METALS & NON-METALS

IONIC BONDS
Metal loses e^-, non-metal gains; hard, high m.p., soluble in water; conduct electricity only in molten or aqueous form

EXTRACTION
Low: heat alone (HgS, Cu_2S); middle: roast or calcine, then reduce (C, Al); high: electrolysis of molten compound

REFINING, CORROSION & ALLOYS ★★★
Refining: anode impure, cathode pure strip, anode mud below. Rusting needs air + water; silver turns black (Ag_2S); copper turns green. Brass Cu + Zn, bronze Cu + Sn, solder Pb + Sn

Must-know equations: write them daily

$2Na_{(s)} + 2H_2O_{(l)} \rightarrow 2NaOH_{(aq)} + H_{2(g)}\uparrow + \textbf{Heat}$
Na, K react violently with cold water: stored in kerosene

$Al_2O_{3(s)} + 2NaOH_{(aq)} \rightarrow 2NaAlO_{2(aq)} + H_2O_{(l)}$
amphoteric oxide reacts with a base too (sodium aluminate)

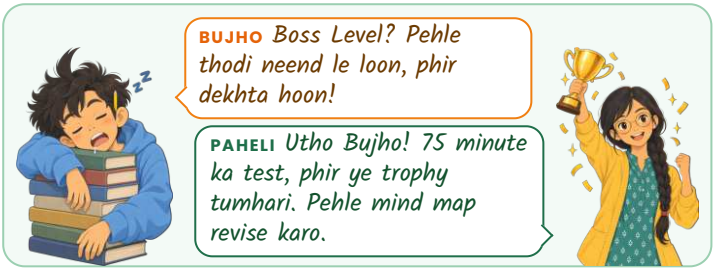
$Fe_{(s)} + CuSO_{4(aq)} \rightarrow FeSO_{4(aq)} + Cu_{(s)}$
displacement: Fe is above Cu in the series

$2ZnS_{(s)} + 3O_{2(g)} \xrightarrow{\Delta} 2ZnO_{(s)} + 2SO_{2(g)}\uparrow$
roasting: sulphide ore heated in excess air

$ZnCO_{3(s)} \xrightarrow{\Delta} ZnO_{(s)} + CO_{2(g)}\uparrow$
calcination: carbonate ore heated in limited air

$ZnO_{(s)} + C_{(s)} \xrightarrow{\Delta} Zn_{(s)} + CO_{(g)}$
reduction of the oxide with carbon (coke)

$Fe_2O_{3(s)} + 2Al_{(s)} \rightarrow 2Fe_{(l)} + Al_2O_{3(s)} + \textbf{Heat}$
thermit reaction: molten iron joins railway tracks



BOSS LEVEL RULES

- ✔ **Time:** 75 minutes. **Maximum marks:** 40. Close the book and set a timer.
- ✔ **A:** 14 MCQs × 1 = 14. **B:** 5 short answers = 12. **C:** 2 long answers × 5 = 10. **D:** 1 case-based question = 4.
- ✔ Write every equation balanced, with state symbols. Draw diagrams in pencil and label them.

Phone side mein rakho, timer lagao. Boss se ladna hai, scroll nahi karna!



2 SECTION A: MCQS

1. Which equations form an amphoteric oxide, and what are A and B? (1) 2Pb + O_2 → A PbO (2) B Zn + O_2 → 2ZnO ★★☆

(a) only 1; A = 1, B = 1 (b) both; A = 2, B = 2
(c) only 2; A = 2, B = 1 (d) both; A = 1, B = 2

2. Match (A) forms an amphoteric oxide, (B) melts on the palm, (C) gives H_2 with very dilute HNO_3, (D) cannot displace H_2 from acids, with (i) Ga (ii) Au (iii) Al (iv) Mn. Code for A, B, C, D: ★★☆

(a) iii, i, iv, ii (b) i, iii, ii, iv
(c) iii, iv, i, ii (d) iv, i, iii, ii

3. Metal X reacts with cold water, Y only with hot water and Z only with steam. Their increasing order of reactivity is: ★★★

(a) X < Y < Z (b) Z < Y < X
(c) Y < Z < X (d) Z < X < Y

4. An element with atomic number ____ will form a basic oxide. CBSE SQP 2023 ★★☆

(a) 7 (2, 5) (b) 17 (2, 8, 7)
(c) 14 (2, 8, 4) (d) 11 (2, 8, 1)

5. In the reaction $3MnO_2 + 4Al \rightarrow 3Mn + 2Al_2O_3$ + heat, the substance oxidised is: CBSE SQP 2023 ★★★

(a) MnO_2 (b) Al
(c) Mn (d) Al_2O_3

6. When magnesium chloride is formed, a magnesium atom: ★★★

(a) gains two electrons (b) loses two electrons
(c) shares two electrons (d) loses one electron

7. Mercury is obtained from its ore cinnabar (HgS) by: ★★☆

(a) heating it in air to form HgO, then heating the HgO further
(b) reducing it with carbon in a blast furnace
(c) electrolysis of molten cinnabar
(d) heating it strongly with aluminium powder

8. Zinc carbonate is changed into zinc oxide by heating it strongly in limited air. This process is: ★★★

(a) roasting (b) calcination
(c) refining (d) electrolysis

9. In the electrolytic refining of copper, the cathode is a: ★★★

(a) block of impure copper (b) thin strip of pure copper
(c) graphite rod (d) zinc plate

10. In stainless steel, iron is mixed with: NCERT Exemplar ★★☆

(a) nickel and chromium (b) copper and chromium
(c) nickel and copper (d) copper and gold

11. On hammering, substance M flattens into a sheet and rings; substance N breaks into powder. So: CBSE SQP 2024 ★★☆

(a) both are metals (b) both are non-metals
(c) M metal, N non-metal (d) M non-metal, N metal

12. Silver articles turn black in air. The black coating is of: ★★☆

(a) silver oxide (b) silver sulphide
(c) silver nitrate (d) silver carbonate

A–R dekh ke dimaag ghoom gaya? Pehle A check karo, phir R, phir dono ka link!



13. **Assertion (A):** Ionic compounds have high melting and boiling points.
Reason (R): A large amount of energy is needed to break the strong electrostatic forces between oppositely charged ions. ★★★

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

14. **Assertion (A):** Gold is found in the free state in nature.
Reason (R): Gold is a highly reactive metal. ★★☆

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

How to attack Sections B, C and D

- ✔ 2 marks = two clear points; 3 marks = three points or an equation with a reason.
- ✔ Every equation: balanced, with (s), (l), (g) or (aq), and Δ on the arrow where heat is needed.
- ✔ Time plan: A in 15 min, B in 25 min, C and D in 30 min, 5 min to recheck.

3 SECTION B: SHORT ANSWERS

15. State reasons: (A) Zinc oxide is an amphoteric oxide. (B) Calcium starts floating when it is put in water. CBSE 2024 [2]

16. During the electrolytic refining of silver: (A) What are the cathode and the anode made of? (B) What is anode mud? CBSE Compartment 2025 [2]

17. How does galvanisation protect iron from rusting? Why does a galvanised article stay protected even if its zinc coating is scratched? [2]

18. Name the reaction between iron(III) oxide and aluminium. Write its balanced equation. Why is it called a displacement reaction? Give one use of this reaction. CBSE 2024 [3]

19. A student is given copper(I) sulphide (Cu_2S) and copper(I) oxide (Cu_2O). Show with balanced equations how copper can be obtained from them. Why is no reducing agent like carbon needed here? CBSE SQP 2024 [3]

4 SECTION C: LONG ANSWERS

20. (A) Write the electronic configurations of sodium (Z = 11), magnesium (Z = 12) and chlorine (Z = 17). (B) Show the formation of NaCl and $MgCl_2$ by the transfer of electrons. (C) Why do ionic compounds have high melting points? Why do they conduct electricity in the molten state but not as solids? [5]

21. A reddish-brown metal X, when heated in air, gives a black compound Y. When Y is heated in hydrogen gas, X is obtained back. X is refined by electrolysis and the refined X is used in electrical wiring. (A) Identify X and Y and write both equations. (B) Draw a labelled diagram of the electrolytic refining of X. (C) Write the reactions at the anode and the cathode. Based on CBSE SQP 2023 [5]

5 SECTION D: CASE-BASED

CASE: THE GREEN DOMES

Many old buildings have domes made of thin copper sheets. Freshly fitted copper is reddish-brown and shiny, but over the years the domes turn green. Iron does not turn green; it slowly rusts and crumbles. CBSE SQP 2025

1. Why does a copper dome turn green? Name the green substance. [1]
2. Should the copper domes be replaced by iron domes to avoid the change of colour? Give a reason. [1]
3. Why did old builders choose copper for domes? Give two properties of copper that help. [2]



★★★ *NaCl aur magnesium chloride ka electron-dot, thermit equation aur refining diagram: ye pakka aayenge, Nitesh Bhaiya ki guarantee!*

6 ANSWER KEY

Answer key: Section A (MCQs)

Q	Ans	Why
1	(b)	PbO and ZnO are amphoteric; A = B = 2.
2	(a)	Al, Ga, Mn, Au in that order.
3	(b)	Na (cold) > Mg (hot) > Fe (steam).
4	(d)	2, 8, 1 is sodium, a metal: basic Na_2O.
5	(b)	Al gains oxygen, so it is oxidised; MnO_2 is reduced.
6	(b)	Mg (2, 8, 2) gives 2 e^- → Mg^{2+}.
7	(a)	2HgS → 2HgO → 2Hg; low-reactivity metal.
8	(b)	Carbonate + limited air = calcination.
9	(b)	Pure Cu deposits on the cathode strip.
10	(a)	Fe + Ni + Cr (with a little C).
11	(c)	Malleable and sonorous: metal. Brittle: non-metal.
12	(b)	Ag reacts with sulphur in air: black Ag_2S.
13	(a)	Strong ionic forces need much energy; R explains A.
14	(c)	Gold is very unreactive, so R is false.

Displacement wala sawaal galat hua? Series dekho: Zn aur Fe dono Cu ko uske salt se hata dete hain, Cu kisi ko nahi!



Class Question: one non-metal, two oxides

Q. Non-metal A forms two oxides: B is neutral and poisonous (incomplete burning); C is acidic and causes global warming. Identify A, B and C.

- A = **carbon**; B = **carbon monoxide, CO** (neutral oxide); C = **carbon dioxide, CO_2** (acidic oxide, greenhouse gas)
- $2C + O_2 \rightarrow 2CO$ (limited air); $C + O_2 \rightarrow CO_2$ (plenty of air)

Score card: how did you fight the boss?

Marks	Result
36–40	BOSS DEFEATED: chapter mastered
28–35	STRONG WIN: fix the small slips
20–27	ALMOST THERE: revise weak days
Below 20	RETRY: re-read Days 10–13



- **Where to revise:** Q1–4, 11, 15 → Day 10; Q6, 13, 20 → Day 11; Q5, 7–9, 14, 16, 18, 19, 21 → Day 12; Q10, 12, 17, 22 → Day 13.

36 se upar? Bujho ki tarah naacho, boss khatam!

Boss Level tracker: write your percentage

Chapter 1 Boss (Day 4) ____ %
Chapter 2 Boss (Day 9) ____ %
Chapter 3 Boss (Day 14) ____ %

Model answers: Section B

15 Reasons: ZnO and calcium

- (A) ZnO reacts with both acids and bases to give salt and water.

$$ZnO(s) + 2HCl(aq) \rightarrow ZnCl_2(aq) + H_2O(l)$$

$$ZnO(s) + 2NaOH(aq) \rightarrow Na_2ZnO_2(aq) + H_2O(l)$$

sodium zincate

- (B) Hydrogen bubbles formed stick to the calcium, so it floats.

16 Refining of silver

- (A) Cathode: thin strip of pure silver. Anode: block of impure silver.
- (B) Insoluble impurities that settle at the bottom, below the anode, are called anode mud.

17 Galvanisation

- A thin coat of zinc keeps air and moisture away from the iron.
- Zinc is more reactive than iron, so even if the coat is scratched, zinc reacts first and the iron stays protected.

Model answers: Section B (3 marks)

18 Thermit reaction

- Name: thermit reaction (highly exothermic).

$$Fe_2O_3(s) + 2Al(s) \rightarrow 2Fe(l) + Al_2O_3(s) + \text{Heat}$$

Fe 2 = 2, Al 2 = 2, O 3 = 3 ✔

- Al is more reactive than Fe, so it displaces Fe from its oxide.
- Use: the molten iron joins railway tracks and cracked machine parts.

19 Copper from Cu_2S

- Roasting: part of the sulphide ore is heated in air.

$$2Cu_2S(s) + 3O_2(g) \xrightarrow{\Delta} 2Cu_2O(s) + 2SO_2(g)\uparrow$$

Cu 4 = 4, S 2 = 2, O 6 = 6 ✔

$$2Cu_2O(s) + Cu_2S(s) \xrightarrow{\Delta} 6Cu(s) + SO_2(g)\uparrow$$

Cu 6 = 6, O 2 = 2, S 1 = 1 ✔

- Cu is low in the series, so heating alone reduces it.

Model answers: Section C

20 Ionic bonding (5 marks)

- (A) Na: 2, 8, 1; Mg: 2, 8, 2; Cl: 2, 8, 7.
- (B) Na gives 1 e^- to Cl; Mg gives 1 e^- to each of two Cl atoms. Each ion gets a stable octet.

FORMATION OF SODIUM CHLORIDE



FORMATION OF MAGNESIUM CHLORIDE



- (C) A lot of energy is needed to break the strong attraction between oppositely charged ions, so the melting point is high.
- In a solid the ions are fixed in place. When molten, they move freely and carry the current.

Model answers: Sections C and D

21 Copper and its refining (5 marks)

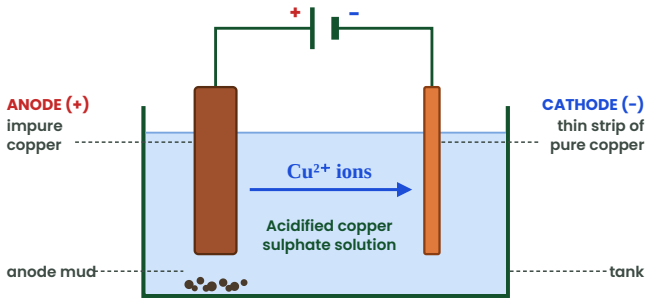
- (A) X = copper (Cu); Y = copper(II) oxide (CuO), black.

$$2Cu_{(s)} + O_{2(g)} \xrightarrow{\Delta} 2CuO_{(s)}$$

$$CuO_{(s)} + H_{2(g)} \xrightarrow{\Delta} Cu_{(s)} + H_2O_{(l)}$$

CuO is reduced; H_2 is oxidised

(B) ELECTROLYTIC REFINING OF COPPER



- (C) Anode: $Cu(s) \rightarrow Cu^{2+}(aq) + 2e^-$ (impure copper dissolves).
- Cathode: $Cu^{2+}(aq) + 2e^- \rightarrow Cu(s)$ (pure copper deposits).

22 Case: the green domes

- (1) Copper reacts with moist carbon dioxide in air and gets a green coat of basic copper carbonate.
- (2) No. Iron rusts, and rust flakes off, so the dome would get weaker; the green coat on copper protects the metal below it.
- (3) Copper is highly malleable, so thin sheets can be shaped into domes; it also resists corrosion and lasts long.

Most asked in Chapter 3: Eduspark PYQ analysis

Topic	Day	Rating
Reactivity series and displacement	10	★★★
Amphoteric oxides (Al_2O_3, ZnO)	10	★★★
Thermit reaction	12	★★★
Corrosion and its prevention	13	★★★
Electron-dot formation of NaCl, $MgCl_2$	11	★★☆
Electrolytic refining, anode mud	12	★★☆
Roasting vs calcination	12	★★☆
Alloys and carats of gold	13	★★☆

Class Question: golden flame and Z = 17

Q. Element A gives a golden-yellow flame; B has atomic number 17. Identify A and B, their compound and its type of bonding.

- A = **sodium** (Na, 2, 8, 1); B = **chlorine** (Cl, 2, 8, 7)
- Na gives one electron to Cl: $Na \rightarrow Na^+ + e^-$, $Cl + e^- \rightarrow Cl^-$
- Compound **NaCl**, **ionic (electrovalent) bond**: high melting point, conducts electricity when molten or in water

Bonus rapid-fire round: quick answers

Best conductor of heat and electricity: **silver**. Non-metal liquid at room temperature: **bromine**. Metals that melt on your palm: **gallium, caesium**. Non-metal that conducts: **graphite**. Mixture that dissolves gold: **aqua regia (3 : 1)**. Coating of zinc on iron: **galvanisation**.

Common mistakes that cost marks

× WRONG	✓ RIGHT
Metal + HNO_3 gives H_2	HNO_3 oxidises the H_2 formed to water
Al_2O_3 is a basic oxide	Al_2O_3, ZnO: amphoteric
Cathode = impure metal	Anode = impure; cathode = thin strip of pure metal
Solid NaCl conducts	Only molten or dissolved NaCl conducts
Roasting uses limited air	Roasting: excess air (sulphides); calcination: limited air (carbonates)

Day 14 checklist: tick before you sleep

- ☐ I revised the mind map and all 7 must-know equations
- ☐ I attempted all 14 MCQs without looking at the answer key
- ☐ I wrote every short answer with balanced equations and state symbols
- ☐ I drew the electron-dot and refining diagrams neatly from memory
- ☐ I scored myself honestly and revised the days of my wrong answers

My score: ____ / 5

5/5 = Jalwa hai! 4 se kam? Revision skip kiya toh Boss Level pe cycle gir jaayegi!



LEVEL 14 COMPLETE **CHAPTER 3 COMPLETE**

+200 XP earned. Next: Level 15



BUJHO *Samajh gaya! Reactivity series yaad hai, toh extraction aur displacement dono yaad hain!*

PAHELI *Boss defeated! Kal se Chapter 4: carbon ki duniya.*

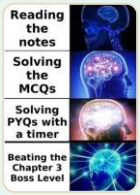
Coming up: Day 15

Carbon and its compounds: covalent bonds, why carbon is so versatile, and electron-dot structures of methane.



NITESH SIR SAYS

Chapter 3 khatam! Galtiyan note karo aur Sunday ko dobara test do. ***Consistency hi jalwa hai!***



CHEMISTRY | CHAPTER 3 COMPLETE

CHAPTER 3 SWAHAA!

Metals and Non-metals | Days 10–14



BUJHO *Sodium kerosene mein, reactivity series zubaan pe, rust se bachna bhi yaad. Chapter 3 Swahaa!*

PAHELI *Kamaal kar diya! Ab carbon ka jaadu seekhte hain, Day 15 se.*

CHAPTER XP EARNED
+625 XP

LEVELS CLEARED
5 levels (Days 10–14)

BOOK PROGRESS
Level 14 / 60 1750 XP total

WHAT YOU HAVE MASTERED

- ✔ Physical properties of metals and non-metals
- ✔ Reactivity series and amphoteric oxides
- ✔ Extraction, roasting, calcination, refining
- ✔ Reactions with oxygen, water and acids
- ✔ Ionic bonds and electron-dot structures
- ✔ Corrosion, its prevention and alloys

NEXT CHAPTER Chapter 4: Carbon and its Compounds (Day 15) *Bas ek chapter aur!*

DAY 15 · LEVEL 15

CHEMISTRY | CHAPTER 4 | PART 1 · ASTRA SERIES

CARBON & ITS COMPOUNDS

Goal: covalent bonds, electron-dot structures, allotropes.



by Nitesh Prakash

01 Covalent Bond | 02 Electron-dot | 03 Bond Types | 04 Allotropes | 05 Versatile Carbon | +100 XP today | Level 15 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 CARBON: THE BASICS

Carbon at a glance

Symbol, atomic number	**C, 6**
Electronic configuration	**K 2, L 4**
Valence electrons	**4** (valency 4); a non-metal
Earth's crust	Only 0.02% (carbonates, hydrogencarbonates, coal, petroleum)
Atmosphere	0.03% as CO_2
Found in	Food, clothes, medicines, fuels and our own body

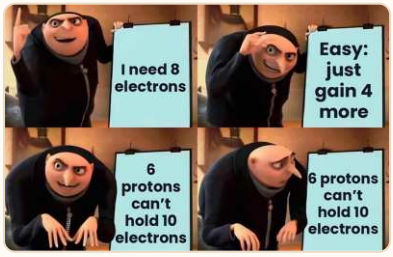
Where carbon occurs

Free state (as element)
- Diamond
- Graphite
- Fullerenes

Combined state (as compounds)
- CO_2 in air
- Carbonates: marble, limestone
- Fossil fuels; all living things

2 WHY CARBON SHARES ELECTRONS ★★★

GAIN 4 ELECTRONS	LOSE 4 ELECTRONS
Would form C^{4-} anion	Would form C^{4+} cation
× Nucleus with 6 protons cannot hold 10 electrons	× Removing 4 electrons needs a very large amount of energy



MEME BREAK
*Plan fail! Isliye carbon ne **sharing** chuni: 4 electrons share karo, octet complete.*

COVALENT BOND

- A bond formed by the **sharing of an electron pair** between two atoms
- Both atoms reach a **noble gas configuration** (octet; H gets a duplet)
- Carbon shares its 4 valence electrons: it is **tetravalent**

3 ELECTRON-DOT STRUCTURES ★★★ MOST ASKED

1. Write valence electrons: H 1, C 4, N 5, O 6, Cl 7.
2. Share pairs until every atom has 8 electrons (H has 2).
3. One shared pair = one bond (–). Two pairs (=), three pairs (≡).



Hydrogen, H_2
H–H: single bond



Chlorine, Cl_2
Cl–Cl: single bond



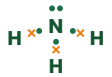
Oxygen, O_2
O=O: double bond



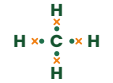
Nitrogen, N_2
N≡N: triple bond



Water, H_2O
2 single bonds, 2 lone pairs



Ammonia, NH_3
3 single bonds, 1 lone pair



Methane, CH_4
4 single C–H bonds



Carbon dioxide, CO_2
O=C=O: 2 double bonds

• electrons of one atom × electrons of the other atom. In each molecule, every atom ends with a full outer shell.

Solved Example: Drawing carbon dioxide

1. Valence electrons: C has 4, each O has 6.
2. C shares 2 electrons with each O, and each O shares 2 with C: two shared pairs per side.
3. Check: C has 4 pairs = 8 ✔; each O has 2 shared + 2 lone pairs = 8 ✔. Result: **O=C=O**.



★★★ *Methane aur carbon dioxide ka electron-dot structure har saal aata hai: Nitesh Bhaiya ki guarantee, roz ek baar bana ke dekho!*



BUJHO N_2 *mein bhi* O_2 *jaisa double bond hota hai na?*

PAHELI *Nahi! N ke 5 valence electrons, octet ke liye 3 share karta hai: N≡N, triple bond.*



Properties of covalent compounds

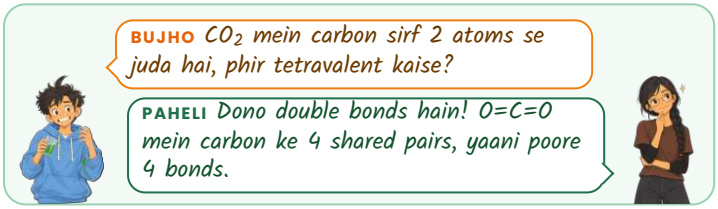
Compound	m.p. (K)	b.p. (K)
Acetic acid (CH_3COOH)	290	391
Chloroform ($CHCl_3$)	209	334
Ethanol (CH_3CH_2OH)	156	351
Methane (CH_4)	90	111

- **Low melting and boiling points:** forces between molecules are weak
- **Poor conductors of electricity:** no ions are formed

4 SINGLE, DOUBLE AND TRIPLE BONDS ★★☆

Bond	Shared pairs	Shared e^-	Examples
Single (–)	1	2	H–H, Cl–Cl, C–H in CH_4
Double (=)	2	4	O=O, O=C=O
Triple (≡)	3	6	N≡N

Between the same two atoms, strength grows with shared pairs: **triple** > **double** > **single**.



Solved Example: Counting covalent bonds

Q. How many covalent bonds are there in a molecule of pentane, C_5H_{12}?

- C–C bonds in a chain of 5 carbons = **4**
- C–H bonds = number of H atoms = **12**
- Total = 4 + 12 = **16 covalent bonds**

Shortcut for an alkane C_nH_{2n+2}: bonds = **3n + 1**. CH_4 4, C_2H_6 7, C_3H_8 10, C_5H_{12} 16 ✔



MEME BREAK
*Nahi Bujho! CH_4 mein koi ion nahi bante, electrons **share** hote hain. Yahi covalent compound hai.*

Exam tips

- ✔ Show only the **outer-shell (valence)** electrons in a dot structure.
- ✔ Show every shared pair **between** the two atoms; lone pairs stay on their own atom.
- ✔ Count after drawing: each atom has 8 electrons around it (H has 2).
- ✔ Write the line formula beside the dot structure: H–H, O=O, N≡N.

5 ALLOTROPES OF CARBON ★★★ MOST ASKED

ALLOTROPES

- Different **physical forms** of the same element
- Physical properties differ; **chemical properties are the same**
- Crystalline: **diamond, graphite, fullerene (C-60)**; amorphous: **coal, coke, charcoal, lampblack (soot)**

$C_{(s)} + O_{2(g)} \rightarrow CO_{2(g)} + \textbf{Heat}$

Diamond, graphite or fullerene: each burns to give only CO_2. Same element, same chemistry.

Property	Diamond	Graphite
Bonding	Each C bonded to **4** other C atoms	Each C bonded to **3** other C atoms
Structure	Rigid 3-D network of carbon atoms	Hexagonal layers held loosely, one above another
Hardness	**Hardest** natural substance known	Soft and slippery
Electricity	Non-conductor	**Good conductor**
Uses	Cutting and drilling tools, jewellery	Pencil lead, dry lubricant, electrodes



DIAMOND
Har carbon ke 4 dost, 3-D lock: isliye sabse hard!

GRAPHITE
Sirf 3 dost, layers phisal jaati hain: pencil mein ghis jaata hoon.

Fullerene (C-60): 60 carbon atoms in a **football** shape made of **20 hexagons + 12 pentagons**; named after the architect Buckminster Fuller.

Why graphite conducts but diamond does not

- Graphite: each C uses only 3 of its 4 valence electrons in bonds; the 4th is free to move, so it conducts
- Diamond: all 4 electrons are locked in bonds, no free electrons



MEME BREAK
*Jab pata chale ki chamakta diamond aur kaala pencil lead, dono **pure carbon** hain!*

Organic compounds: a historic surprise

- Carbon compounds were once thought to need a **"vital force"** found only in living things
- In 1828, **Friedrich Wöhler** made urea, an organic compound, from ammonium cyanate in the lab

$NH_4CNO \xrightarrow{\Delta} NH_2CONH_2$

ammonium cyanate → urea: same atoms (N 2, H 4, C 1, O 1), rearranged ✔

6 VERSATILE NATURE OF CARBON ★★★ MOST ASKED

CATENATION

- Carbon's unique ability to **form bonds with other carbon atoms**, giving large molecules
- The C–C bond is short and **very strong**, so long chains, branches and rings are stable
- **Class Question:** why less catenation in silicon? Si atoms are bigger, so the Si–Si bond is long and **weak**: Si–H chains stop at about **7–8 Si atoms** and are very reactive

LONG CHAIN	BRANCHED	RING
C–C–C–C–C	C–C–C–C | C	

TETRAVALENCY

- Valency 4: one carbon bonds with **4 other atoms**: C, H, O, N, S, Cl and more
- Small atom: the nucleus holds shared pairs strongly, so the bonds are **strong and stable**
- Result: about **three million** known carbon compounds



MEME BREAK

*Partner dhoondhne ka tension hi nahi: carbon **khud se hi** bond bana leta hai, isi ko kehte hain catenation!*

Common mistakes

× WRONG	✓ RIGHT
CO_2 drawn as O–C–O	O=C=O, two double bonds
N=N in nitrogen	N≡N, a triple bond
Diamond conducts too	Only graphite conducts
C forms C^{4+}/C^{4-} ions	C shares electrons (covalent)

TRY YOURSELF: DRAW THE ELECTRON-DOT STRUCTURES

(i) HCl
(ii) Br_2
(iii) Ethane, C_2H_6
(iv) Chloromethane, CH_3Cl

Short answer questions

1. Draw the electron-dot structures of (a) CH_4, (b) CO_2 and (c) N_2. Name the type of bond in each. NCERT
2. Why does carbon form covalent bonds? Give two reasons why covalent compounds have low melting points and do not conduct electricity. NCERT
3. Diamond and graphite are both pure carbon. Why is diamond hard and a non-conductor while graphite is soft and a good conductor?
4. Draw the electron-dot structure of a sulphur molecule made up of eight sulphur atoms. NCERT

7 PRACTICE MCQS

Neend aa rahi hai? Pehle ye 10 MCQs, answer key page 4 par hai!



1. The number of valence electrons in a carbon atom (atomic number 6) is:
(a) 2 (b) 4
(c) 6 (d) 8

2. The bond in a nitrogen molecule is: ★★★
(a) single covalent (b) double covalent
(c) triple covalent (d) ionic

3. Shared electron pairs in a CO_2 molecule: ★★★
(a) 2 (b) 3
(c) 4 (d) 6

4. Which of these is a good conductor of electricity? ★★☆
(a) Diamond (b) Graphite
(c) Sulphur (d) Phosphorus

5. The number of covalent bonds in pentane (C_5H_{12}) is: ★★☆
(a) 5 (b) 12
(c) 16 (d) 17

6. Carbon forms covalent bonds because it: ★★★
(a) can neither gain nor lose 4 electrons easily
(b) is a metal
(c) has a very large atom
(d) forms ions easily

7. The property of carbon atoms to link with one another to form long chains is called:
(a) tetravalency (b) catenation
(c) isomerism (d) allotropy

8. In diamond, each carbon atom is bonded to how many other carbon atoms?
(a) 2 (b) 3
(c) 4 (d) 6

9. **Assertion (A):** Diamond is the hardest natural substance.
Reason (R): In diamond, each carbon atom is bonded to four other carbon atoms, forming a rigid three-dimensional structure. ★★★
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

10. **Assertion (A):** Covalent compounds are generally poor conductors of electricity.
Reason (R): Covalent compounds have strong forces of attraction between their molecules. ★★☆
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

Answer key: MCQs

Q	Ans	Why
1	(b)	Configuration 2, 4: four outer electrons.
2	(c)	N shares 3 pairs: N≡N.
3	(c)	Two double bonds = 4 shared pairs.
4	(b)	Graphite has free electrons.
5	(c)	4 C–C + 12 C–H = 16.
6	(a)	C^{4-} and C^{4+} are both unstable.
7	(b)	Self-linking of carbon = catenation.
8	(c)	4 bonds each: rigid 3-D network.
9	(a)	The rigid network explains hardness.
10	(c)	Forces between molecules are weak.

Model answers: short answer

2 Why covalent, and why low m.p. and non-conducting

- C cannot gain 4 e^- (6 protons cannot hold 10 electrons) or lose 4 e^- (needs too much energy), so it **shares**
- Weak forces between molecules → **low m.p. and b.p.**
- No ions formed → **poor conductors**

3 Diamond vs graphite

- Diamond: each C bonded to 4 C in a rigid 3-D network, no free electrons → hard, non-conductor
- Graphite: each C bonded to 3 C in layers that slide; one free electron per C → soft, good conductor

4 S_8 molecule

- Eight S atoms form a **crown-shaped ring**
- Each S shares one pair with each neighbour (2 single bonds) and keeps **2 lone pairs**: 8 electrons around every S

Q1: see the electron-dot structures of CH_4 (4 single), CO_2 (2 double) and N_2 (triple) on page 1.

Chapter 4 hit-list: Eduspark PYQ analysis

Topic	Day	Rating
Electron-dot structures (CH_4, CO_2, N_2)	15	★★★
Diamond vs graphite	15	★★★
Catenation and tetravalency	15	★★★
Homologous series, isomers	16	★★★
Functional groups, naming	17	★★★
Ethanol, ethanoic acid, esters	18	★★★
Soaps, micelles, hard water	19	★★☆

Real life: carbon at work

- Diamond-tipped saws and drills cut stone and glass
- Graphite rods are electrodes in dry cells; graphite powder lubricates locks
- Your pencil "lead" is graphite mixed with clay, with no lead at all

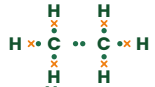
★ How we starred (Eduspark PYQ analysis): we counted how often each topic has come in Chapter 4 board and sample-paper questions. Year tags appear only where the paper year is known.

Try-yourself answers

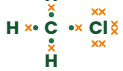


(i) H–Cl: 1 shared pair, 3 lone pairs on Cl

(ii) Br–Br: like Cl_2, 1 shared pair and 3 lone pairs on each Br



(iii) 7 single bonds: 1 C–C + 6 C–H



(iv) 3 C–H + 1 C–Cl; 3 lone pairs on Cl

Day 15 checklist: tick before you sleep

- ☐ I can explain why carbon forms covalent bonds
- ☐ I can draw electron-dot structures of H_2, O_2, N_2, CH_4 and CO_2
- ☐ I can tell single, double and triple bonds by shared pairs
- ☐ I can compare diamond, graphite and fullerene
- ☐ I solved all 10 MCQs and checked them with the key

My score: ____ / 5

5/5 = Jalwa hai! 4 se kam? Page 1–2 phir se padho.

LEVEL 15 COMPLETE +100 XP earned. Next: Level 16



BUJHO *Samajh gaya! Carbon sharing karta hai, aur diamond-graphite dono carbon hi hain!*

PAHELI *Shabaash! Kal Level 16 mein milte hain hydrocarbons se.*

Day 15 mind map: revise in 2 minutes

COVALENT BOND
Sharing of electron pairs; octet (H: duplet)

ELECTRON-DOT ★★★
Shared pairs shown as dots; CH_4, CO_2 and N_2 asked most

CARBON: PART 1

BOND TYPES
Shared electrons: single 2, double 4, triple 6

ALLOTROPES ★★★
Diamond (4 bonds), graphite (3 bonds), C-60

VERSATILE CARBON
Catenation + tetravalency = about three million compounds

Coming up: Day 16

Hydrocarbons: saturated and unsaturated, chains, branches and rings, isomers and the homologous series. Carbon's big family!



NITESH SIR SAYS
*Electron-dot structure roz ek baar khud banao. **Practice hi jalwa hai!***

DAY 16 | LEVEL 16

CHEMISTRY | CHAPTER 4 | PART 2 | ASTRA SERIES

HYDROCARBONS & HOMOLOGOUS SERIES

Goal: hydrocarbon types, isomers, homologous series.



by Nitesh Prakash

01 Saturated | 02 Unsaturated | 03 Chains & Rings | 04 Isomers | 05 Homologous Series | +100 XP today | Level 16 / 60

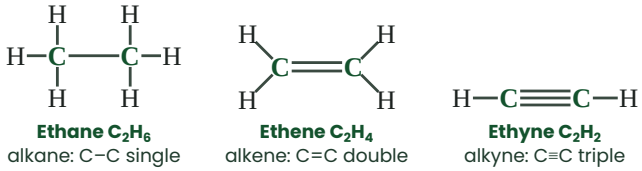
STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 HYDROCARBONS: SATURATED VS UNSATURATED

DEFINITION

- **Hydrocarbons** are compounds made of **carbon and hydrogen only**
- **Saturated:** carbon atoms joined by **single bonds only**
- **Unsaturated:** at least one carbon–carbon **double or triple bond**

The C_2 family: one, two or three bonds



Ethane C_2H_6 — alkane: C–C single | **Ethene C_2H_4** — alkene: C=C double | **Ethyne C_2H_2** — alkyne: C≡C triple

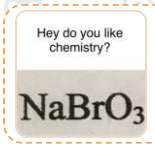
Every carbon still makes **4 bonds**. More bonds between the two carbons means **fewer hydrogens**.

Electron-dot link (Day 15): single bond = 1 shared pair (2 electrons), double = 2 pairs (4), triple = 3 pairs (6).

CLASS TRICK: COUNT THE BONDS

Double bond = 2, triple = 3. **Alkane = C + H − 1, Alkene = C + H, Alkyne = C + H + 1**. Check: C_2H_6 = 7, C_2H_4 = 6 (4 C–H + 2 in C=C), C_2H_2 = 5 (2 C–H + 3 in C≡C).

SATURATED ★★★	UNSATURATED
Only **single** C–C bonds	At least one **C=C** or **C≡C** bond
Alkanes: C_nH_{2n+2}	Alkenes C_nH_{2n}; alkynes C_nH_{2n-2}
Less reactive; substitution reactions	More reactive; addition reactions
Burn with a **clean blue** flame	Burn with a **sooty yellow** flame
Methane, ethane, propane	Ethene, propene, ethyne



★★★ *Alkane, alkene, alkyne ke general formula pakka aayenge, Nitesh Bhaiya ki guarantee hai! Yaad nahi kiya toh jawab hoga "Nah bro"!*

2 ALKANES, ALKENES AND ALKYNES

First five members: names and formulae

C atoms	Alkane	Alkene	Alkyne
Formula	C_nH_{2n+2}	C_nH_{2n}	C_nH_{2n-2}
1 (meth)	Methane CH_4	none	none
2 (eth)	Ethane C_2H_6	Ethene C_2H_4	Ethyne C_2H_2
3 (prop)	Propane C_3H_8	Propene C_3H_6	Propyne C_3H_4
4 (but)	Butane C_4H_{10}	Butene C_4H_8	Butyne C_4H_6
5 (pent)	Pentane C_5H_{12}	Pentene C_5H_{10}	Pentyne C_5H_8

Names: prefix for C count + ane / ene / yne.

Class Question: Why no methene or methyne? C=C or C≡C needs **two carbon atoms**; methane has only one. So the first alkene is **ethene (C_2H_4)**, the first alkyne **ethyne (C_2H_2)**.

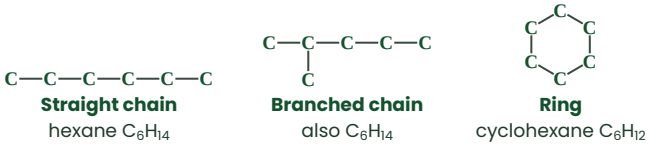
Solved Example 1: Formula from the general formula

(a) Alkyne with 5 C: n = 5, so H = 2(5) − 2 = 8. Formula: **C_5H_8** (pentyne).
(b) Which is saturated, C_3H_6 or C_4H_{10}? For n = 4, 2n + 2 = 10, so C_4H_{10} is a **saturated** alkane. C_3H_6 fits C_nH_{2n}: an alkene.
(c) Covalent bonds in an alkane = **3n + 1**. Ethane: 1 C–C + 6 C–H = **7**; propane 10; butane 13. ★★☆

3 CHAINS, BRANCHES AND RINGS

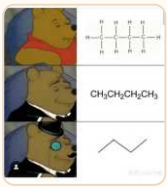
- Catenation lets carbon atoms join into **straight chains**, **branched chains** and **rings** (closed chains)
- Chains and rings can be saturated or unsaturated

Three ways six carbons can link



Straight chain hexane C_6H_{14} | **Branched chain** also C_6H_{14} | **Ring** cyclohexane C_6H_{12}

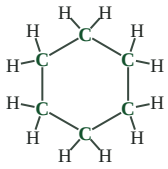
Carbon skeletons only; each C gets enough H to make 4 bonds.



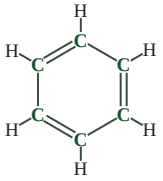
MEME BREAK

Butane ke teen roop: full structure, condensed $CH_3CH_2CH_2CH_3$ aur zig-zag. Molecule ek hi hai, bas likhne ka style alag!

Rings: cyclohexane vs benzene



Cyclohexane C_6H_{12}
all single bonds: saturated



Benzene C_6H_6
3 alternate C=C: unsaturated

- Cyclohexane: **2 H on every C**; saturated rings are **cycloalkanes**, formula C_nH_{2n}
- Benzene: **1 H on every C**; unsaturated rings like benzene are called **aromatic** compounds
- Cyclopentane C_5H_{10}: a five-carbon ring with only single bonds
- **Cycloalkenes C_nH_{2n-2}** (cyclohexene C_6H_{10}); **cycloalkynes C_nH_{2n-4}**



BUJHO *Cyclohexane mein ring hai, toh ye unsaturated hoga na?*

PAHELI *Nahi Bujho! Ring ka matlab unsaturated nahi. Cyclohexane mein sab single bonds hain, toh saturated. Benzene mein double bonds hain, isliye unsaturated.*

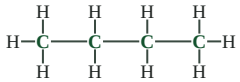


4 STRUCTURAL ISOMERS

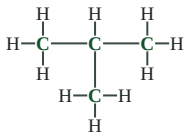
DEFINITION

- Compounds with the **same molecular formula** but **different structures** are **structural isomers**
- They have different physical properties, such as boiling points

Butane C_4H_{10}: two isomers



$CH_3–CH_2–CH_2–CH_3$
n-Butane (straight chain)



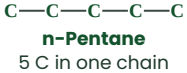
$CH_3–CH(CH_3)–CH_3$
Isobutane (2-methylpropane)

Both have 4 C and 10 H, but the carbons are **arranged differently**.

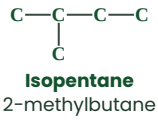


MEME BREAK
Isobutane, n-butane se: "Kaun? Main? Haan bhai, main bhi C_4H_{10} hoon, bas branch wala!"

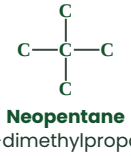
Pentane C_5H_{12}: three isomers ★★★



n-Pentane
5 C in one chain



Isopentane
2-methylbutane



Neopentane
2,2-dimethylpropane

Alkane	CH_4, C_2H_6, C_3H_8	C_4H_{10}	C_5H_{12}	C_6H_{14}
Isomers	1 each (none)	2	3	5

Isomerism in alkanes starts at butane (4 carbon atoms).

× WRONG	✔ RIGHT
Bending the chain into an L or zig-zag makes a new isomer	A bent drawing is the **same** compound; check the longest chain and the branches



BUJHO *Maine toh pentane ke 5 isomers bana diye, ab toh topper pakka?*

PAHELI *Do toh same nikle, bas chain mod di thi. Longest chain gino: 5 C, 4 C, 3 C. Pentane ke sirf 3 isomers!*



5 HOMOLOGOUS SERIES

DEFINITION

- A **homologous series** is a family of compounds with the **same functional group** (or same general formula) and similar chemical properties
- Successive members differ by a **$–CH_2–$** unit

Characteristics of a homologous series ★★★

- ✔ Same **general formula** for all members
- ✔ Successive members differ by **CH_2** and by **14 u** in mass
- ✔ **Similar chemical properties**: decided by the same functional group
- ✔ **Gradation in physical properties**: melting and boiling points rise as mass rises; solubility in a solvent changes in order
- ✔ Members can be made by similar general methods

Mass ladder: watch the 14 u steps

Alkane	Mass (u)	Alcohol	Mass (u)
CH_4	16	CH_3OH	32
C_2H_6	30	C_2H_5OH	46
C_3H_8	44	C_3H_7OH	60
C_4H_{10}	58	C_4H_9OH	74

Each step adds $–CH_2–$: 12 + 2 × 1 = **14 u**. (C = 12 u, H = 1 u, O = 16 u)

ISOMERS	HOMOLOGUES
Same molecular formula	Formulae differ by **CH_2**
Same molecular mass	Masses differ by **14 u**
n-butane and isobutane	Methane, ethane, propane

Solved Example 2: Series and next member

(a) Group these: CH_4, C_2H_4, C_3H_8, C_4H_8, C_2H_6.
- C_nH_{2n+2}: CH_4, C_2H_6, C_3H_8 (alkanes)
- C_nH_{2n}: C_2H_4, C_4H_8 (alkenes)

(b) Next homologue of C_3H_6 → add CH_2 → **C_4H_8**; of C_2H_5OH → **C_3H_7OH**.

(c) Trap: the **4th alkene** is pentene **C_5H_{10}**, not butene, because the alkene series starts at ethene. ★★☆

Formula finder: name the type in 3 steps

1. Count carbons: that number is **n**
2. Compare H with **2n + 2** (alkane), **2n** (alkene or cycloalkane), **2n − 2** (alkyne)
3. Example: C_4H_6 has n = 4 and 2n − 2 = 6, so it fits the **alkyne** formula (butyne)

TRY YOURSELF

Pehle khud karo, answer key page 4 par hai!



1. Write the formula of the alkene with 6 carbon atoms.
2. Pick the saturated ones: C_5H_{12}, C_5H_{10}, C_2H_2, C_6H_{12} (cyclohexane).
3. How many covalent bonds are there in pentane, C_5H_{12}?
4. Write the next homologue of CH_3OH and its molecular mass.

Common mistakes: avoid these

- "Ring means unsaturated": wrong; cyclohexane is saturated
- C_nH_{2n} fits **alkenes and cycloalkanes**; read the structure, not only the formula
- There is no "methene" or "methyne": these series start at 2 C
- Isomers have the **same** formula; homologues differ by CH_2

Ye galtiyan ki toh marks katenge, aur phir yahi face banega!



Exam tips

- ✔ Draw **every bond and every H**; check that each carbon shows exactly 4 bonds
- ✔ For isomers, give the **structure and the name** of each one
- ✔ Show the working for 14 u: $CH_2 = 12 + 2 \times 1$
- ✔ Write "C≡C" with three lines and "C=C" with two; a missing line changes the compound

Day 16 mind map: revise in 2 minutes

SATURATED
Alkanes C_nH_{2n+2} with single bonds only

UNSATURATED
Alkenes C_nH_{2n} and alkynes C_nH_{2n-2} have C=C or C≡C bonds

HYDROCARBONS

CHAINS & RINGS
Straight, branched, ring; C_6H_{12} and C_6H_6

ISOMERS ★★★
Same formula, new structure; butane 2, pentane 3

HOMOLOGOUS SERIES
Differ by CH_2 = 14 u; similar chemical, graded physical properties

6 PRACTICE MCQS

Padhai nahi ho rahi? 10 MCQs solve karo, dimaag apne aap start ho jayega!



1. The general formula of alkynes is: ★★★
(a) C_nH_{2n} (b) C_nH_{2n+2}
(c) C_nH_{2n-2} (d) C_nH_n

2. Which of these is an unsaturated hydrocarbon? ★★★
(a) C_3H_8 (b) C_2H_6
(c) C_3H_4 (d) C_4H_{10}

3. How many structural isomers does pentane have? NCERT ★★★
(a) 2 (b) 3
(c) 4 (d) 5

4. Formulae of cyclohexane and benzene, respectively: ★★☆
(a) C_6H_{12}, C_6H_6 (b) C_6H_6, C_6H_{12}
(c) C_6H_{14}, C_6H_6 (d) C_6H_{12}, C_6H_{12}

5. Which pair belongs to the same homologous series? ★★★
(a) CH_4, C_2H_4 (b) C_2H_6, C_3H_8
(c) C_2H_2, C_2H_4 (d) C_3H_6, C_3H_8

6. Which is NOT a feature of a homologous series? ★★☆
(a) Same general formula (b) Similar chemical properties
(c) Same physical properties (d) Members differ by CH_2

7. Number of single covalent bonds in propane, C_3H_8: ★★☆
(a) 8 (b) 10
(c) 11 (d) 12

8. n-Butane and isobutane are: ★★★
(a) homologues (b) structural isomers
(c) allotropes (d) identical compounds

9. **Assertion (A):** Benzene is an unsaturated hydrocarbon.
Reason (R): Cyclohexane has only single bonds in its ring. ★★☆
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

10. **Assertion (A):** Successive members of a homologous series differ in mass by 14 u.
Reason (R): They differ from each other by a $–CH_2–$ unit. ★★★
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

Short and long answer questions

1. How many structural isomers can you draw for pentane? Draw them and name each. (3 marks) NCERT
2. Give three differences between saturated and unsaturated hydrocarbons, with one example of each. (3 marks)
3. What is a homologous series? Write two of its characteristics. Give the formula of the fourth member of the alkene series. (3 marks)

Answer key: MCQs

Q	Ans	Why
1	(c)	C_nH_{2n-2}: C_2H_2, C_3H_4 and so on.
2	(c)	C_3H_4 fits C_nH_{2n-2}: propyne, a triple bond.
3	(b)	n-pentane, isopentane, neopentane.
4	(a)	Cyclohexane C_6H_{12}; benzene C_6H_6.
5	(b)	Both alkanes; they differ by one CH_2.
6	(c)	Physical properties change gradually; they are not the same.
7	(b)	2 C–C + 8 C–H = 10 (3n + 1 with n = 3).
8	(b)	Same formula C_4H_{10}, different structures.
9	(b)	Both true; benzene is unsaturated due to its C=C bonds, not because of cyclohexane.
10	(a)	CH_2 = 12 + 2 = 14 u explains the mass gap.

Model answers: short and long

1 Pentane, C_5H_{12}: three isomers

- **n-Pentane:** $CH_3–CH_2–CH_2–CH_2–CH_3$
- **Isopentane (2-methylbutane):** $CH_3–CH(CH_3)–CH_2–CH_3$
- **Neopentane (2,2-dimethylpropane):** $C(CH_3)_4$, four CH_3 groups on one central carbon

2 Saturated vs unsaturated

- **Bonds:** single C–C bonds only vs at least one C=C or C≡C
- **Formula:** C_nH_{2n+2} vs C_nH_{2n} or C_nH_{2n-2}
- **Flame and reactivity:** clean blue flame, less reactive vs sooty yellow flame, more reactive (addition reactions)
- **Examples:** ethane C_2H_6 vs ethene C_2H_4

3 Homologous series

- Compounds with the same functional group or general formula, where successive members differ by $–CH_2–$
- Characteristics: masses differ by 14 u; similar chemical properties, graded physical properties
- Alkenes: ethene, propene, butene, **pentene C_5H_{10}** is the 4th member

Day 16 hit-list: Eduspark PYQ analysis

Topic	Rating
Isomers of butane and pentane	★★★
Homologous series: definition, CH_2, 14 u	★★★
General formulae of alkanes, alkenes, alkynes	★★★
Saturated vs unsaturated (flame test)	★★☆
Cyclohexane and benzene structures	★★☆
Counting covalent bonds	★☆☆

Coming up: Day 17

Functional groups (–OH, –CHO, –COOH and more) and IUPAC naming: how propanol, propanal, propanone and propanoic acid get their names.

Day 16 checklist: tick before you sleep

- [ ] I can tell saturated from unsaturated and draw ethane, ethene and ethyne
- [ ] I can write alkane, alkene and alkyne formulae from the general formulae
- [ ] I can draw straight, branched and ring structures, including benzene
- [ ] I can draw the 2 isomers of butane and the 3 isomers of pentane
- [ ] I solved all 10 MCQs and checked them with the key

My score: ____ / 5

5/5 = Jalwa hai! 4 se kam? Page 1–2 phir se padho.

LEVEL 16 COMPLETE +100 XP earned. Next: Level 17



BUJHO *Samajh gaya! Ring ho ya chain, sirf single bonds hain toh saturated. Same formula, alag structure: isomers!*

PAHELI *Shabaash! Kal Level 17: functional groups aur IUPAC naming ki baari.*

Try-yourself answers

1. Hexene, C_6H_{12} (n = 6 in C_nH_{2n})
2. Saturated: C_5H_{12} (alkane) and cyclohexane C_6H_{12} (single bonds in a ring)
3. Pentane: 3(5) + 1 = **16** bonds (4 C–C + 12 C–H)
4. C_2H_5OH (ethanol): 24 + 5 + 16 + 1 = **46 u**

Real life: hydrocarbons around you

- **CNG** is mainly methane; **LPG** is mainly butane (with propane)
- **Petrol**, kerosene and **candle wax** are mixtures of alkanes
- **Ethyne** (acetylene) burns in oxygen in welding torches
- Kekulé proposed the benzene ring structure in 1865

EXTRA KNOWLEDGE (HIGH LEVEL)

Not in current CBSE syllabus. Types of isomers:

- **Chain:** butane / 2-methylpropane (C_4H_{10})
- **Position:** propan-1-ol / propan-2-ol (C_3H_8O)
- **Functional:** ethanol C_2H_5OH / dimethyl ether CH_3OCH_3
- **Stereo:** same bonds, different 3-D arrangement



NITESH SIR SAYS

*Roz 5 minute structures haath se draw karo. **Jo draw karega, wahi yaad rakhega!** Kal milte hai Day 17 pe.*

★ How we starred (Eduspark PYQ analysis): we counted how often each topic has come in Chapter 4 board and sample-paper questions. Year tags appear only where the paper year is known.

DAY 17 · LEVEL 17

CHEMISTRY | CHAPTER 4 | PART 3 · ASTRA SERIES

FUNCTIONAL GROUPS & IUPAC NAMES

Goal: spot the group, then name the compound.



by Nitesh Prakash

01 Heteroatoms | 02 Group Table | 03 Word Roots | 04 Naming Rules | 05 Examples | +100 XP today | Level 17 / 60

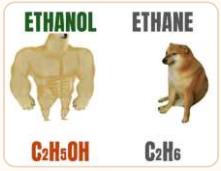
STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 HETEROATOMS & FUNCTIONAL GROUPS

DEFINITION

- **Heteroatom:** any atom other than C and H (**O, N, S, Cl, Br**) that replaces H in a carbon chain
- **Functional group:** the heteroatom or group of atoms that gives the compound its **characteristic chemical properties**
- It joins the chain through its **free valency** (the line in –OH), in place of one H

HYDROCARBON $CH_3–CH_3$ ethane (a gas) → replace one H by –OH → **ALCOHOL** $CH_3–CH_2–OH$ ethanol (a liquid)



MEME BREAK

*Ek H hatao, –OH lagao: gas wala ethane ban gaya liquid ethanol! Naya group matlab **naye properties**.*

Key idea: what the group decides

- ✔ **Same functional group → similar chemical properties.** Methanol, ethanol and propanol all react with sodium.
- ✔ **Chain length** changes mainly physical properties, such as melting and boiling points.



BUJHO *Ethanol mein –OH hai, toh ye NaOH jaisa base hoga?*

PAHELI *Nahi! Yahan –OH carbon se covalent bond se juda hai, hydroxide ion nahi deta. Ethanol litmus ka rang nahi badalta.*



Common names you will hear

IUPAC name	Common name	Met in
methanal	formaldehyde	formalin
ethanol	ethyl alcohol	sanitisers
ethanoic acid	acetic acid	vinegar (5–8% solution)
propanone	acetone	nail-polish remover
methanoic acid	formic acid	ant stings

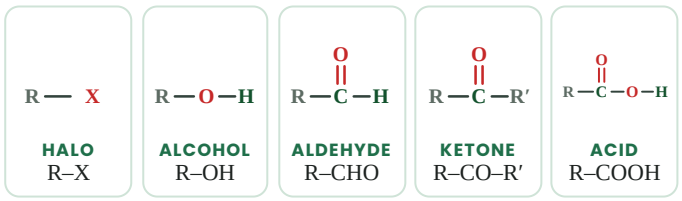
2 FUNCTIONAL GROUPS TABLE ★★★ MOST ASKED

The five groups you must know

Class	Group	Ending	Example
Halo	**–Cl, –Br**	**chloro-, bromo-**	CH_3Cl, chloromethane
Alcohol	**–OH**	**-ol**	CH_3OH, methanol
Aldehyde	**–CHO**	**-al**	HCHO, methanal
Ketone	**>C=O**	**-one**	CH_3COCH_3, propanone
Carboxylic acid	**–COOH**	**-oic acid**	CH_3COOH, ethanoic acid
Alkene, alkyne	**C=C, C≡C**	**-ene, -yne**	$CH_2=CH_2$, ethene

Halo is written as a prefix; all the others are suffixes. C=C and C≡C are not heteroatoms, but are named with an ending in the same way.

How each group is drawn



R, R′ = rest of the carbon chain; X = Cl or Br. Aldehyde: C=O at the **end** of the chain (bonded to H). Ketone: C=O in the **middle** (bonded to two carbons).



★★★ *Group, ending aur example: ye table exam ka favourite hai, Nitesh Bhaiya ki guarantee!*

SHORTCUT: THE ENDING TELLS THE FAMILY

-ol alcohol, -al aldehyde, -one ketone, -oic acid carboxylic acid.

3 WORD ROOTS: COUNT THE CARBONS

C atoms	1	2	3	4	5
Root	Meth	Eth	Prop	But	Pent
C atoms	6	7	8	9	10
Root	Hex	Hept	Oct	Non	Dec

Memory trick: "**M**onkeys **E**at **P**eeled **B**ananas" for 1 to 4 carbons. From 5 onwards the roots are number words you already know: pentagon, hexagon, octopus, decade.

4 IUPAC NAMING RULES ★★★ MOST ASKED

IUPAC (International Union of Pure and Applied Chemistry) names have four parts:

PREFIX	+	WORD ROOT	+	BOND	+	SUFFIX
chloro-, bromo- for halogens		meth, eth, prop… (no. of C)		-ane, -ene, -yne (single, =, ≡)		-ol, -al, -one, -oic acid

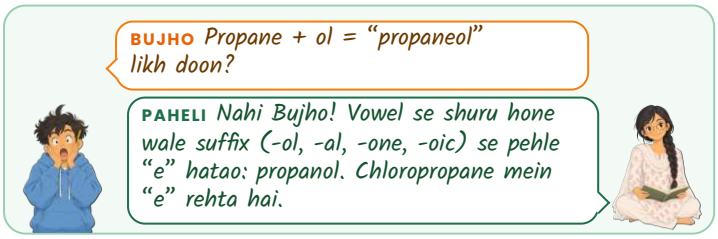
chloro + **prop** + **ane** = chloropropane | **prop** + **an** + **ol** = propanol

Read every name backwards: suffix = group, middle = bond type, root = carbon count

Four steps to name any compound

1. Count the carbons in the **longest chain**, including the C of –CHO or –COOH. 3 C → root **prop**.
2. Check the bonds: all single → **-ane**; one C=C → **-ene**; one C≡C → **-yne**.
3. Add the group. A suffix starting with a vowel **drops the final "e"**: propan**e** + ol = **propanol**. A halogen goes in front as a prefix: **chloro**propane.
4. Number the chain from the end **nearer the group**, e.g. propan-1-ol, 1-chloropropane. For Class 10 the plain name is enough unless asked.

× WRONG	✔ RIGHT
propaneol, propaneal	propanol, propanal (drop the "e")
CH_3CHO = methanal	ethanal: the CHO carbon counts, so 2 C
methanone, ethanone	propanone is the simplest ketone (3 C)
propanechloro	chloropropane (halogen = prefix)



Exam tips

- ✔ The carbon of –CHO and –COOH is **part of the chain**: HCOOH is methanoic acid, CH_3COOH is ethanoic acid.
- ✔ A ketone needs **at least 3 carbons**: its C=O carbon must sit between two carbons.
- ✔ Halogens are always a **prefix**; all other groups here are a **suffix**.
- ✔ When a structure is asked, write the condensed formula ($CH_3–CH_2–CHO$), not just the molecular formula (C_3H_6O).

REMEMBER: THE SIMPLEST MEMBER

Methanol, methanal, methanoic acid and chloromethane have 1 C; ethene and ethyne 2 C; **propanone needs 3 C**.

5 WORKED NAMES: THE PROPANE FAMILY

Solved Example 1: Formula → name ★★★

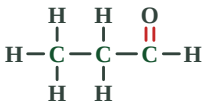
Formula	Group	Build-up	IUPAC name
$CH_3–CH_2–CH_2–Cl$	Halo	chloro + propane	**chloropropane**
$CH_3–CH_2–CH_2–OH$	Alcohol	propan(e) + ol	**propanol**
$CH_3–CH_2–CHO$	Aldehyde	propan(e) + al	**propanal**
$CH_3–CO–CH_3$	Ketone	propan(e) + one	**propanone**
$CH_3–CH_2–COOH$	Carboxylic acid	propan(e) + oic acid	**propanoic acid**
$CH_3–CH=CH_2$	Alkene	prop + ene	**propene**
$CH_3–C≡CH$	Alkyne	prop + yne	**propyne**
$CH_3–CH_2–Br$	Halo	bromo + ethane	**bromoethane**



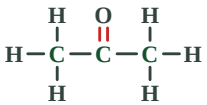
MEME BREAK

*Propanal (CH_3CH_2CHO) aur propanone (CH_3COCH_3): dono ka formula **C_3H_6O**! Group alag, isliye naam aur properties alag.*

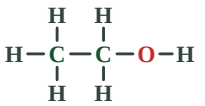
Solved Example 2: Full structural formulae ★★★



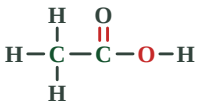
PROPANAL
$CH_3–CH_2–CHO$



PROPANONE
$CH_3–CO–CH_3$



ETHANOL
$CH_3–CH_2–OH$



ETHANOIC ACID
$CH_3–COOH$

Every carbon shows 4 bonds and every H one bond. Propanal and propanone have the same formula, C_3H_6O, but different groups.

Solved Example 3: Name → structure ★★★

Q. Draw the structures of ethanoic acid, bromopentane, butanone and hexanal. Are structural isomers possible for bromopentane? NCERT

Name	Condensed formula	Molecular
Ethanoic acid	$CH_3–COOH$	$C_2H_4O_2$
Bromopentane	$CH_3–CH_2–CH_2–CH_2–CH_2–Br$	$C_5H_{11}Br$
Butanone	$CH_3–CO–CH_2–CH_3$	C_4H_8O
Hexanal	$CH_3–CH_2–CH_2–CH_2–CH_2–CHO$	$C_6H_{12}O$

Isomers: yes: 1-, 2- and 3-bromopentane (Br position changes), plus branched chains (how to name them: Extra Knowledge box, page 4).



MEME BREAK

*–CHO aur –COOH ka carbon bhi chain mein gina jaata hai! Isliye CH_3COOH = 2 C = **ethanoic acid**.*

Day 17 mind map: revise in 2 minutes

HETEROATOM
O, N, S, Cl or Br in place of H; the group sets the reactions

WORD ROOT
Meth 1, Eth 2, Prop 3, But 4, Pent 5 … Dec 10

FUNCTIONAL GROUPS & NAMES

ENDINGS ★★★
-ol (OH), -al (CHO), -one (C=O), -oic acid (COOH)

PREFIX
chloro-, bromo- in front: chloropropane, bromoethane

BONDS
-ane (single), -ene (C=C), -yne (C≡C); drop the final "e" before a vowel suffix: propanol, propanal, propanone

Families: one group, many members ★★☆

C	Alcohols (-ol)	Aldehydes (-al)	Acids (-oic acid)
1	methanol	methanal	methanoic acid
2	ethanol	ethanal	ethanoic acid
3	propanol	propanal	propanoic acid
4	butanol	butanal	butanoic acid
5	pentanol	pentanal	pentanoic acid

Each member differs from the next by **CH_2** (14 u): a homologous series. The 2 C members: C_2H_5OH, CH_3CHO, CH_3COOH. Ketones start at 3 C: propanone CH_3COCH_3, butanone $CH_3COC_2H_5$.

Solved Example 4: Name these ★★☆

Q. How would you name: (i) $CH_3–CH_2–Br$ (ii) HCHO (iii) $HC≡C–CH_2–CH_2–CH_2–CH_3$? NCERT

- (i) 2 C, halogen Br → **bromoethane**
- (ii) 1 C, aldehyde group → **methanal**
- (iii) 6 C, one triple bond → **hexyne** (hex-1-yne), C_6H_{10}

Also: $CH_3CH_2CH_2COOH$ (4 C, –COOH) is **butanoic acid**, and $CH_3COCH_2CH_3$ (4 C, C=O in the middle) is **butanone**.

TRY YOURSELF: NAME OR DRAW

(i) $CH_3CH_2CH_2CH_2OH$ (ii) $CH_3CH_2CH_2CHO$
(iii) $CH_3CH_2CH_2CH_2COOH$ (iv) $CH_3CH_2CH_2CH_2Cl$
(v) $CH_3CH_2CH_2Br$ (vi) Draw ethanal
(vii) Draw pentanal (viii) Draw methanoic acid

Answers are on page 4, below the model answers.

Short and long answer questions

1. What is a functional group? Name the group present in, and give the IUPAC name of: (i) $CH_3CH_2CH_2OH$ (ii) CH_3CH_2COOH (iii) $CH_3COCH_2CH_3$ (iv) CH_3CH_2CHO **(3 marks)**
2. Write the condensed structural formulae of: (i) propene (ii) propyne (iii) chloroethane (iv) pentanoic acid. **(2 marks)**
3. Why is CH_3COCH_3 named propanone and not propanal? Explain why the simplest ketone must have three carbon atoms. **(3 marks)**

6 PRACTICE MCQS

Kyun nahi ho rahi padhai? Phone side mein rakho, 10 MCQs khud solve karo!



1. The functional group present in propanal is: ★★★
(a) –OH (b) –CHO
(c) >C=O (d) –COOH

2. The IUPAC name of CH_3COCH_3 is: ★★★
(a) propanal (b) propanone
(c) propanol (d) ethanone

3. The name ending used for a carboxylic acid is: ★★☆
(a) -ol (b) -al
(c) -one (d) -oic acid

4. The IUPAC name of $CH_3–CH_2–Br$ is: NCERT ★★★
(a) bromomethane (b) bromoethane
(c) bromopropane (d) ethanebromo

5. Which of the following compounds is a ketone? ★★☆
(a) CH_3CHO (b) CH_3COOH
(c) $CH_3COCH_2CH_3$ (d) CH_3CH_2OH

6. How many carbon atoms does one hexanal molecule have? ★☆☆
(a) 4 (b) 5
(c) 6 (d) 7

7. Propanal and propanone have the same: ★★☆
(a) functional group (b) molecular formula, C_3H_6O
(c) name ending (d) chemical properties

8. The correct formula of ethanoic acid is: ★★★
(a) HCOOH (b) CH_3COOH
(c) CH_3CH_2COOH (d) CH_3CHO

9. **Assertion (A):** Methanol and ethanol show similar chemical properties.
Reason (R): Both contain the same functional group, –OH. ★★★
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

10. **Assertion (A):** The IUPAC name of CH_3CHO is methanal.
Reason (R): The carbon atom of the –CHO group is counted in the parent chain. ★★☆
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

Case-based question (4 marks)

A teacher wrote four compounds on the board: **A** CH_3CH_2OH, **B** CH_3CHO, **C** CH_3COCH_3 and **D** CH_3COOH.

1. Name the functional group present in B and in C. **(1)**
2. Write the IUPAC names of A and D. **(1)**
3. Which compound, as a 5–8% solution in water, is sold as vinegar? Write the formula and IUPAC name of its next higher homologue. **(2)**

Answer key: MCQs

Q	Ans	Why
1	(b)	–CHO is the aldehyde group; propan**al**.
2	(b)	C=O in the middle: ketone, propan**one**.
3	(d)	Acids end in -oic acid, e.g. ethanoic acid.
4	(b)	2 C + Br as prefix: bromoethane.
5	(c)	C=O between two carbons: butanone.
6	(c)	Hex = 6; the CHO carbon is included.
7	(b)	Both are C_3H_6O; groups differ.
8	(b)	2 C including the COOH carbon.
9	(a)	Same group → similar properties.
10	(d)	CH_3CHO is ethanal; R is true.

Model answers: short and long answer

1 Functional group and names

The atom or group of atoms that decides the chemical properties of an organic compound.

Compound	Group	IUPAC name
$CH_3CH_2CH_2OH$	alcohol, –OH	**propanol**
CH_3CH_2COOH	carboxylic acid, –COOH	**propanoic acid**
$CH_3COCH_2CH_3$	ketone, >C=O	**butanone**
CH_3CH_2CHO	aldehyde, –CHO	**propanal**

2 Condensed formulae

(i) $CH_3–CH=CH_2$ (ii) $CH_3–C\equiv CH$
(iii) $CH_3–CH_2–Cl$
(iv) $CH_3–CH_2–CH_2–CH_2–COOH$, pentanoic acid

3 Propanone, not propanal

- In $CH_3–CO–CH_3$, the C=O carbon is **in the middle**, bonded to two carbons: that makes it a **ketone** (-one)
- In propanal, $CH_3–CH_2–CHO$, the C=O is at the **end** and bonded to H: an **aldehyde** (-al)
- A ketone's C=O carbon needs a carbon on **each side**, so at least 3 C: propanone is the simplest

Try-yourself answers

(i) butanol (ii) butanal
(iii) pentanoic acid (iv) chlorobutane
(v) bromopropane (vi) $CH_3–CHO$
(vii) $CH_3–CH_2–CH_2–CH_2–CHO$ (viii) H–COOH

Case-based answers

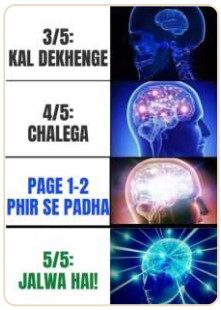
- (i) B: aldehyde group, –CHO; C: ketone group, >C=O
- (ii) A: ethanol; D: ethanoic acid
- (iii) D, ethanoic acid. Next homologue: CH_3CH_2COOH, **propanoic acid**

Coming up: Day 18

Chemical properties: combustion, oxidation, addition and substitution, plus the two stars of the chapter, ethanol and ethanoic acid.

Day 17 checklist: tick before you sleep

- ☐ I can define a heteroatom and a functional group
- ☐ I can write the five groups with their formula and name ending
- ☐ I know the word roots meth to dec for 1 to 10 carbons
- ☐ I can name a compound from its formula and draw it from its name
- ☐ I solved all 10 MCQs and checked them with the key



My score: ____ / 5

5/5 = Jalwa hai! 4 se kam? Page 1–2 phir se padho.

LEVEL 17 COMPLETE +100 XP earned. Next: Level 18



BUJHO *Samajh gaya! Root se carbon gino, ending se group pehchano: propanal, propanone, propanoic acid, sab clear!*

PAHELI *Shabaash! Kal Level 18: chemical properties of carbon compounds.*

Day 17 hit-list: Eduspark PYQ analysis

Topic	Rating
Functional group table (group, ending, example)	★★★
Naming propanal, propanone, propanol, propanoic acid	★★★
Drawing a structure from its name	★★★
Halo prefix: chloro-, bromo-	★★☆
Word roots and -ane, -ene, -yne	★★☆
Common names: acetone, acetic acid, formalin	★☆☆
Simplest ketone needs 3 C (propanone)	★★☆

Extra Knowledge (High Level) not in current CBSE syllabus

Naming branched alkanes: (1) longest carbon chain = parent; (2) number from the end nearest a branch (lowest locants); (3) branch name with its position, di-/tri- for repeats, different branches in alphabetical order.

$CH_3–CH(CH_3)–CH_3$	**2-methylpropane**
$CH_3–CH(CH_3)–CH(CH_3)–CH_3$	**2,3-dimethylbutane**
$CH_3–CH(CH_3)–CH(C_2H_5)–CH_2–CH_3$	**3-ethyl-2-methylpentane**



NITESH SIR SAYS
*Naam ratna nahi, banana seekho: root + bond + ending. **Kal milte hai Day 18 pe!***

★ How we starred (Eduspark PYQ analysis): we counted how often each topic has come in Chapter 4 board and sample-paper questions. Year tags appear only where the paper year is known.

DAY 18 · LEVEL 18

CHEMISTRY | CHAPTER 4 | PART 4 · ASTRA SERIES

PROPERTIES OF CARBON COMPOUNDS

Goal: four key reactions, ethanol and ethanoic acid.



by Nitesh Prakash

01 Combustion | 02 Oxidation | 03 Addition | 04 Substitution | 05 Ethanol & Ethanoic Acid | +125 XP today | Level 18 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 COMBUSTION ★★★ MOST ASKED

DEFINITION

- Carbon and its compounds **burn in oxygen** to give **$CO_2 + H_2O$**
- A lot of **heat and light** is released, so most carbon compounds are used as **fuels**

$$C_{(s)} + O_{2(g)} \rightarrow CO_{2(g)} + \text{Heat and light}$$
C 1 = 1, O 2 = 2 ✔

$$CH_{4(g)} + 2O_{2(g)} \rightarrow CO_{2(g)} + 2H_2O_{(l)} + \text{Heat and light}$$
C 1 = 1, H 4 = 4, O 4 = 4 ✔

$$CH_3CH_2OH_{(l)} + 3O_{2(g)} \rightarrow 2CO_{2(g)} + 3H_2O_{(l)} + \text{Heat}$$
ethanol burns with a clean flame, giving heat and light
C 2 = 2, H 6 = 6, O 7 = 7 ✔

SATURATED (e.g. CH_4)	UNSATURATED (e.g. C_2H_2)
✔ Clean **blue flame**	× **Yellow, sooty** flame with black smoke
Burn completely in air	Burn incompletely: higher carbon share
Gas stove, LPG, CNG	Saturated ones also go sooty when air is limited

Flame facts: why vessels turn black

- Black base of a cooking vessel = **air holes blocked**, fuel wasted; clean them for a blue flame
- Yellow glow = **unburnt carbon particles** heated till they glow
- A **flame forms only when gases burn**: wax and LPG give a flame, coal and charcoal just glow red
- Incomplete burning also gives **poisonous CO**; sulphur and nitrogen in coal give acidic oxides

TRY YOURSELF: BALANCE THESE COMBUSTIONS

(i) $C_2H_6 + O_2 \rightarrow CO_2 + H_2O$ (ii) $C_3H_8 + O_2 \rightarrow CO_2 + H_2O$

Balance C, then H, then O last. Answers on page 4.



★★★ *Thanos Baba ka gyaan: combustion ki equation bhi perfectly balanced honi chahiye. Ye pakka aayega, Nitesh Bhaiya ki guarantee!*

2 OXIDATION ★★★

DEFINITION

- Controlled addition of oxygen: **alcohol → carboxylic acid**
- **Oxidising agents** are substances that **give oxygen** to others: alkaline $KMnO_4$, acidified $K_2Cr_2O_7$

$$CH_3CH_2OH + 2[O] \xrightarrow[\text{or acidified } K_2Cr_2O_7 + \text{Heat}]{\text{Alkaline } KMnO_4 + \text{Heat}} CH_3COOH + H_2O$$
ethanol → ethanoic acid; [O] = oxygen from the oxidising agent
C 2 = 2, H 6 = 6, O 3 = 3 ✔

Activity: warm ethanol + alkaline permanganate

- Add 5% alkaline $KMnO_4$ drop by drop to warm ethanol
- **Purple colour disappears** at first: $KMnO_4$ is used up in oxidising ethanol
- Excess added: colour **stays**, no ethanol left to oxidise
- Acidified $K_2Cr_2O_7$: **orange → green**

3 ADDITION (HYDROGENATION) ★★★

DEFINITION

- **Unsaturated** compounds (C=C or C≡C) **add hydrogen** to become saturated
- Needs a **catalyst: nickel (Ni) or palladium (Pd)**

$$CH_2{=}CH_2 + H_2 \xrightarrow[\text{Heat}]{\text{Ni catalyst}} CH_3{-}CH_3$$
ethene → ethane: C 2 = 2, H 6 = 6 ✔

Use: vegetable oil → vanaspati ghee

- Vegetable oils (long unsaturated chains) + H_2, Ni catalyst → **solid vanaspati ghee**
- **Catalyst:** changes the rate of a reaction without being used up itself
- Animal fats are **saturated** and bad for the heart; vegetable oils (unsaturated) are the healthier choice for cooking



MEME BREAK

*Iron Man = Fe-male. Toh hydrogenation ka superhero? **Ni-Man!** Nickel catalyst ke bina tel kabhi ghee nahi banta.*

4 SUBSTITUTION ★★☆

DEFINITION

- Saturated hydrocarbons are **fairly unreactive**
- In **sunlight**, chlorine **replaces hydrogen atoms one by one**; excess Cl_2 gives CH_2Cl_2, $CHCl_3$, CCl_4

$$CH_4 + Cl_2 \xrightarrow{\text{Sunlight}} CH_3Cl + HCl$$

methane → chloromethane: C 1 = 1, H 4 = 4, Cl 2 = 2 ✔



MEME BREAK
Substitution = displacement ka carbon version: Cl andar aaya, H bahar. Bechara H bhi ab Cu ki tarah side mein baitha hai!

Four reactions at a glance

Reaction	Who reacts	Condition	Product
Combustion	almost all	O_2, burn	$CO_2 + H_2O$ + heat
Oxidation	alcohols	alk. $KMnO_4$ or acid. $K_2Cr_2O_7$, heat	carboxylic acid
Addition	unsaturated	H_2, Ni or Pd	saturated
Substitution	saturated	Cl_2, sunlight	$CH_3Cl + HCl$



BUJHO *Combustion aur oxidation alag kaise? Dono mein oxygen hi toh hai!*

PAHELI *Combustion = jalna, $CO_2 + H_2O$ + heat. Oxidation = $KMnO_4$ ki madad se ethanol → ethanoic acid, bina jale!*



5 ETHANOL ★★★

Properties and uses of ethanol

- Colourless **liquid** at room temperature; **soluble in water** in all proportions
- Active ingredient of alcoholic drinks; good **solvent** in tincture of iodine, cough syrups and tonics
- Neutral to litmus; burns with a clean flame, so it is **added to petrol** as a cleaner fuel

Reaction 1: ethanol + sodium

$$2Na + 2CH_3CH_2OH \rightarrow 2CH_3CH_2ONa + H_2\uparrow$$

sodium ethoxide + hydrogen: Na 2 = 2, C 4 = 4, H 12 = 12, O 2 = 2 ✔

Test: the gas burns with a **"pop" sound** (hydrogen). Ethanol gives **no** fizz with $NaHCO_3$.

Reaction 2: dehydration (removal of water)

$$CH_3CH_2OH \xrightarrow[443\ K]{\text{Hot conc. } H_2SO_4} CH_2{=}CH_2 + H_2O$$

ethanol → ethene: C 2 = 2, H 6 = 6, O 1 = 1 ✔

Conc. H_2SO_4 is a **dehydrating agent**: it pulls water out of ethanol.

Harmful effects and denatured alcohol

- Small amounts cause drunkenness; regular drinking leads to **serious health problems**
- **Methanol** (CH_3OH) even in small amounts can kill: in the liver it is oxidised to **methanal**, which coagulates cell protoplasm and damages the optic nerve (**blindness**)
- **Denatured alcohol** = industrial ethanol + methanol (poison) + **blue dye**, so that it cannot be misused for drinking

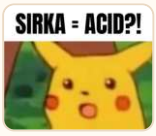


MEME BREAK
*Methanol ka ek ghoont: aankhein aur jaan dono khatre mein. Ethanol ki aadat: liver aur dil dono bolein **ouch!***

6 ETHANOIC ACID (ACETIC ACID) ★★★

Key facts

- Common name **acetic acid**; a **weak acid** (ionises only partly), turns blue litmus **red**
- **Vinegar = 5–8% ethanoic acid in water**: preservative in pickles and a flavouring
- **Other uses:** making cellulose acetate (rayon, photographic film), dyes, plastics and esters for perfumes; coagulating rubber latex
- Pure acid: melting point **290 K**, so it freezes in cold weather: **glacial acetic acid**



MEME BREAK
*Sirke ka khatta taste asal mein 5–8% ethanoic acid hai? **Acha!?** Aur pura acid sardi mein barf ki tarah jam jaata hai!*

Esterification: acid + alcohol → ester

$$CH_3COOH + CH_3CH_2OH \xrightarrow[\text{warm}]{\text{Conc. } H_2SO_4} CH_3COOCH_2CH_3 + H_2O$$

ethyl ethanoate (ester) + water: C 4 = 4, H 10 = 10, O 3 = 3 ✔

- Esters are **sweet-smelling**: used in **perfumes** and as **flavouring agents**
- The reaction is **slow and reversible**; conc. H_2SO_4 speeds it up

Saponification: ester + base → back to alcohol

$$CH_3COOCH_2CH_3 + NaOH \rightarrow CH_3CH_2OH + CH_3COONa$$

ethanol + sodium ethanoate: C 4 = 4, H 9 = 9, O 3 = 3, Na 1 = 1 ✔

Saponification = the reaction used to **make soap** (Day 19).

Ethanoic acid with bases and carbonates

$$CH_3COOH_{(aq)} + NaOH_{(aq)} \rightarrow CH_3COONa_{(aq)} + H_2O_{(l)}$$

neutralisation: sodium ethanoate (sodium acetate) + water ✔

$$2CH_3COOH_{(aq)} + Na_2CO_{3(s)} \rightarrow 2CH_3COONa_{(aq)} + H_2O_{(l)} + CO_{2(g)}\uparrow$$

C 5 = 5, H 8 = 8, O 7 = 7, Na 2 = 2 ✔

$$CH_3COOH_{(aq)} + NaHCO_{3(s)} \rightarrow CH_3COONa_{(aq)} + H_2O_{(l)} + CO_{2(g)}\uparrow$$

C 3 = 3, H 5 = 5, O 5 = 5, Na 1 = 1 ✔

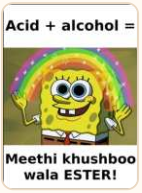
Brisk effervescence; the CO_2 turns lime water milky. This is how you tell **ethanoic acid from ethanol**.



BUJHO *Acid + alcohol se ester bana, toh ye neutralisation hua na?*

PAHELI *Nahi! Neutralisation = acid + **base**. Acid + alcohol = **esterification**, conc. H_2SO_4 catalyst, meethi khushboo!*





MEME BREAK

*Chapter 1: toxic gas + toxic solid = namak. Aaj ka version: khatta acid + alcohol = **meethi khushboo wala ester!** ★★★ Esterification aur saponification pakke, Nitesh Bhaiya ki guarantee!*

7 EXAM TIPS AND COMMON MISTAKES

× WRONG	✓ RIGHT
Saturated → sooty flame	**Unsaturated** → sooty flame
Ethanol + Na → ethanoate	Sodium **ethoxide** + H_2
Hydrogenation catalyst: Fe	**Ni or Pd**
Ester + NaOH = esterification	**Saponification**
$CH_4 + Cl_2$ in the dark	Needs **sunlight**

Exam tips

- ✔ Always write the **condition over the arrow**: Ni, sunlight, conc. H_2SO_4 + 443 K, alkaline $KMnO_4$ + heat.
- ✔ Organic conversions (oxidation, addition, substitution, esterification) are written NCERT style without state symbols; the condition and product names carry the marks.
- ✔ "Distinguish ethanol and ethanoic acid": $NaHCO_3$ test (fizz with acid only) or blue litmus (red with acid only).
- ✔ Ester names: ethanoic acid + ethanol → **ethyl ethanoate**.

8 MIND MAP: REVISE IN 2 MINUTES

Day 18 mind map

COMBUSTION ★★★
$CO_2 + H_2O$ + heat; saturated blue flame, unsaturated sooty

OXIDATION
Ethanol → ethanoic acid; alkaline $KMnO_4$ or acidified $K_2Cr_2O_7$

CHEMICAL PROPERTIES OF CARBON COMPOUNDS

ADDITION
Unsaturated + H_2, Ni or Pd; oil → vanaspati ghee

SUBSTITUTION
Saturated + Cl_2 in sunlight; CH_3Cl + HCl

ETHANOL AND ETHANOIC ACID
Ethanol + Na → ethoxide + H_2; conc. H_2SO_4, 443 K → ethene. Acid + alcohol → ester (esterification); ester + NaOH → alcohol + salt (saponification); acid + $NaHCO_3$ → CO_2 fizz

Quick quiz: name the reaction

(i) Oil → ghee (ii) $CH_4 \rightarrow CH_3Cl$ (iii) Ethanol → acid
(iv) Ester + NaOH (v) Ethanol → ethene (vi) LPG burning

Answers: (i) addition (ii) substitution (iii) oxidation (iv) saponification (v) dehydration (vi) combustion

Extra from Class: ethanol as a fuel

Molasses (from sugarcane juice) is fermented by **yeast**: $C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2$ (enzyme zymase). Ethanol burns with a clean blue flame ($C_2H_5OH + 3O_2 \rightarrow 2CO_2 + 3H_2O$), so it is blended with petrol as a cleaner fuel (India's **E20**; Brazil).

9 PRACTICE MCQS

Pehle khud socho! Answer key page 4 par hai.



1. Which of these burns with a yellow, sooty flame? ★★★
(a) Methane (b) Ethane
(c) Ethyne (d) Propane

2. The conversion of ethanol into ethanoic acid by alkaline $KMnO_4$ is an example of: ★★★
(a) addition (b) substitution
(c) oxidation (d) reduction

3. Which catalyst is used in the hydrogenation of vegetable oils to make vanaspati ghee? ★★★
(a) Iron (b) Nickel
(c) Conc. H_2SO_4 (d) Platinum oxide

4. $CH_4 + Cl_2 \rightarrow CH_3Cl + HCl$ in sunlight is a: ★★☆
(a) addition reaction (b) substitution reaction
(c) combustion reaction (d) neutralisation

5. Sodium metal reacts with ethanol to give: ★★★
(a) sodium ethoxide + H_2 (b) sodium ethanoate + H_2
(c) sodium ethoxide + O_2 (d) sodium ethanoate + CO_2

6. Ethanol is heated at 443 K with excess hot conc. H_2SO_4. The product is: ★★☆
(a) ethane (b) ethene
(c) ethyne (d) ethanal

7. Vinegar is a solution of: ★★☆
(a) 50–60% ethanoic acid (b) 5–8% ethanoic acid
(c) 5–8% ethanol (d) pure ethanoic acid

8. Ethyl ethanoate is heated with NaOH solution to give ethanol and sodium ethanoate. This reaction is called: ★★★
(a) esterification (b) saponification
(c) hydrogenation (d) dehydration

9. **Assertion (A):** Unsaturated hydrocarbons burn in air with a yellow, sooty flame.
Reason (R): Unsaturated hydrocarbons have a higher share of carbon, so they do not burn completely in air. ★★★
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

10. **Assertion (A):** Methane reacts readily with chlorine in the dark to form chloromethane.
Reason (R): Substitution of hydrogen by chlorine in saturated hydrocarbons takes place in the presence of sunlight. ★★☆
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

Short and long answer questions

1. How would you distinguish experimentally between an alcohol and a carboxylic acid? (2 marks) NCERT
2. Why is the conversion of ethanol to ethanoic acid an oxidation reaction? Name two oxidising agents. (2 marks) NCERT
3. What is esterification? Describe an activity to show it, with the equation. What happens when the ester formed is heated with NaOH solution? Name this reaction. (5 marks)

Answer key: MCQs and try-yourself

Q	Ans	Why
1	(c)	Ethyne C_2H_2 is unsaturated (triple bond).
2	(c)	Oxygen is added: alcohol → acid.
3	(b)	Ni (or Pd) catalyses hydrogenation.
4	(b)	Cl replaces one H atom.
5	(a)	$2Na + 2C_2H_5OH \rightarrow 2C_2H_5ONa + H_2$
6	(b)	Dehydration gives ethene, $CH_2{=}CH_2$.
7	(b)	Vinegar = 5–8% ethanoic acid.
8	(b)	Ester + base → alcohol + salt.
9	(a)	More carbon, incomplete burning: soot.
10	(d)	No reaction in the dark; sunlight needed.
Try	(i) $2C_2H_6 + 7O_2 \rightarrow 4CO_2 + 6H_2O$ (ii) $C_3H_8 + 5O_2 \rightarrow 3CO_2 + 4H_2O$	

Scored 8 or more? Move on. Below 6? Revise sections 1 to 6 once more, then retry.

Model answers: short and long answer

1 Alcohol vs carboxylic acid

- Add a pinch of **$NaHCO_3$** to each: brisk **effervescence** of CO_2 (turns lime water milky) only with the **acid**
- Blue litmus turns red only with the acid; the alcohol shows no change

$$CH_3COOH_{(aq)} + NaHCO_{3(s)} \rightarrow CH_3COONa_{(aq)} + H_2O_{(l)} + CO_{2(g)}\uparrow$$

2 Ethanol → ethanoic acid is oxidation

- Oxygen is **added** (and hydrogen removed): $CH_3CH_2OH \rightarrow CH_3COOH$
- Oxidising agents: **alkaline $KMnO_4$** and **acidified $K_2Cr_2O_7$**, on heating

3 Esterification and saponification

- **Esterification:** a carboxylic acid reacts with an alcohol in the presence of an acid catalyst to form a sweet-smelling **ester**
- **Activity:** 1 mL ethanol + 1 mL glacial acetic acid + a few drops of conc. H_2SO_4; warm in a water bath for 5 min; pour into 20–50 mL water: **fruity smell**

$$CH_3COOH + CH_3CH_2OH \xrightarrow[\text{warm}]{\text{Conc. } H_2SO_4} CH_3COOCH_2CH_3 + H_2O$$

- Heated with NaOH, the ester gives back the alcohol and the sodium salt of the acid: **saponification** (used to make soap)

$$CH_3COOCH_2CH_3 + NaOH \rightarrow CH_3CH_2OH + CH_3COONa$$

Day 18 hit-list: Eduspark PYQ analysis

Topic / reaction	Rating
Esterification + saponification equations	★★★
Ethanol → ethanoic acid (oxidising agents)	★★★
Hydrogenation of oils (Ni catalyst)	★★★
Clean vs sooty flame	★★★
Ethanol + Na; acid + $NaHCO_3$	★★☆
Substitution of methane in sunlight	★★☆

Day 18 checklist: tick before you sleep

- [ ] I can write combustion equations and explain clean vs sooty flames
- [ ] I can name the oxidising agents for ethanol → ethanoic acid
- [ ] I can explain hydrogenation (Ni or Pd) and substitution (sunlight)
- [ ] I can write ethanol + Na, dehydration, esterification and saponification
- [ ] I solved all 10 MCQs and checked them with the key

My score: ____ / 5

5/5 = Jalwa hai! 4 se kam? Page 1–2 phir se padho, phir hi sona.



LEVEL 18 COMPLETE +125 XP earned. Next: Level 19



BUJHO *Samajh gaya! Acid + alcohol se meetha ester, aur ester + NaOH se wapas alcohol!*

PAHELI *Shabaash! Kal Level 19: soaps and detergents.*

Coming up: Day 19

Soaps and detergents: how micelles trap oily dirt and why soap fails in hard water.

USES AT A GLANCE

Ethanol: medicines and fuel additive; ethanoic acid: vinegar and pickles; esters: perfumes and flavours; hydrogenation: vanaspati ghee; saponification: soap.



NITESH SIR SAYS

Har reaction ki condition arrow ke upar likho, wahi marks dilati hai. ***Kal milte hai Day 19 pe!***

★ How we starred (Eduspark PYQ analysis): we counted how often each reaction or topic has come in Chapter 4 board and sample-paper questions. Year tags appear only where the paper year is known.

DAY 19 · LEVEL 19

CHEMISTRY | CHAPTER 4 | PART 5 · ASTRA SERIES

SOAPS & DETERGENTS

Goal: explain micelles, scum and soap vs detergent.



by Nitesh Prakash

01 Soap Structure | 02 Micelles | 03 Hard Water | 04 Soap vs Detergent | 05 Chapter Recap | +125 XP today | Level 19 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 SOAPS AND DETERGENTS

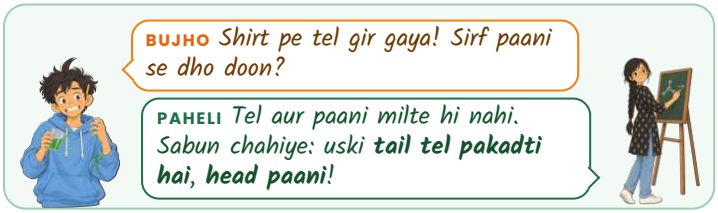
DEFINITION

- **Soap:** the **sodium or potassium salt** of a **long-chain carboxylic acid** (fatty acid)
- Examples: sodium stearate $C_{17}H_{35}COO^-Na^+$, sodium palmitate $C_{15}H_{31}COO^-Na^+$
- **Detergent:** generally the **sodium salt of a sulphonic acid** (or an ammonium salt with chloride or bromide ions), also with a long hydrocarbon chain

Structure of a soap molecule ★★★



- **Tail** (long hydrocarbon chain): **hydrophobic**, water-hating, dissolves in oil and grease
- **Head** (ionic $-COO^-Na^+$ end): **hydrophilic**, water-loving, dissolves in water



Recap: how soap is made (saponification) ★★☆

$$(C_{17}H_{35}COO)_3C_3H_5(s) + 3NaOH(aq) \xrightarrow{\Delta} 3C_{17}H_{35}COONa(s) + C_3H_5(OH)_3(aq)$$

fat + caustic soda → soap + glycerol; C 57, H 113, O 9, Na 3 on each side ✔

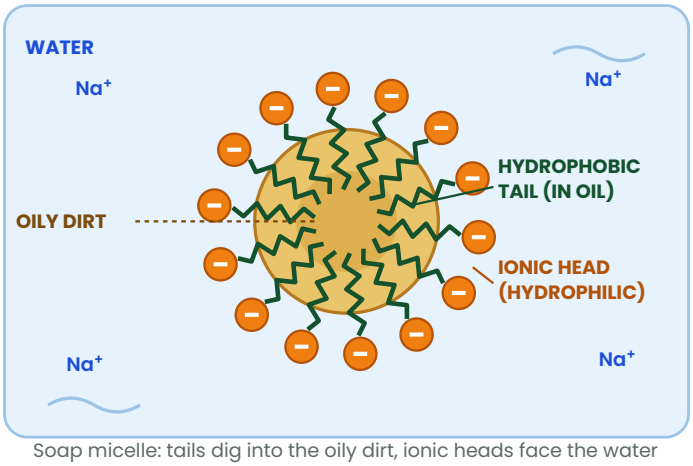
Alkaline hydrolysis of an ester (fat or oil) with NaOH gives soap; KOH gives soft (potassium) soap.



MEME BREAK

Soap molecule to oil and water both: "Humare liye kaam karoge?" Tail bole tel, head bole paani. ***Dono se dosti!***

2 MICELLES AND CLEANSING ACTION ★★★ MOST ASKED



Soap micelle: tails dig into the oily dirt, ionic heads face the water

DEFINITION

- **Micelle:** a **spherical cluster** of soap molecules in water, with **tails inside** and **ionic heads outside**

How soap cleans: 4 steps

1. In water, soap molecules arrange so the **tails stay away from water** and the heads stay in it.
2. The tails dissolve in the **oily dirt**; the dirt is trapped at the **centre of the micelle**.
3. The ionic heads on the outside **repel each other**, so micelles stay suspended (an emulsion) and do not come together.
4. Rinsing, scrubbing or beating the cloth **washes the micelles away** with the dirt.



★★★ *Micelle ka labelled diagram + cleansing action pakka aayega: Nitesh Bhaiya ki guarantee!*

Why does soap solution look cloudy? ★★☆

- Micelles are **large enough to scatter light**, so a soap solution looks cloudy (it is a colloid)
- In **ethanol**, soap dissolves completely (the tail is soluble in ethanol), so **no micelles form**

3 HARD WATER AND SCUM ★★★ MOST ASKED

SOFT WATER	HARD WATER
Few or no Ca^{2+}/Mg^{2+} salts	Contains **calcium and magnesium salts** (hydrogencarbonates, chlorides, sulphates)
Rain water, distilled water	Well water, hand-pump water
✔ Soap gives plenty of **lather**	× Little lather; white curdy **scum** forms

Scum: the soap-wasting precipitate ★★★

- **Scum** = insoluble **calcium or magnesium salts of fatty acids**, formed when soap reacts with Ca^{2+} and Mg^{2+}

$$2C_{17}H_{35}COONa_{(aq)} + CaCl_{2(aq)} \rightarrow (C_{17}H_{35}COO)_2Ca_{(s)}\downarrow + 2NaCl_{(aq)}$$

scum (calcium stearate). C 36 = 36, H 70 = 70, O 4 = 4, Na 2 = 2, Ca 1 = 1, Cl 2 = 2 ✔

$$2C_{17}H_{35}COONa_{(aq)} + MgSO_{4(aq)} \rightarrow (C_{17}H_{35}COO)_2Mg_{(s)}\downarrow + Na_2SO_{4(aq)}$$

magnesium salts of hard water give scum the same way ✔

- Soap is **wasted** (it reacts first, cleans later); scum **sticks to clothes**, buckets and tiles



MEME BREAK
Soap to the Ca^{2+} and Mg^{2+} of hard water: "Kya tapleek hai aapko? Mera saara jhaag kha gaye!"

Activity: Test for hard water ★★☆

1. Take 10 mL distilled (soft) water in test tube A and 10 mL well or hand-pump (hard) water in test tube B.
2. Add 5 drops of **soap solution** to each and shake for the same time.
3. A: **lots of lather**. B: little lather, **curdy white scum**. Soap needed for lather in B is much more.
4. Repeat with **detergent**: lather forms in **both**. So a detergent cannot tell hard water from soft.

Result: to check hardness, always use **soap**, not detergent.

Exam tips: score full marks ★★★

1. Micelle diagram: draw **tails inside, heads outside** and write all **4 labels**.
2. Write "**insoluble** calcium or magnesium salts" for scum; the word insoluble carries the mark.
3. Name the ions: Ca^{2+} and Mg^{2+}, not "salts" alone.
4. In soap vs detergent, give **ionic group** and **hard water** points first.
5. Soap solution is **slightly basic** (salt of a weak acid and a strong base): red litmus turns blue.

4 DETERGENTS: WHY THEY WIN IN HARD WATER

- A detergent also has a **long hydrocarbon tail** and an **ionic head**, such as $-SO_3^-Na^+$
- Its charged end does **not form insoluble salts** with Ca^{2+} and Mg^{2+}, so **no scum** forms
- So detergents **lather well even in hard water**; they clean by the same **micelle** action
- Uses: washing powders, **shampoos**, products for cleaning clothes and dishes



BUJHO *Aachoo! Itna detergent kyun? Sabun kaafi nahi?*

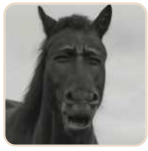
PAHELI *Hand-pump ka paani hard hai: sabun scum banata hai. **Detergent ko calcium-magnesium se farak nahi padta!***



5 SOAP VS DETERGENT ★★★

Point	Soap	Detergent
Chemical nature	Na/K salt of a long-chain **carboxylic** (fatty) acid	Na salt of a long-chain **sulphonic** acid (or ammonium salt)
Ionic group	$-COO^-Na^+$	$-SO_3^-Na^+$
Hard water	× Forms scum, little lather	✔ No scum, lathers well
Made from	Vegetable oils, animal fats	Petroleum hydrocarbons
Cleansing action	Relatively weaker	Stronger
Biodegradable?	✔ Yes	Some are **not**; foam pollutes rivers and ponds

Detergent, jab sabun hard water mein scum banata hai: Ha ha ha! Mera to jhaag ban raha hai!



Spot the type: soap or detergent?

Formula	Type
$C_{17}H_{35}COO^-Na^+$ (sodium stearate)	**Soap**
$C_{15}H_{31}COO^-K^+$ (potassium palmitate)	**Soap**
$C_{12}H_{25}C_6H_4SO_3^-Na^+$ (sodium dodecylbenzene sulphonate)	**Detergent**

Rule: $-COO^-$ head means soap; $-SO_3^-$ head means detergent.

Link to Chapter 2: softening hard water

- **Washing soda** (sodium carbonate) removes **permanent hardness**: it precipitates Ca^{2+} and Mg^{2+} as insoluble carbonates
- After softening, soap lathers easily again

Real life: where you meet them

- Bathing soap and laundry bar: **soap**; shampoo, washing powder, dishwash liquid: **detergents**

6 CHAPTER 4 QUICK RECAP

Chapter 4 recap mind map: revise in 3 minutes

CARBON AND ITS COMPOUNDS

BONDING (DAY 15)
C = 6 (2, 4): tetravalent; forms **covalent bonds** by sharing; H–H, O=O, N≡N; CH_4 four single bonds; O=C=O

ALLOTROPES
Diamond: each C to 4 C, hardest, non-conductor. Graphite: 3 C in layers, conductor. Fullerene C-60

VERSATILE CARBON
Catenation (long chains, branches, rings) + **tetravalency** → millions of compounds

HYDROCARBONS (DAY 16)
Alkane C_nH_{2n+2}, alkene C_nH_{2n}, alkyne C_nH_{2n-2}; benzene C_6H_6, cyclohexane C_6H_{12}

ISOMERS AND SERIES
Butane 2 isomers, pentane 3. Homologous series: next member + CH_2 = 14 u

FUNCTIONAL GROUPS (DAY 17)
–Cl/–Br halo; –OH -ol; –CHO -al; >C=O -one; –COOH -oic acid; C=C -ene; C≡C -yne

REACTIONS (DAY 18)
Combustion; oxidation (alkaline $KMnO_4$ or acidified $K_2Cr_2O_7$); addition (Ni or Pd); substitution (Cl_2, sunlight)

ETHANOL AND ETHANOIC ACID
Na gives H_2; hot concentrated H_2SO_4 (443 K) gives ethene; vinegar is 5–8% ethanoic acid; esterification and saponification

SOAPS AND DETERGENTS (DAY 19)
Ionic head + hydrocarbon tail; **micelle** traps oily dirt; soap + Ca^{2+}/Mg^{2+} → scum; detergents work in hard water

Must-know equations of Chapter 4

$$CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(l) + \text{Heat}$$
combustion: C 1 = 1, H 4 = 4, O 4 = 4 ✔

$$2Na(s) + 2CH_3CH_2OH(l) \rightarrow 2CH_3CH_2ONa(aq) + H_2(g)\uparrow$$
ethanol + sodium: Na 2 = 2, C 4 = 4, H 12 = 12, O 2 = 2 ✔

$$CH_3COOH(l) + C_2H_5OH(l) \xrightarrow{\text{acid catalyst}} CH_3COOC_2H_5(l) + H_2O(l)$$
esterification

$$CH_3CH_2OH(l) + 2[O] \xrightarrow{\text{alkaline } KMnO_4 + \text{heat}} CH_3COOH(l) + H_2O(l)$$
oxidation: C 2 = 2, H 6 = 6, O 3 = 3 ✔

$$CH_3COOH(aq) + NaHCO_3(s) \rightarrow CH_3COONa(aq) + H_2O(l) + CO_2(g)\uparrow$$
C 3 = 3, H 5 = 5, O 5 = 5, Na 1 = 1 ✔

7 COMMON MISTAKES

× WRONG	✔ RIGHT
Soap = salt of a sulphonic acid	Soap is a carboxylate salt
The ionic head dissolves in oil	Tail in oil, ionic head in water
Use detergent to test hard water	Use soap, not detergent
All detergents are eco-friendly	Some are non-biodegradable
Micelles clump and settle	They repel, stay suspended

8 PRACTICE MCQS

Mr Bean jaisa focus! Khud solve karo, answer key page 4 par hai.



1. Soaps are chemically: ★★★
(a) fatty acid salts (b) sulphonic acid salts
(c) glycerol esters (d) long-chain alcohols

2. In a soap micelle, the hydrocarbon tails point: ★★★
(a) towards the water outside
(b) towards the oily dirt at the centre
(c) in random directions
(d) towards the air above

3. Soap does not lather well in hard water because it contains: ★★★
(a) Na^+ and K^+ ions
(b) Ca^{2+} and Mg^{2+} ions
(c) Cl^- ions only
(d) dissolved oxygen

4. The scum formed with hard water is: ★★☆
(a) sodium chloride
(b) glycerol
(c) Ca or Mg salt of a fatty acid
(d) a detergent

5. Which substance would you use to check whether a water sample is hard? ★★☆
(a) detergent (b) soap
(c) ethanol (d) common salt

6. A soap solution looks cloudy because: ★☆☆
(a) soap is insoluble in water (b) micelles scatter light
(c) it contains scum (d) it is acidic in nature

7. The hydrophilic end of a soap molecule is: ★★☆
(a) $–COO^-Na^+$ (b) $–SO_3^-Na^+$
(c) $–CH_3$ (d) $–C_{17}H_{35}$

8. Soap forms no micelles in ethanol because: ★★☆
(a) soap reacts with ethanol to form an ester
(b) soap dissolves completely in ethanol
(c) ethanol contains Ca^{2+} ions
(d) ethanol is an ionic liquid

9. **Assertion (A):** Detergents are better than soaps for washing clothes in hard water.
Reason (R): The calcium and magnesium salts of detergents are soluble in water. ★★★
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

10. **Assertion (A):** Soap removes oily dirt from clothes.
Reason (R): The ionic end of soap dissolves in oil, and the hydrocarbon end dissolves in water. ★★★
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

Short and long answer questions

1. Explain the formation of scum when hard water is treated with soap. Write a balanced equation. (2 marks) NCERT
2. Would you be able to check if water is hard by using a detergent? Give a reason. (2 marks) NCERT
3. Why does micelle formation take place when soap is added to water? Will a micelle be formed in ethanol also? (3 marks) NCERT
4. Draw a labelled diagram of a soap micelle and explain the cleansing action of soap. Why are detergents preferred in hard water? (5 marks) ★★★

Answer key: MCQs

Q	Ans	Why
1	(a)	Soap = Na/K salt of a fatty acid.
2	(b)	Tails are hydrophobic and dissolve in oil.
3	(b)	Ca^{2+}, Mg^{2+} make hard water.
4	(c)	Insoluble Ca or Mg stearate, for example.
5	(b)	Detergent lathers in both waters.
6	(b)	Micelles are big enough to scatter light.
7	(a)	Carboxylate head; (b) is a detergent head.
8	(b)	Tail dissolves in ethanol, so no clustering.
9	(a)	Soluble Ca/Mg salts, so no scum.
10	(c)	R is reversed: tail in oil, head in water.

Model answers

1 Scum formation

- Hard water has Ca^{2+} and Mg^{2+} ions; soap reacts with them to give an **insoluble, curdy precipitate** (scum)
- Lather forms only after these ions are used up, so soap is wasted

$$2C_{17}H_{35}COONa_{(aq)} + CaCl_{2(aq)} \rightarrow (C_{17}H_{35}COO)_2Ca_{(s)}\downarrow + 2NaCl_{(aq)}$$

2 Detergent and hard water

- **No.** A detergent lathers well in both hard and soft water, because its Ca and Mg salts are soluble; only soap shows the difference

3 Why micelles form

- The hydrocarbon tail is **insoluble in water**, so the molecules cluster with tails inside and ionic heads facing the water
- **Not in ethanol:** the tail is soluble in ethanol, so soap simply dissolves

4 Cleansing action (5 marks)

- Draw the micelle from page 1 with all 4 labels: oily dirt, hydrophobic tail, ionic head, water
- Tails trap oily dirt at the centre, heads face water; rinsing removes the micelles with the dirt
- Detergents form no scum with Ca^{2+}/Mg^{2+}, so they clean well in hard water

SUGGESTED MARKING FOR Q4

Diagram with 4 labels: 2 marks; micelle formation: 1; removal of trapped dirt: 1; detergents in hard water: 1.

Day 19 checklist: tick before you sleep

- ☐ I can draw a soap molecule and label its head and tail
- ☐ I can draw and label a micelle and explain how it removes oily dirt
- ☐ I can write the scum equation and explain why soap fails in hard water
- ☐ I can give four differences between soaps and detergents
- ☐ I solved all 10 MCQs and checked them with the key

My score: ____ / 5

5/5 = Jalwa hai! 4 se kam? Kyun nahi ho rahi padhai? Page 1–2 phir se padho.



LEVEL 19 COMPLETE +125 XP earned. Next: Level 20



BUJHO *Samajh gaya! Tail tel pakadti hai, head paani; aur hard water mein detergent hi chalega!*

PAHELI *Shabaash! Kal Level 20 hai: Boss Level aur poora Chemistry mock test.*

Chapter 4 hit-list: Eduspark PYQ analysis

Topic	Day	Rating
Electron-dot structures (CH_4, CO_2, N_2)	15	★★★
Homologous series and isomers	16	★★☆
Functional groups and IUPAC names	17	★★★
Ethanol and ethanoic acid reactions	18	★★★
Micelles and cleansing action	19	★★★
Soap vs detergent; hard water	19	★★★
Allotropes of carbon	15	★★☆

Coming up: Day 20

Boss Level for Chapter 4 plus a full Chemistry mock test covering Days 1–20, with complete answers. Revise today's mind map and the must-know equations first!



NITESH SIR SAYS

*Micelle ka diagram teen baar banao, exam mein haath khud chalega. **Consistency hi jalwa hai!***

★ How we starred (Eduspark PYQ analysis): we counted how often each topic has come in Chapter 4 board and sample-paper questions. NCERT tags mark NCERT in-text questions.

DAY 20
LEVEL 20

CHEMISTRY | CHAPTER 4 | BOSS LEVEL + MOCK TEST

BOSS LEVEL + CHEMISTRY MOCK TEST

Goal: clear Chapter 4, then test all four chemistry chapters.



by Nitesh Prakash

01 Boss Level | 02 Mock Test | 03 Answer Key | 04 Marking Points | 05 Checklist | +200 XP today | Level 20 / 60

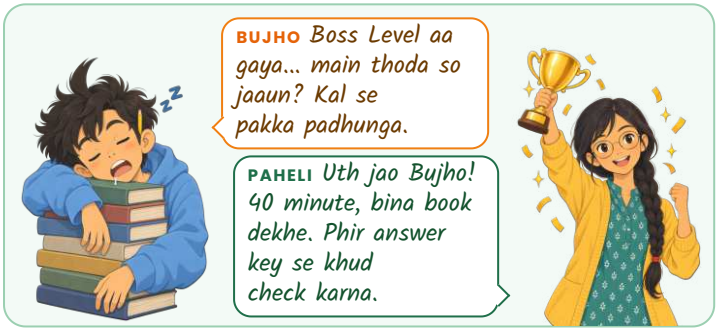
STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 CHAPTER 4 BOSS LEVEL

Time: 40 min | **Max. marks:** 24 | **Questions:** 14
Q1–10: 1 mark each; Q11–13: 3 marks each; Q14: 5 marks

TIME PLAN FOR 40 MINUTES

Q1–10 in 10 min, Q11–13 in 15 min, Q14 in 10 min, then 5 min to re-check equations and state symbols.



SECTION A: OBJECTIVE — 10 × 1 = 10 marks

1. The total number of covalent bonds in a molecule of propane, C_3H_8, is: ★★★
(a) 8 (b) 10 (c) 11 (d) 13

2. Which belongs to the same homologous series as C_3H_6? ★★★
(a) C_4H_8 (b) C_3H_8 (c) C_2H_2 (d) C_4H_{10}

3. The functional group present in propanal is: ★★☆
(a) –OH (b) –CHO (c) –COOH (d) >C=O

4. The number of electron pairs shared between the two atoms in a nitrogen molecule, N_2, is: ★★☆
(a) 1 (b) 2 (c) 3 (d) 6

5. Which of these is an unsaturated hydrocarbon? ★★☆
(a) C_2H_6 (b) C_3H_8 (c) C_3H_4 (d) C_4H_{10}

6. Ethanol reacts with sodium metal to form: ★★★
(a) sodium ethanoate + H_2 (b) sodium ethoxide + H_2
(c) sodium ethoxide + O_2 (d) sodium ethanoate + O_2

7. Vegetable oils are converted into vanaspati ghee by: ★★★
(a) oxidation (b) substitution
(c) hydrogenation (Ni catalyst) (d) saponification

8. In a soap micelle, the hydrocarbon tails point: ★★☆
(a) outwards, into the water (b) inwards, into the oily dirt
(c) in random directions (d) along the water surface

ASSERTION–REASON OPTIONS: Q9, Q10

(a) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
(b) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of Assertion (A).
(c) Assertion (A) is true, but Reason (R) is false.
(d) Assertion (A) is false, but Reason (R) is true.

9. **Assertion (A):** Diamond is a non-conductor of electricity, whereas graphite is a good conductor.
Reason (R): In diamond each carbon atom is bonded to four other carbon atoms, while in graphite each carbon atom is bonded to only three, leaving electrons free to move. ★★☆

10. **Assertion (A):** Detergents can be used for washing clothes even in hard water.
Reason (R): Detergents are sodium salts of long-chain carboxylic acids. ★★☆

SECTION B: SHORT ANSWER — 3 × 3 = 9 marks

11. (a) Why can carbon form neither C^{4+} nor C^{4-} ions? (b) Name the two properties of carbon that explain the huge number of carbon compounds. [3] ★★★

12. Write the IUPAC name and the functional group of:
(i) CH_3CH_2Br (ii) CH_3COCH_3 (iii) CH_3CH_2CHO [3] ★★★

13. How will you distinguish between ethanol and ethanoic acid by a chemical test? Write the balanced equation of the reaction. [3] ★★☆

SECTION C: LONG ANSWER — 1 × 5 = 5 marks

14. An organic compound **A** (C_2H_6O) is used as a solvent and in cough syrups. Heated with excess hot concentrated H_2SO_4 at 443 K, it gives a gas **B** that decolourises bromine water. Oxidised with alkaline $KMnO_4$, **A** gives **C**, which reacts with **A** in the presence of an acid to give a sweet-smelling compound **D**. Identify A, B, C and D and write the three equations. Name the reaction that forms D, and state what D gives when heated with sodium hydroxide. [5] ★★★



EXAM RULE

Mr Bean wali copy-paste nahi chalegi! Pehle khud likho, answer key **page 3** *par hai.*